Current CV

Professional Profile

I am a sophomore at UTSA with a major in Physics and minors in Mathematics and Astronomy. Currently, I work in the UTSA MAE Group's lab at the University of Texas at San Antonio (UTSA), led by Dr. Chris Packham, where we research active galactic nuclei (AGN) and infrared imaging of tori. Through this, I am learning techniques such as deconvolution and FortesFit for galaxy imaging, as well as how to map torus structures and refine data to uncover critical phenomena of AGN.
As of late, I have been lucky enough to begin working on a project by using AI to model AGN deconvolution, find parameters within data, and fit blackbody curves. Hopefully through this automation, new properties of AGN can be found.

Core Skills

Skill Area	**Details / Notes**
Technical	Python, Linux, Programming, Machine Learning
Communication	Presentation, Mentoring, Coaching
Leadership	Leadership, Upholding Standards, Coaching
Research	Data Analysis, Machine Learning, Research Methods
Project Management / Planning	Project Management, Project Planning, Upholding Standards

Education

- Highschool Diploma | James Madison Highschool, San Antonio TX | Aug 2020 - May 2024
  GPA: 4.3 on 4.0 scale

- UTSA | August 2024 - Present
  GPA: 4.0 on 4.0 scale

Research

April 30th 2025:
Presented at the UTSA Honors ELF.

May 7th 2025:
Presented the poster at Cosmic Horizons conference hosted by NSF-Simons Cosmic AI Institute.

May 5th-9th 2025:
Attended the Astro-AI Boot Camp hosted by NSF-Simons Cosmic AI Institute.

Summer 2025:
First Project! Mapping Polar Dust Properties while accounting for Emission Line Contamination. Completed: Presented:

Career Summary

Jun 2024 - Aug 2024

NEISD, San Antonio Custodian (Full Time)
Outline:
I worked full time delivering excellence to the school facilities. As a new custodian I worked with diligence bringing smiles to kids of Stahl Elementary. Cleanliness allows others to do their jobs, and proper workflow must be maintained.
Key Responsibilities

- Supporting team of 5 and project managing delivery of tasks for school procedures.
- Meeting and greeting students, ensuring they feel welcome and comfortable
- Providing a safe learning environment, safeguarding students and notifying relevant colleagues of any signs of students who may need external support
- Movement of School property to ensure correct timelines and workplaces for students and teachers.
- Cleaning teacher work areas for great efficiency

Jun 2023 - Aug 2023

Whataburger, San Antonio
Whataburger Food Production Assistant (Part Time)
Outline:
Worked in food production learning cleanliness techniques for food, time management, as well as customer relations.

Affiliations

Additional

Hobbies & Interests

Coaching, Track & Field, Sprinting, Games, Yu-Gi-Oh, Nutrition, Reading

Awards

Scholarships & Awards

- NSF APPEX Scholarship — $8,000/year (2024–2028)
- Distinguished Presidential Scholarship — $5,000/year (2024–2028)
- Honors College Collaboration Housing Scholarship — $1,500 (2024)
- South Texas AAU Janice Henderson Scholarship — $1,250 (2024)
- James Madison Hall of Honor Inductee – Track & Field (2024)
- Academic All-District Team – NEISD (2020–2024)

Microcredentials

Athletics

Track & Field Sprinter

- James Madison High School (2021–2024)
- 3D Athletics MMXC Club (2021–2023)
- Randolph Roadrunners Club (2023–2024)

Coaching

- UTSA Sprint Group (2024-present)
- Randolph Roadrunners (2025-present)

References

- Dr. Aaron Cassil — aaron.cassil@utsa.edu
  UTSA APPEX | Mentor
- Dr. Chris Packham — chris.packham@utsa.edu
  PI | Head of UTSA MAE group
- Dr. Gail Taylor — gail.taylor@utsa.edu
  UTSA APPEX | Mentor | Retired
- Mrs. Erica Fisher — erica.p.carson@gmail.com
  Roadrunner Coach | Mentor

CV

William Olson

Aspiring astrophysicist researching Active Galactic Nuclei using machine learning and mid-infrared imaging.

Email: william.olson@my.utsa.edu | **Phone:** 210-274-3958 | **Location:** San Antonio, TX
GitHub:https://github.com/2xwlll | linkedin | linktree

EDUCATION

University of Texas at San Antonio (UTSA) — *B.S. in Physics, Minors in Mathematics and Astronomy*
Aug 2024 – Present | **GPA:** 4.0 / 4.0

- **Honors College** and **NSF-APPEX Scholar**
- Focus areas: Astrophysics, Machine Learning, Data Science
- Classes

Dont care anymore

James Madison High School, San Antonio, TX
Aug 2020 – May 2024 | **GPA:** 4.3 / 4.0

- **James Madison Hall of Honor** – Track & Field (2024)
- **Academic All-District Team** (2020–2024)
-
- Add classes What I've done well in:

RESEARCH EXPERIENCE

Undergraduate Researcher — UTSA Mid-Infrared Astronomy and Education (MAE) Group

Supervisor: Dr. Chris Packham | **Aug 2024 – Present**

- Investigating **Active Galactic Nuclei (AGN)** and **infrared imaging of dusty tori**.
- Applying **deconvolution** and **FortesFit** to refine mid-infrared galaxy imaging data.
- Developing an **AI pipeline** to automate AGN deconvolution, parameter estimation, and blackbody curve fitting.
- Exploring how machine learning can uncover hidden features in AGN structures.

Current tasks:

- Developing 'Ionization Cones Automated' — a UNet-based ML model for detecting ionization cones in AGN images.
- Helping Integrate

Presentations:

- *UTSA Honors ELF Symposium* — April 2025
- *NSF–Simons Cosmic AI Institute "Cosmic Horizons" Conference* — Poster Presentation, May 2025
- *Astro-AI Boot Camp, NSF–Simons Cosmic AI Institute* — May 2025

TECHNICAL SKILLS

Programming: Python, NumPy, SciPy, Matplotlib, scikit-learn, TensorFlow, Linux
Data & Imaging: FITS data handling, deconvolution, image segmentation, 2D modeling
Machine Learning: UNet architectures, data augmentation, supervised learning
Research Tools: Git/GitHub, LaTeX, VS Code, Neovim
Other Skills: Presentation Design, Research Communication, Independent Project Planning

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AWARDS & SCHOLARSHIPS

NSF APPEX Scholar — $8,000/year (2024–2028)

Distinguished Presidential Scholarship — $5,000/year (2024–2028)

Honors College Collaboration Housing Scholarship — $1,500 (2024)

South Texas AAU Janice Henderson Scholarship — $1,250 (2024)

Honors College Student Advisory Board Stipend — $500/semester (2025–2026)

James Madison Hall of Honor – Track & Field (2024)

PROFESSIONAL EXPERIENCE

Custodian (Full-Time) — North East ISD, Stahl Elementary

Jun 2024 – Aug 2024

- Managed workflow and task delegation within a 5-person team ensuring facility safety and cleanliness.

- Developed leadership and planning skills by coordinating day-to-day operations for school readiness.

Food Production Assistant (Part-Time) — Whataburger, San Antonio, TX

Jun 2023 – Aug 2023

- Maintained food safety and quality control in a high-paced environment.
- Strengthened teamwork and customer communication skills.

AFFILIATIONS/ LEADERSHIP/ SERVICE

Coach — UTSA Sprint Group
Supervisor: Self-directed | Aug 2024 – Present

- Organize and lead sprint training sessions for university athletes, focusing on technique, acceleration, and consistency.
- Mentor peers interested in competitive track and strength training programs.

Coach — Randolph Roadrunners Track & Field Club
Head*: Coach Erica Fisher | San Antonio, TX | Aug 2025 – Present*

- Coached ~100 youth athletes (ages 5–18), improving sprint mechanics and confidence through individualized form analysis.
- Streamlined meet logistics, reducing total event duration through improved scheduling and coordination.

Sprinter — 3D Athletics MMXC Club *(2021–2023)*

Sprinter — Randolph Roadrunners Club *(2023–2024)*

Member — Society of Physics Students, UTSA *(2025–Present)*

Undergraduate Member — UTSA Honors College Student Advisory Board
Supervisor: Dr. Jill Fleuriet | Aug 2025 – Present

- Analyze student event participation data (~40% attendance rate) to identify barriers to engagement.
- Collaborate with a 4-person team to develop and propose data-driven strategies for increasing student involvement.
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- *James Madison HS Outreach Speech —*

ADDITIONAL INTERESTS

Astrophysics, Artificial Intelligence, Track & Field Sprinting, Nutrition, Yu-Gi-Oh, Mentorship

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REFERENCES

- **Dr. Chris Packham** — PI, UTSA MAE Group | chris.packham@utsa.edu
- **Dr. Aaron Cassil** — Mentor, UTSA APPEX | aaron.cassil@utsa.edu
- **Dr. Gail Taylor** — Mentor, UTSA APPEX | gail.taylor@utsa.edu
- **Mrs. Erica Fisher** — Coach, Randolph Roadrunners | erica.p.carson@gmail.com

Personal Statements

Active galactic nuclei (AGN) are among the universe's most extreme laboratories. Powered by accretion onto supermassive black holes, they release enough energy to reshape their host galaxies—driving outflows, regulating star formation, and influencing cosmic evolution. Yet despite modern telescopes like the James Webb Space Telescope, many of the signatures of AGN feedback remain difficult to identify and quantify. The data are complex, noisy, and often interpreted by eye. Early in my research career, I became fascinated not only by AGN themselves, but by this disconnect between observational capability and analytical methodology—and I began asking whether artificial intelligence could bridge that gap.

I joined the MAE Group at the University of Texas at San Antonio to explore this question through the study of ionization cones: conical regions of gas illuminated by AGN radiation that encode the geometry and orientation of the central engine. These structures are faint, asymmetric, and frequently obscured, making them notoriously difficult to identify reliably. While astronomers traditionally locate them through manual inspection of emission-line maps, this approach limits reproducibility and prevents large-scale statistical studies. I wondered whether a machine could be taught to recognize the same morphological patterns that astronomers learn through experience—and perhaps do so more consistently.

Under the mentorship of Dr. Chris Packham, I designed and implemented a complete machine-learning pipeline to address this problem. Starting from raw JWST-style data formats, I built a Python-based system to preprocess, normalize, and augment spectral imaging data. Because real observations are limited and heterogeneous, I generated physically motivated synthetic [O III] emission maps that simulated realistic ionization cones, including variations in opening angle, orientation, asymmetry, dust attenuation, and noise. This allowed me to retain precise ground truth while exposing the model to the messy conditions encountered in real observations.

Using this dataset, I trained a two-dimensional U-Net convolutional neural network to segment ionization cone structures automatically. The model achieved strong performance, accurately identifying cone morphologies even when emission was faint or partially obscured. More importantly, it generalized well from synthetic data to real observations, demonstrating that physically informed simulations can effectively train models for astronomical discovery. Through iterative testing and failure analysis, I learned how dataset design, loss functions, and architecture choices shape scientific outcomes—a lesson that now informs how I approach any computational problem.

This project marked my transition from student to independent researcher. Before it, I had never trained a neural network or built a data pipeline from scratch. By the end, I had written every line of code, designed evaluation metrics, diagnosed failure modes, and refined the model through targeted experimentation. Just as importantly, I learned to translate physical intuition into algorithmic structure—to teach a computer to "see" what astronomers see.

My approach to research mirrors my approach to athletics. As a track athlete, I break down each stride to improve efficiency and performance. In research, I apply the same mindset: analyze bottlenecks, remove wasted motion, and build systems that channel effort effectively. Artificial

intelligence, to me, is not a replacement for physical insight, but a force multiplier—allowing human curiosity to scale to datasets and complexities we could never analyze alone.

Looking ahead, I aim to extend this work to multi-wavelength and three-dimensional spectroscopic data, enabling automated measurements of AGN outflow geometry and kinematics across large surveys. More broadly, I plan to pursue a Ph.D. in astrophysics focused on the integration of machine learning with observational astronomy, with applications ranging from AGN feedback to exoplanet characterization and time-domain surveys. My long-term goal is to lead research that develops transparent, physically grounded AI tools that enhance—not obscure—scientific understanding.

From early lessons in persistence taught by my family to my current work at the intersection of astrophysics and artificial intelligence, I have learned that discovery depends as much on how we ask questions as on the tools we use to answer them. With the right foundation, curiosity becomes structure, and structure becomes insight. I am excited to continue building on that foundation through further research and training, contributing to a future where we not only observe the universe in greater detail, but understand it more deeply.

Hey sam, thanks so much for doing this. My best and newest draft is at the end of the docs. You can read through my helpful stuff and my other drafts if you want to, but you dont have to :) thanks so much again.

REU info - Google Docs

Bing Videos

Don’t type too much autobiography, but really turn it around to research.
Persuasive story on why they should take you. PASIOANTE YE!! REALLY FOCUS ON THE

1. Know the parameters
    a. Font, spacing margins
    b. word/character/page limits
    c. Prompt
    d. Do they really want a P.S.? (or a statement of intent or an essay)
2. Who are you?
    a. Think outside the box
    b. Make your story interesting in a fun little way (some girl compared her story to shopping at costco)
    c. Once you “wake them up”, you have an advantage
    d. Separate random information from a good story
    e. Make lists of questions for everything (who, what, when, where, why, how for this)
        1. Why astrophysics
        2. What is astrophysics to you?

3. When did you get into astrophysics ect…

f. Ask them non-typical questions and give non typical answers
    1. What experience in my life has turned me into an astrophysicist?
g. DON'T TELL THEM FACTS TELL THEM A STORY
h. Get a hook, but don't plead, beg, give facts, justify, or excuse.

Why should you be in this program.

I believe I should be in this program because I bring the bang to the buck.

Personal Statement:

Spirals, I can't get them out of my head. An explosion of energy, winding, coiling back to the center, ending coldly at a dead stand still. This is a spiral as I know it and I can't help but see it in everything, existing, staring right at me. Whether it's the circular form of leg cycles when sprinting, the pattern of galaxies or my daily schedule, everything I seemed to do had something to do with those darn spirals.

Era 1: 2006-2012 The Normality

The spiral starts with an explosion of energy.

From when I was born, my life was a Mormon's dream. "Family home-evening" every Monday, reading the bible with one paragraph for every family member while reflecting on it, and going to church and fasting those first Sundays.
It was fun.
I would never berate Mormons. I respect what they do.
They try to be morally good people, while following Mormon Belief. That is their complete end goal for life.
We stopped this pattern though.
There was a lot of backlash from our almost entirely Mormon family, claiming we would spend eternity in "the worser heaven" or the telestial kingdom because we stopped believing with our utmost power in the three powers: God, Jesus, and the Holy Ghost.
To this, my Dad couldn't do anything to this, but stop and ask why.
More explosions, the beginning of the spiral.

Era 2: (2012-2018) Change

A spiral continues by unfurling

I could easily lift a lot of weight compared to others in middle school. My strength per weight ratio was unrealistically high for never lifting before. Before it was my belief in God that made

me fast and strong. Now it was the steps I took to get there. On the other hand, my academics exploded.

I forgive myself. With everything passing by in time, once something is done, it is done. That is how the world works. Going back in time is deemed: “A pipe dream” by many esteemed people. I have to live in the present and work hard. So with my “hard-nose”, pedal to the metal attitude, I forgive myself for basking in sweet relaxation.

“It’s ok”, I reassure myself. “This is a strategic retreat from working hard, because with proper rest you will be able to work even better later”. I get anxious. I need to be doing something. Homework,

Quizzical:

A stream of skepticality crept across my brother's face. I had just told him how the red hulk was stronger than the original green hulk and explained why that was. However, green hulk was the same hulk we had both watched (in lack of better words) “whoop some butt” on television. So to him, it was highly questionable that this new “off brand” red hulk would be stronger than the original. I explained to him in every way the red hulk could be stronger than the original green hulk: red hulk can absorb radiation (just some marvel weirdness they invented to make him different). My younger brother was a bit more convinced when I told him this, but he still had some uneasiness. Skepticism is very important in decision making and the scientific process as a scientist, but for me it always seemed like a daily part of my life most likely because of my Dad.

We used to be mormon, but with how my dad was always skeptical and cautious about every decision, we eventually broke away. Asking “why” in the Mormon church did not give logical answers. There was major backlash from our Mormon dominant family, but my Dad simply didn’t find what the church said logical. From us “quitting” the Mormon church and onwards, my Dad heavily stressed questioning everything. From what we did in school, to our daily habits, he wanted us to ask why we were doing what we were doing and if it made sense.

At the same time, I was obsessed with space. How could a cosmos so big, not have any other forms of life, or even other intelligent life forms? Why isn’t everyone always looking for life in space, as wouldn’t the payoff of finding another species be invaluable? But then my Dad’s skepticism came into play. Space is big, far, and hard to study. But then the opposite question arises. Why even study astrophysics at all, when the payoff is usually more helpful on earth?

Tone down religion, its this thing of I don’t know why but I would just stare out at the stars and it would fill me with curiosity. To make them believe I have a legit a met reason to participate in their program.

I

Too much on the poster and not too much on the statement. Why are you interested in the program? Why should they invest in you? Altruism and why will I benefit them? Still sort of a

selfish thing. US will be a better country because of more scientists. What is their greed that makes them interested in you and how can I match that?

Little Dad intro about him asking and us breaking away form the church.

I want to write a story about alltruism: selfishness versus selflessness. The true faces of evil. Go to a lot of office hours grind office hours. Don't ask society what to do. Ethiopian food

Ologiies

Biology

Draft 3:

My father is a quizzical person. He just asks questions, always wanting answers. This led to our family eventually parting ways with the Mormon church, a huge turning point in my life. The sudden, drastic change in lifestyle also had me asking questions. Before I knew it, my fathers nature of questioning had influenced me. Asking questions eventually led me to the realm of science. Further still, I was perplexed by space and its properties.

Space had always seemed mysterious to me. How could something so big, still be an obscure mystery to all people? Dark matter, "little red dots", and extraterrestrial life, there are so many topics of interest for me. In my introductory Physics classes, both in highschool and college, I have started to explore the world of physical sciences and am eager to apply my knowledge to cutting edge research in astronomy. I have started to "shadow" in a lab at UTSA and I go to the lab meetings every week and have slowly started to learn deconvolution, FortesFit, and other galaxy imaging tools/techniques. I find it incredible that we can see an entire galaxy mapped out on a laptop and alter the galaxy so it gives us the accurate data we want. While mapping a torus doesn't quite match up with [whatever the REU is doing], this wonderful REU program will be the springboard for my future, propelling me into a pathway of excellence. Graduate school is the perfect match, because I could dive into a deep corner of research delving into everything specific in an area. To really be set up for graduate school, I'll need other experience which is where this [REU specifics] comes in.

Why not another program though? If I truly am interested in space like I say, then why not another REU that could be easier to get into or another option entirely like summer research for my university? The answer is simple. [Specific practice the REU does] and [people in charge of the program] puts this program above the others. The [Professor of the specific program] has prestige in [what they are good at] and both them and [another professor in charge] are wonderful at facilitating growth and learning. On the other hand, [the practice the REU will explore] might not be super in-line with the torus research I will be doing over the course of my undergrad, but it is important to give me both experience in another field when I apply to some of the more crazy graduate programs.

By participating in this REU of [specific area the REU practices], I hope to expand my own horizons. Not just for myself, but for the people, the astrophysicist, and the undergraduate research program I was a part of. I will become a better scientist to more fully explore space and its mysteries.

My dad loves asking questions. He taught me to reason, question, and hunt for answers. He always said, like a construction worker, I need to have a solid foundation of learning and

reasoning before I can begin building ideas. I was not always a top student: I “failed” kindergarten and more recently, received 75 percent on a college chemistry test. But, I always kept working: I passed 1st grade with flying colors and got a perfect score on my college chemistry final. Combining my work ethic with my curious nature, I continue to establish my foundation of learning and reasoning. Over time, I found that science offers the best framework for seeking answers. Among all its mysteries, space captures my imagination the most, due to its vastness and complexity.

My academic journey has been shaped by a deep interest in astrophysics. As a young kid, I was fascinated with Neil DeGrasse Tyson’s version of Cosmos, which blossomed my thoughts on space and its properties. Later, in my introductory physics courses in high school, I began exploring the fundamental principles of physical sciences, which sparked my desire to contribute to astronomy research in college. Currently, I am shadowing the GATOS Collaboration lab at UTSA, where we currently research active galactic nuclei (AGN) and infrared imaging and learning techniques such as deconvolution and FortesFit for galaxy imaging. I recently learned mapping torus structures in galaxies, refining data to reveal critical phenomena like AGN (active galactic nuclei). This hands-on experience has taught me the importance of precision, collaboration, and curiosity in scientific research. Hopefully, I can bring these skills to both improve and utilize them on [the REU’s projects].

The reason I am applying to this REU is my ultimate goal of earning a Ph.D. in astrophysics with a focus in the usage of AI in telescopic instruments and determining exoplanets. This REU will provide the perfect opportunity to broaden my research experience, particularly in [specific area the REU focuses on], and continue to develop the skills needed for my doctoral degree like data analysis through programming, interpersonal skills, and effective communication.

What sets this REU apart is [specific practice or feature of the REU program, e.g., a focus on computational astrophysics, observational techniques, or interdisciplinary collaboration]. In particular, I am drawn to the work of [Professor X], whose research on [specific topic, e.g., dark matter distribution or star formation] resonates with my interests. Their recent publication on [specific paper or project] highlights the area of interest for the university, (somehow say you liked the article and what you liked about it) and I am eager to contribute to similar projects during this program. The opportunity to work with [another faculty member or group] further enhances this program's appeal, as their expertise in [specific area] aligns closely with the torus/AGN research at UTSA.

By participating in this REU, I aim to expand my scientific horizons and contribute meaningfully to the program’s research efforts. This experience will not only refine my technical and analytical skills but also prepare me for the rigors of graduate study and a future career in astrophysics. I am excited to immerse myself in the challenges and rewards this program offers, as the next crucial step, in my journey to explore the mysterious universe.

Cover Letter:

Dear,

- HAWAII

My dad loves asking questions. He taught me to reason, question, and hunt for answers. He always said, like a construction worker, I need to have a solid foundation of learning and reasoning before I can begin building ideas. My dad also emphasized hard work. I was not always a top student: I “failed” kindergarten and more recently, received 75 percent on a college chemistry test. But, I always kept working: I passed 1st grade with flying colors and got a perfect score on my college chemistry final. Combining my work ethic with my curious nature, I continue to establish my foundation of learning and reasoning. Over time, I found that science offers the best framework for seeking answers. Among all its mysteries, space captures my imagination the most, due to its vastness and complexity.

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By participating in this REU, I aim to expand my scientific horizons and contribute meaningfully to the program’s research efforts. This experience will not only refine my technical and analytical skills but also prepare me for the rigors of graduate study and a future career in astrophysics. I am excited to immerse myself in the challenges and rewards this program offers, as the next crucial step, in my journey to explore the mysterious universe.

Has artificial intelligence (AI) eroded hard work? It seems that the general consensus is around a little bit. Some argue yes, because the AI itself creates the work done, while others argue that cheating has always been available and AI is no exception. I like AI. Specifically, I like AI in space. Since I had watched Neil DeGrasse Tyson's version of Cosmos, I have been fascinated with the stars. What is out there? What is it made of? I started- slowly but surely- started learning about the universe.

When I learned of AI in early 2021, I dismissed it, skeptical of another "this is the next future". Next when I really took a look at it was ChatGPT. I asked extremely obscure Yu-Gi-Oh deck building questions and the language model failed miserably. I again dismissed the entire idea of AI (here I probably should have researched more about the topic. Everyone knows that the third time's the charm, and this round I had begun shadowing a lab at the University of Texas at San Antonio. The PI, Dr. Chris Packham, had mentioned about the surgance of AI in astronomy. I typed up papers on it. AI was being used in medicine (Halicin at MIT), biology, and was also starting to gain major traction in astrophysics. I was perplexed, what is AI then and how is it useful in the field of research? How could it be bad for research?

Has artificial intelligence (AI) eroded hard work? Growing up with the Church of Latter Day Saints hard work was emphasized. Paying 10% of your income, reliably reading the scriptures, and fasting the first Sunday of every month. The Mormon Church is not easy to take practice in, but the community it creates is truly wonderful because the practices do set up the person doing them to be a better person. However, later we left this community. Stemming from my dad's questioning of the ways of the church, and the practicality of it, we eventually left the church- a huge turning point in my life.

This can similarly be compared to AI. When I left the church, I didn't change the person I was, just the identity. AI can never replace humans, or the work we do. AI can make predictive guesses off of data, but can't perfectly replicate what humans can do or create. Leaving the mormon church made me question theories and thoughts more, but it didn't pull me away from the values it taught and certainly not away from

"You can get the boy out of church, but you can't bring the church out the boy." My friend once said this to me after learning I used to be Mormon. I hold no qualms with Mormons. I actually quite respect them for standing up for what they believe in. The community that they create is so wonderful. But, why did my friend give the remark? Am I exceptionally upright? No, I don't think so. So I realized moments later, that she was just making a random remark. I think too hard sometimes. Seriously though, it is in my blood. My dad was the one who prompted our departure from Mormonism. He asked questions to his "Brothers", looked online, pondered some more, and eventually- we just left. Our family (almost entirely Mormon) was not necessarily sad, or angry, but rather very confused on why we left. All we had done was simply asked questions, and not really gotten answers. But time passed. Both sides eventually moved

on and kind of just lived normally. We still love each other very much. I feel that when I moved away from the Mormon church, my focus in life became more academic. I asked questions as well, because my dad stressed it to me and my siblings so much. I began to be hungry for answers. Maybe even more so than my dad.

My friend once said to me, 'You can get the boy out of church, but you can't take the church out of the boy.' She was joking about my past as a Mormon, but her words made me reflect on how questioning and seeking understanding have shaped me. Overall, Mormonism is quite nice. Mormons stand up for what they believe in, form a close knit community, and practice good virtues to one another and non-church members. These were heavily stressed to me, and I took it to habit. Growing up, my family left the Mormon church not out of anger or "rebellion", but because we had questions that couldn't be answered. While our decision was surprising to some of our outer family, we still love and support each other, but go different ways when it comes to religion. Still, this was a major shift in my life and it was not easy to leave the church, friends I made, and mentors I loved.

As I stepped away from religion, I found myself focusing more on academics in life. Maybe add something here where we talk about carrying over those good attributes In elementary, I was a competitive little guy. Competing with others on multiplication tables among other things helped prop my first financial plan as a kid. Like my uncle and grandfather, I wanted to be an actuary. But as time went on, I found that math is quite fictional to me. I don't mean that it isn't useful, but rather I didn't find math to be the most practical.

From this foundation, I have gotten into research at UTSA through shadowing the GATOS lab directed by Dr. Chris Packham, and currently aspire to get a P.h.D. in AI in instrumental research of exoplanets, speeding up our search for life in space.

What sets this REU apart is [specific practice or feature of the REU program, e.g., a focus on computational astrophysics, observational techniques, or interdisciplinary collaboration]. In particular, I am drawn to the work of [Professor X], whose research on [specific topic, e.g., dark matter distribution or star formation] resonates with my interests. Their recent publication on [specific paper or project] highlights the area of interest for the university, (somehow say you liked the article and what you liked about it) and I am eager to contribute to similar projects during this program. The opportunity to work with [another faculty member or group] further enhances this program's appeal, as their expertise in [specific area] aligns closely with the torus/AGN research at UTSA.

By participating in this REU, I aim to expand my scientific horizons and contribute meaningfully to the program's research efforts. This experience will not only refine my technical and analytical skills but also prepare me for the rigors of graduate study and a future career in astrophysics. I am excited to immerse myself in the challenges and rewards this program offers, as the next crucial step, in my journey to explore the mysterious universe.

COVER LETTER -

My friend once said to me, "You can get the boy out of church, but you can't take the church out of the boy." She was joking about my past as a Mormon, but her words got me thinking: maybe she had a point. Growing up Mormon taught me a lot about the importance of

community, integrity, and helping others. These weren't just ideas—they were habits. I still admire the way Mormons stand up for what they believe in and support one another so strongly.

When my family left the church, it wasn't because we were angry or rebellious. We just had questions—big ones—that didn't seem to have clear answers. The decision wasn't easy. I said goodbye to friends and mentors who had shaped my early life, and our choice left some family members scratching their heads. But we stayed close, and I came away from that experience with two things: a deep respect for my roots and an insatiable hunger for answers. If I had to pin down what makes me tick, it's that drive to keep asking questions—and not just settling until I get to the truth.

That same curiosity found its way into every corner of my life, especially academics. As a kid, I was fiercely competitive. I raced through multiplication tables like they were Olympic events and even put together my first "financial plan" after learning about actuaries from my uncle and grandfather. I thought numbers would always be my path forward, but over time, I realized math felt incomplete to me. It was a great tool, sure, but I wanted something more—something that connected abstract ideas to the mysteries of the real world.

That's when I discovered astrophysics. The universe is a puzzle that combines math, science, and imagination, and every answer we uncover just raises more exciting questions. Through shadowing the GATOS lab directed by Dr. Chris Packham at UTSA, I've explored topics like torus/AGN research and discovered how much I love the process of using data to understand the cosmos. Now, I hope to pursue a Ph.D. in AI-driven instrumentation for exoplanet research, using machine learning to accelerate the search for life beyond Earth.

What sets this REU apart is [specific feature of the REU program, e.g., the emphasis on interdisciplinary collaboration or unique research facilities]. I'm especially drawn to the work of [Professor X], whose research on [specific topic] aligns closely with my interests. Their recent publication on [specific paper/project] caught my attention, particularly [something specific you liked about it]. I would love to contribute to similar projects during this program and gain valuable experience in [specific research skill or topic].

By participating in this REU, I aim to expand my scientific horizons and tackle challenging questions alongside talented peers and mentors. This experience will not only sharpen my technical and analytical skills but also bring me closer to my goal of understanding the universe—and maybe even helping humanity take its next big step toward finding life beyond Earth.

Personal

Active galactic nuclei (AGN) are among the universe's most extreme environments. Powered by accretion onto supermassive black holes, they release enough energy to reshape their host galaxies by driving outflows, heating gas, and influencing star formation. Understanding how this energy couples to its surroundings, known as AGN feedback, remains one of the central challenges of modern astrophysics. What first drew me to this problem was not only the physics itself, but a deeper question: despite revolutionary telescopes like the James Webb Space Telescope, why do we still rely so heavily on subjective visual inspection to interpret these systems?

I encountered this tension while working with the MAE Group at the University of Texas at San Antonio. Ionization cones, conical regions of gas illuminated by AGN radiation, offer critical insight into the geometry and orientation of the central engine. Yet in practice, these structures are faint, asymmetric, and often obscured. Identifying them typically involves manually scanning emission line maps, galaxy by galaxy. As I worked with these data, I began to see this approach as a bottleneck. If ionization cones follow physical patterns, then in principle those patterns should be learnable. The question became whether a machine could be taught to recognize them in a consistent and scalable way.

Answering that question required rethinking how the problem was framed. Rather than starting with existing observations, which are limited and heterogeneous, I focused on how ionization cones should appear under realistic conditions. I developed physically motivated synthetic [O III] emission maps that captured the variability expected in real AGN, including changes in orientation, opening angle, asymmetry, dust attenuation, and noise. These simulations provided a controlled environment in which structure, rather than brightness alone, carried the signal.

To test whether this structure could be learned, I trained a two-dimensional U Net convolutional neural network to segment ionization cones automatically. The model learned to identify cone morphologies even when emission was faint or partially obscured, and it generalized well when applied to real observations. More important than the performance metrics was what the failures revealed. When the model struggled, it exposed gaps in the training data or unrealistic assumptions in the simulations. Improving the results meant revisiting the physics, expanding the dataset, and refining how information was presented to the network. Progress came not from a single solution, but from an iterative cycle of questioning, testing, and revision.

This process reshaped how I see research. Early on, I had never built a data pipeline or trained a neural network from scratch. Confronted with an open ended problem, I learned to decompose it, identify what was missing, and rebuild until the system behaved in a physically meaningful way. The project became less about applying tools and more about developing judgment, learning how to decide what mattered and why.

That mindset extends beyond the lab. As a track athlete, improvement comes from analyzing inefficiencies and making small, deliberate adjustments. Research demands the same discipline. Artificial intelligence, in this context, is not a replacement for physical understanding,

but a way to extend it. When grounded in realistic assumptions, it allows questions to be asked at scales that would otherwise be unreachable.

I plan to pursue a Ph.D. in astrophysics focused on integrating machine learning with observational astronomy, particularly in data intensive problems like AGN feedback. By developing transparent and physically informed AI tools, I hope to help move astronomical analysis from subjective pattern recognition toward reproducible discovery. This project taught me that progress begins not with answers, but with the willingness to ask better questions and rebuild when the first approach falls short.

 Goldwater :(

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For my lab, I was really intimidated at first at the terminology. As my lab is all graduate students, it was hard for me to follow all of the abbreviations, acronyms, etc. As time went on and I kept learning, I once walked away from a lab meeting and thought to myself “Hey I could follow everything that was going on!”. This scholarship is all about potential. ALL about potential. How independent am I? Am I presenting? Resume is not a checklist. They look for potential, like a Conference. I presented at CosmicAI. Emphasize outreach. Join a tutoring club. Very forward thinking about how I think about research and what my plans are and their potential. **Due December 20th**

onca@utsa.edu
2nd Year
Major:
Physics (Concentration in Astronomy)
Minor in:
Astrophysics
Mathematics
GPA:
4.0
Highest Degree Anticipated:
Ph.D
Career Interest Statement:
I currently hope to become a tenure professor. I like to think I am very gifted in teach
Potential Recommendation Letters:
Dr. Chris Packham, tenure professor, member of aura.
Dr. Aaron Cassil, tenure, Assistant head of Biology, PI of the APPEX Research Fund
My New Japanese Physics Professor…? Really got to talk to him in office hours. Dr. Konno MTW 2:30pm - 3:30pm AET 3.376
Talk to Andrew Chapman about recommenders. Ask before Nov 1.

- Single-spaced
- 12-point Arial font
- 1-inch margins on all sides
- Page length, including bibliographic information, must not exceed 3 pages.
- **In the header at the top of each page, you must include your name and the name of your academic institution.**
- **The essay must include a title.**
- Format your Research Essay in either one or two columns on a page.
- As appropriate, utilize graphs, tables, and figures in the essay to explain or clarify results.
    - Figures, table captions and bibliographic citations may use 10-point Arial font.
    - Research essays that use mathematical expressions may use LaTeX or other document preparation software to enable proper formatting.
- **Do not include hyperlinks. Materials that are hyperlinked to your research essay will not be read.**
- Citation Format – For the Research Essay, use the citation format commonly used in your discipline. For space reasons, you may abbreviate titles, journal names, etc.
- Save and upload your Research Essay as a PDF file.

The Artificial Intelligence (AI) bandwagon is increasingly controversial. Some scientists choose to refute the use and importance of AI for three big reasons.

Describe your Career goal and professional aspirations. Will be used online.

What are your career goals and professional aspirations? Indicate which areas of math, science, or engineering you are considering pursuing in your research career and specify how your current academic program and your overall education plan will assist you in achieving this.

Intro: Statement of my research and why I like it.
Show my understanding, its body of research and where it is going
Show them how I have developed, or connect dots show them where I have gotten to now.
I will list all research experiences later so think broadly
Future directions

Describe any social and/or economic impacts I have encountered that have influenced in my research career.

Quality of research projects over quantity.

List up to 5 research projects you consider to be your most significant work associated with your interests in the sciences, math, or engineering. Be very explicit, “the data tells” “we hypothesize”, “etc”. Use I when applicable and “we” when applicable.

First project was shadowing FortesFit and learning my way around AGN and how they work/ are important to astrophysics AND human society on earth.

Listing research skills:

MAST browsing

Data manipulation

Data Unbiasing?

Advice for essay:
Be clear. Be concrete. Be concise. Can be a bit Jargon-y

December 20th for sure. Packham no konno no.

Only undergraduate classes for goldwater.
Show thought process. This is what I was thinking, this is why I did this… etc.

Goldwater Essay

Introduction

Active Galactic Nuclei (AGN) are among the universe's most powerful engines, driven by accretion onto supermassive black holes that can outshine their host galaxies. These central black holes release enormous energy through radiation and outflows. it is this energy that shapes gas dynamics, star formation, and the long-term evolution of galaxies. Understanding how AGN transfer this energy to their surroundings, a process known as AGN feedback, remains one of modern astrophysics' central challenges.

A particularly revealing signature of this feedback is the ionization cone: a conical region of excited gas illuminated by the AGN's intense radiation field. These cones reveal the geometry and orientation of the central engine and trace how radiation escapes through the surrounding dusty torus. Their morphology encodes the coupling between the nucleus and the interstellar medium, making them key to understanding the co-evolution of galaxies and black holes. Yet identifying ionization cones in spectral imaging data remains notoriously difficult. The emission is faint, asymmetric, and often obscured by dust or instrumental noise. Traditionally, astronomers have relied on tedious manual inspection (painstakingly examining [O III] emission maps, one galaxy at a time) to infer cone structures.

When I joined the MAE Group at the University of Texas at San Antonio, I was struck by a paradox: despite having cutting-edge telescopes such as JWST, capable of detecting the faintest galactic structures, our analysis methods still depend on subjective visual recognition. This limitation prevents large-scale statistical studies of AGN feedback, leading me to wonder whether artificial intelligence could overcome that bottleneck. Could a computer learn to identify ionization cones automatically, even when they are faint or irregular?

I hypothesized that training deep learning models on realistic, synthetic [O III] emission maps—data simulating the messy conditions observed by telescopes—could yield an algorithm capable of detecting and segmenting ionization cones autonomously. This approach would not only accelerate discovery but also make results reproducible and scalable across vast datasets. The project thus became a synthesis of two passions: astrophysics and artificial intelligence. My goal was to translate the interpretive power of the human eye into a quantitative, automated process capable of revealing patterns invisible to traditional analysis.

Methods

I conducted this research under the mentorship of Dr. Chris Packham, whose expertise in AGN infrared imaging and torus modeling guided the astrophysical foundation of the work. My focus, however, lay in designing and implementing the computational system from the ground up. I built a Python-based data pipeline to preprocess JWST Integral Field Unit (IFU) data cubes in FITS format. This pipeline automates normalization, denoising, resizing, and augmentation, producing consistent inputs for machine-learning analysis. Each stage was designed for reproducibility, with custom scripts tracking parameters and outputs.

Because real JWST datasets are limited in number and often vary in quality, I generated synthetic [O III] emission maps to form the core training data. Each map contained one or more ionization cones represented by two-dimensional Gaussian profiles with variable opening angles, orientations, asymmetries, and noise levels. I introduced additional physically motivated distortions—dust attenuation, pixel noise, and partial obscuration—to mimic realistic observation conditions. This strategy preserved precise ground-truth labels while teaching the network to recognize the morphological diversity expected in real AGN.

To model spatial structures, I implemented a two-dimensional U-Net convolutional neural network, widely used in biomedical image segmentation for its ability to capture both global context and fine-scale features. The architecture consists of an encoder that compresses spatial information and a decoder that reconstructs detailed segmentation masks. I adapted it for astrophysical imagery by fine-tuning kernel sizes, normalization layers, and activation functions for optimal feature recovery. Training used a Dice loss function, which measures overlap

between predicted and true cone regions—a critical choice for imbalanced datasets where cone pixels occupy only a small fraction of the image.

The final dataset comprised roughly 1,100 samples: 1,000 synthetic [O III] maps, 20 real JWST examples, and 80 altered real augmentations in which true images were slightly modified to increase variability. I adopted a 90–10 data split, emphasizing synthetic data for controlled learning while reserving real data to prevent overfitting to telescope-specific artifacts. Training employed the Adam optimizer (learning rate = 10^{-4}) over 30 epochs, with dynamic augmentations such as rotation and flipping to promote generalization.

Throughout the project, I took full ownership of development and analysis. I wrote every preprocessing, training, and evaluation script; designed visualization tools for inspecting segmentation outputs; and implemented logging systems to monitor learning behavior. To assess performance, I computed Dice coefficient, precision, recall, and Intersection-over-Union (IoU), and visualized overlays between predicted masks and ground-truth cones to diagnose systematic errors.

Results

The U-Net exhibited stable and efficient learning. Training and validation losses both decreased rapidly during the first ten epochs, then gradually converged around epoch fifteen. The validation Dice coefficient plateaued near 0.91, indicating strong agreement between predicted and true cone regions.

Quantitatively, the final model achieved a Dice score of 0.91, precision of 0.85, recall of 0.99, and IoU of approximately 0.84. This means the algorithm correctly identified nearly all true ionization cone pixels while maintaining a low false-positive rate. In astrophysical terms, it reliably detected extended emission structures without spurious segmentation of surrounding diffuse gas. Such balance is crucial: missing a genuine cone feature would bias morphological statistics, whereas an occasional false pixel can be filtered during post-processing.

Qualitative performance mirrored these metrics. When applied to both synthetic and real data, the network accurately traced asymmetric and partially obscured cones, even under moderate noise levels. Figure overlays showed strong spatial alignment between predicted and true emission regions. Notably, the model learned to infer the directionality of cones—an emergent property not explicitly encoded during training—demonstrating that it captured the underlying geometric structure rather than simply correlating pixel intensities.

A few failure cases offered valuable diagnostic insight. Extremely diffuse emission or dual-cone geometries occasionally led the model to merge structures or truncate cone tips. By generating additional synthetic examples with broader variations in opening angles and intensity gradients, I retrained the network and substantially mitigated these issues. This demonstrates that performance scales directly with dataset diversity.

This experiment provided a proof of concept: deep learning can robustly detect ionization cones under conditions that often frustrate manual inspection. Beyond immediate metrics, it showed that integrating physical realism into synthetic training data enables models to generalize effectively to telescope observations. In essence, the algorithm learned the “morphological language” of AGN outflows—the shapes and patterns that encode how energy propagates through galactic gas.

Future Research

The next phase will extend the current pipeline to fully real JWST datasets, particularly narrow-band [O III] imaging or IFU spectroscopy of nearby Seyfert galaxies. My immediate goal is to test the model’s transferability: how a network trained primarily on synthetic data performs on true observations without manual fine-tuning. Success would mark a crucial step toward automated AGN morphology classification.

To improve realism, I am developing a second generation of synthetic data incorporating hydrodynamically inspired cone structures, irregular gas distributions, and varying dust extinction. These simulations bridge the gap between idealized Gaussians and the complex morphologies observed in nature. I also plan to explore multi-wavelength extensions, training the model on combined [O III], Hα, and [N II] emission maps to identify correlations across different ionization states.

Longer-term, I aim to expand the pipeline to three-dimensional data cubes, applying 3D convolutional networks capable of segmenting cones directly in spectral-spatial space. This would enable measurement of outflow velocities and inclination angles from IFU data. Ultimately, I intend to scale the method to large surveys such as SDSS-V and future JWST programs, facilitating population-level studies of AGN feedback. Such analyses could reveal how black hole activity varies with host galaxy type, environment, and evolutionary stage, providing empirical constraints for cosmological simulations.

Conclusion

Before this project, I had never built a complete data pipeline or trained a neural network on astronomical observations. Over the course of this research, I developed the computational and analytical independence required to design, implement, and validate a complex system from first principles. I learned to translate astrophysical intuition into algorithmic form—to teach a computer to “see” what astronomers see.

Equally important, I gained a deeper understanding of AGN physics, from radiative transfer that excites [O III] emission to the geometrical implications of torus opening angles. Working under Drs. Packham and Lulu Zhang, I learned to connect physical theory, observational data, and computational modeling into a coherent framework.

This experience also reshaped how I approach problem solving. In track and field, I analyze each stride to optimize efficiency; in research, I apply the same mindset to streamline analysis pipelines and model architectures. Efficiency—whether in running or research—is about channeling energy with purpose. I now view AI as a natural extension of that philosophy: an instrument allowing human curiosity to reach further and faster.

The Goldwater Scholarship would enable me to continue pursuing this intersection of astrophysics and machine learning, expanding into topics such as exoplanet characterization and time-domain astronomy. My long-term goal is to earn a Ph.D. in astrophysics and become a professor leading research at the boundary between physical science and artificial intelligence. By developing transparent, reproducible AI tools for astronomy, I hope to help the community uncover hidden structures in the universe and transform how we interpret the cosmos.

This project taught me that with the right combination of mentorship, independence, and perseverance, I can turn an idea into a working scientific contribution. From faint emission lines to clear computational insight, I learned that discovery begins not with what the telescope sees, but with how we choose to look.

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Other Draft

Machine Learning Identification of Ionization Cones in Active Galactic Nuclei

Introduction

Active Galactic Nuclei (AGN) are some of the most powerful phenomena in the universe, powered by supermassive black holes that can outshine entire galaxies. As material spirals inward, the black hole releases enormous amounts of energy that shape the gas, dust, and stars around it. Understanding how that energy interacts with the host galaxy—known as AGN feedback—is one of the biggest challenges in astrophysics today.

A key tracer of this feedback is the ionization cone, a conical region of excited gas illuminated by the AGN's radiation. These cones reveal how energy escapes from the black hole's dusty torus and provide clues about the geometry of the central engine. Unfortunately, detecting them in real data is not simple. [O III] emission from these regions is faint, asymmetric, and often buried under noise or dust. Most identifications still rely on manual inspection of images, one galaxy at a time, which makes large-scale studies nearly impossible.

When I joined the Mid-Infrared Active Galactic Nuclei (MAE) Group at the University of Texas at San Antonio, working under Dr. Chris Packham and Dr. Lulu Zhang, I was introduced to a project that aimed to tackle this challenge using artificial intelligence. Dr. Packham and Dr. Zhang designed the scientific foundation and goal of the work: developing a method to automatically detect ionization cones in JWST/MIRI data. Once given this target, I took full responsibility for building the computational system from the ground up. It was the perfect intersection of my interests in astrophysics and AI.

Methods

I began by designing a complete Python-based data pipeline to preprocess JWST/MIRI Integral Field Unit (IFU) data cubes in FITS format. The pipeline automated normalization, denoising, resizing, and augmentation, producing consistent inputs for training a neural network. Every stage was tracked for reproducibility using custom scripts that logged parameters and outputs. Because real JWST datasets are limited and varied, I generated synthetic [O III] emission maps to form the base of the training set. Each image contained one or more ionization cones represented by two-dimensional Gaussian profiles with randomized opening angles, orientations, and noise. To simulate realistic conditions, I introduced effects like dust attenuation, pixel noise, and partial obscuration. This allowed me to create controlled ground-truth labels while teaching the model to handle the messy reality of telescope data.

To perform the segmentation, I implemented a U-Net convolutional neural network, commonly used in biomedical imaging for its ability to capture both broad spatial context and fine details. I customized its kernel sizes, normalization layers, and activation functions to better suit astrophysical images. The network was trained using a Dice loss function, which measures how well the predicted cone regions overlap with the true ones—a vital choice for imbalanced datasets where cone pixels occupy only a small part of the image.

The dataset ultimately included about 1,100 examples: 1,000 synthetic maps, 20 real JWST samples, and 80 modified versions of the real ones to increase variability. I used a 90–10 training-to-validation split, trained for 30 epochs using the Adam optimizer with a learning rate of 10^{-4}, and included live augmentations like rotations and flips to improve generalization.

Throughout the project, I wrote all preprocessing, training, and evaluation scripts myself. I also built visualization tools to overlay predicted and true masks, and I logged performance metrics such as Dice coefficient, precision, recall, and Intersection-over-Union (IoU).

Results

The U-Net showed strong and stable learning. Training and validation loss dropped sharply during the first ten epochs and then leveled out by around epoch fifteen. The final Dice coefficient reached 0.91, with precision at 0.85, recall at 0.99, and IoU near 0.84. In practical terms, the model successfully identified almost all real cone pixels while keeping false positives low.

Qualitatively, the model performed well on both synthetic and real JWST images. It could trace asymmetric and partially obscured cones, even under moderate noise. Interestingly, it learned to infer the direction of cone outflows—something not explicitly encoded in the data—which showed that it captured underlying geometric structure rather than just brightness patterns.

A few edge cases exposed its limits. When emission was extremely diffuse or the galaxy contained dual-cone systems, the network sometimes merged or truncated regions. I expanded the training data to include broader variations in cone shapes and intensity gradients, which significantly reduced these errors. The takeaway was clear: more realistic, diverse training data directly improved performance.

Overall, the project demonstrated that deep learning can effectively identify ionization cones, even when they're faint or irregular. It also showed that physically realistic synthetic data can teach models to generalize to telescope observations—a major step toward automating AGN morphology studies.

Future Research

The next phase of this project focuses on applying the trained model to fully real JWST/MIRI datasets, particularly those from nearby Seyfert galaxies in the GATOS collaboration. My goal is to evaluate how well a model trained mainly on synthetic data transfers to real observations. Success would mark a turning point toward automated AGN morphology classification across large surveys.

To bridge the gap between simulations and reality, I am developing a second generation of synthetic data that includes hydrodynamically inspired outflows, irregular gas structures, and varying dust extinction. I also plan to train on multiple emission lines—[O III], Hα, and [N II]—to study correlations across ionization states. Long-term, I aim to extend the framework to three-dimensional IFU data cubes using 3D convolutional networks, enabling direct measurement of cone inclination and velocity fields.

Our methods will contribute to an upcoming GATOS collaboration paper analyzing JWST/MIRI AGN data, where I will likely appear as a co-author among more than forty collaborators. It's exciting to know that the tools I've built are already becoming part of a larger scientific effort.

Conclusion

Before this project, I had never built an end-to-end data pipeline or trained a deep learning model on real astronomical data. By taking ownership of the computational design and analysis, I learned how to translate astrophysical ideas into algorithmic form—to teach a computer to "see" what astronomers see.

I also gained a deeper understanding of AGN physics, from how radiation ionizes gas to how geometry and dust shape what telescopes observe. Working under Dr. Chris Packham and Dr. Lulu Zhang taught me how to connect physical models, real data, and computation into a single coherent system.

This experience changed how I approach problem-solving. In track and field, I analyze each stride to find efficiency; in research, I apply the same mindset to streamline pipelines and models. I now see artificial intelligence as an extension of that same philosophy—using precision and focus to push discovery forward.

The Goldwater Scholarship would allow me to keep pursuing this intersection of astrophysics and machine learning, expanding toward exoplanet detection and time-domain astronomy. My goal is to earn a Ph.D. in astrophysics and lead research at the boundary of AI and physical science, building transparent, reproducible tools to help uncover the hidden structures of the universe.

This project taught me that discovery doesn't begin with what the telescope sees, but with how we choose to look.

Draft 3! :'O

Machine Learning Identification of Ionization Cones in Active Galactic Nuclei Using JWST MIRI Data

Introduction

Active Galactic Nuclei (AGN) rank among the most energetic phenomena in the universe. Powered by accretion onto supermassive black holes, they release vast amounts of radiation and drive winds and jets that shape the evolution of their host galaxies. A central question in astrophysics is how this feedback operates—how energy from the nucleus couples to the surrounding gas and regulates star formation.

One of the clearest tracers of this feedback is the ionization cone: a biconical region of photoionized gas that reveals how radiation escapes through the dusty torus surrounding the black hole. The shape and extent of these cones encode key information about the geometry of the central engine and the efficiency of AGN feedback. Yet identifying them remains remarkably difficult. The emission is faint, irregular, and frequently obscured by dust or instrumental noise. Historically, astronomers have relied on manual inspection of emission-line maps to infer cone structures—an approach that is both time-consuming and subjective.

When I joined the Mid-Infrared Active Galactic Nuclei (MAE) group at the University of Texas at San Antonio, led by Dr. Chris Packham and Dr. Lulu Zhang, I was confronted with a paradox: even with the James Webb Space Telescope (JWST), capable of resolving unprecedented detail, our analysis techniques still depended on human recognition. Dr. Packham and Dr. Zhang posed a challenge—to determine whether deep learning could autonomously identify ionization cones in mid-infrared data from JWST's MIRI instrument. They defined the scientific goal; every aspect of the computational design, modeling, and validation became my responsibility.

I set out to build a machine-learning framework that could segment ionization cones automatically and, in doing so, translate the interpretive power of an astronomer's eye into a reproducible, quantitative tool. This project would ultimately serve as the first step toward constructing a catalog of emission-line ratios across the GATOS (Galactic Activity, Torus, and Outflow Survey) sample of NGC galaxies, providing spectral diagnostics of AGN feedback in the mid-infrared.

Methods

Data Pipeline and Preprocessing

I began by designing a complete data pipeline in Python to handle JWST MIRI Integral Field Unit (IFU) data cubes stored in FITS format. The pipeline automates normalization, denoising, and augmentation—standardizing data for neural-network input while preserving the morphology of extended emission features. Each processing stage logs metadata for reproducibility, a design choice inspired by professional survey pipelines.

Because real JWST datasets are limited and often contain variable noise structures, I generated synthetic emission-line maps to provide a robust training base. Each synthetic map simulated one or more ionization cones as two-dimensional Gaussian intensity profiles with randomized orientations, asymmetries, and opening angles. I further introduced realistic observational effects: spatially correlated detector noise, dust attenuation gradients, and partial obscuration. These augmentations ensured the training data captured the diversity of conditions expected in MIRI imaging while maintaining exact ground-truth masks.

Model Architecture

The model architecture was based on a modified U-Net, a convolutional neural network optimized for image segmentation. Its encoder–decoder structure captures both large-scale context and fine-scale structure—essential for identifying cone edges and diffuse emission simultaneously. To tailor the model to astrophysical data, I adjusted kernel sizes and incorporated batch normalization to stabilize training under variable noise.

Most importantly, I implemented a hybrid Binary Cross-Entropy (BCE) + Dice loss to handle the extreme class imbalance between cone pixels and background sky. The loss combines pixel-level accuracy (from BCE) with region-level overlap (from Dice), weighted to emphasize correct segmentation of faint, sparse features. This modification allowed the model to converge stably and avoid the common failure mode of ignoring small, low-contrast cones.

Training Strategy

The training dataset consisted of roughly 1,100 samples: 1,000 synthetic maps, 20 real MIRI observations, and 80 augmented variants that introduced small distortions to real data. I used a 90–10 data split, reserving real data for validation to ensure the model generalized beyond synthetic noise patterns.

Training employed the Adam optimizer with a learning rate of 10−410^{-4}10−4 over 30 epochs. On-the-fly augmentations—rotation, flipping, and scaling—further improved generalization. Throughout, I monitored loss curves, Dice coefficients, precision, recall, and Intersection-over-Union (IoU). Visualization scripts overlaid predicted masks on input images to reveal qualitative behavior and identify systematic biases.

Every component of this framework—from preprocessing to metric analysis—was written and validated by me. My mentor, Dr. John [Ph.D.], provided only limited code review during early development, but the complete design and implementation were my own.

Results

Training produced stable, rapid convergence. Both training and validation losses decreased smoothly and plateaued near epoch 15. The final model achieved a Dice score of 0.91, precision = 0.85, recall = 0.99, and IoU ≈ 0.84, indicating strong agreement between predicted and ground-truth cone regions.

These metrics translated into astrophysically meaningful performance. The model consistently detected extended cone structures, even when partially obscured or asymmetric, while avoiding false segmentation of diffuse background emission. It also learned directional properties—the orientation of outflows—an emergent behavior not explicitly encoded in training, suggesting that it captured genuine geometric information rather than superficial intensity patterns.

A few failure modes offered insight: extremely broad or dual-cone geometries sometimes caused the network to merge features or truncate cone tips. By enriching the synthetic dataset with greater variation in opening angles and intensity gradients, I retrained the model and substantially reduced these errors. This iterative process confirmed that performance scales directly with physical realism in the training set.

Collectively, these results established proof of concept: a deep-learning model can autonomously segment ionization cones in MIRI-like data, reproducing what once required expert manual analysis. More importantly, it demonstrated that incorporating physically informed simulations produces models capable of generalizing to telescope observations—a crucial requirement for scalable astronomical inference.

Building the MIRI Emission-Line Catalog

Having validated the segmentation framework, I am now extending it toward spectral diagnostics. The goal is to build a catalog of emission-line ratios for regions identified by the model across the GATOS NGC galaxies. These ratios will quantify ionization strength, density, and temperature within the cones, enabling physical interpretation of the morphology.

The catalog targets the full suite of mid-infrared lines accessible to JWST MIRI, summarized in Table 1 (Zhang et al. 2025). These include high-ionization species such as [Ne V] (14.32 μm, 97.1 eV) and [Mg VII] (5.50 μm, 186.5 eV), as well as lower-ionization tracers like [Fe II] (25.99 μm, 7.9 eV) and [S III] (18.71 μm, 23.3 eV). By comparing ratios such as [Ne V]/[Ne II] or [O IV]/[Fe II], the catalog will provide empirical measures of ionization parameter and shock contribution across the segmented cone regions.

Each emission line's rest wavelength, ionization potential, and critical electron density (computed using PyNeb) form the foundation for physically motivated ratio diagnostics. Once complete, this catalog will allow cross-comparison of model-identified structures with the physical conditions they trace—transforming pure morphology into quantitative feedback signatures.

Future Work

The next stage of research focuses on applying the trained model directly to JWST MIRI IFU cubes of NGC 1068, NGC 1365, and NGC 4151, three well-studied Seyfert galaxies in the GATOS sample. These systems exhibit clear yet complex outflow morphologies, providing ideal test cases for model validation against published analyses.

To bridge the gap between idealized synthetic data and real observations, I am developing a second-generation simulation framework incorporating hydrodynamically inspired cone shapes, turbulent gas distributions, and wavelength-dependent dust extinction. These improvements will increase realism and further enhance model transferability.

In parallel, I plan to extend the current 2D approach into three-dimensional segmentation using 3D convolutional networks. This will allow the algorithm to analyze IFU cubes directly in spatial-spectral space, extracting not only morphology but also velocity and inclination information from emission-line kinematics. Combined with the upcoming emission-line ratio catalog, these tools will enable measurement of outflow energetics and their dependence on AGN luminosity and host-galaxy structure.

Ultimately, this pipeline will scale to large surveys such as SDSS-V and future JWST legacy programs, supporting population-level studies of AGN feedback. By automating cone detection and physical characterization, the system can reveal correlations between feedback efficiency, galaxy mass, and environment—quantities central to models of cosmic evolution.

Scientific Impact and Independence

Although the project originated from goals set by Drs. Packham and Zhang, every computational and analytical component was executed independently by me. I designed and debugged the full pipeline, optimized the network architecture, implemented the hybrid loss function, and conducted all evaluations. Through this process, I learned to operate at the level of a graduate researcher—bridging astrophysics and machine learning without relying on pre-existing frameworks.

This independence also forced a deeper understanding of the underlying physics. Modeling cones required learning the radiative processes that produce mid-infrared emission lines, from collisional excitation to critical density effects. Constructing synthetic maps demanded integrating those principles into generative functions that reproduced realistic observational signatures. Each technical decision—kernel size, normalization layer, loss weighting—became an astrophysical one, grounded in how ionized gas manifests in MIRI data.

Conclusion

This project represents a step toward uniting two frontiers of science: precision astrophysics and artificial intelligence. By training a modified U-Net to segment ionization cones in JWST MIRI data, and by building an accompanying catalog of mid-infrared emission-line ratios, I aim to provide a reproducible, scalable framework for quantifying AGN feedback across the GATOS NGC galaxies.

Scientifically, the work demonstrates that machine learning can extract physically interpretable structures from complex data, bridging human intuition and computational analysis. Personally, it has shown me that independence in research is not merely about coding alone, but about translating physical insight into algorithmic language.

As I continue this work toward publication with Dr. Zhang and the GATOS collaboration, my goal is to extend these methods to multi-wavelength and 3D analyses, advancing our understanding of how black holes shape their galaxies. With support from the Goldwater Scholarship, I hope to

deepen this intersection of astrophysics and AI—developing transparent, physics-informed neural networks that allow us to see the universe not just as it appears, but as it truly behaves.

Draft 4!!

Active Galactic Nuclei (AGN) are among the most energetic phenomena in the universe, powered by accretion onto supermassive black holes that can outshine their host galaxies. The energy released by these central engines regulates gas dynamics, star formation, and galactic evolution through processes collectively known as AGN feedback. Understanding the mechanisms by which AGN influence their surroundings is a central challenge in extragalactic astrophysics. One of the clearest observational signatures of AGN feedback is the ionization cone: a conical region of gas photoionized by the AGN's intense radiation field. These cones provide direct insight into the geometry and orientation of the nucleus and trace how radiation escapes through the surrounding dusty torus. Their morphology encodes information about the coupling between the black hole and the interstellar medium, making them critical probes for understanding galaxy-black hole coevolution. However, ionization cones are faint, often asymmetric, and can be obscured by dust or instrumental noise, making their identification from spectral imaging data challenging. Historically, this identification has relied on manual inspection of emission line maps, such as narrow-band [O III], on a galaxy-by-galaxy basis. While effective for small samples, this approach limits statistical analyses and hinders large-scale characterization of AGN feedback.

When I joined the MAE Group at the University of Texas at San Antonio, I was tasked by Dr. Chris Packham and Dr. Lulu Zhang with developing a computational approach to overcome this limitation using mid-infrared (MIRI) JWST data. My charge was to construct an automated framework capable of detecting and segmenting ionization cones with minimal human intervention. I designed the project to translate the interpretive power of human visual inspection into an algorithmic pipeline that could generalize across faint and complex AGN structures. The resulting pipeline represents a combination of physically informed synthetic training data, a modified U-Net convolutional neural network, and a weighted combination of Dice and binary cross-entropy (BCE) loss. While the conceptual framework was provided, every element of the pipeline—from data preprocessing to training, evaluation, and visualization—was implemented and optimized independently.

The first step involved building a Python-based data pipeline to process JWST MIRI Integral Field Unit (IFU) data cubes in FITS format. The pipeline automates normalization, denoising, resizing, and augmentation, producing consistent inputs suitable for deep learning. Each processing stage includes custom logging and parameter tracking to ensure reproducibility. Because available JWST datasets are limited and heterogeneous, I generated synthetic emission maps that emulate the properties of real AGN observations. These maps include one or more ionization cones represented by two-dimensional Gaussian profiles with variable opening angles, orientations, asymmetries, and noise levels. To approximate realistic telescope conditions, I introduced physically motivated distortions, including dust attenuation, pixel noise, and partial obscuration. The synthetic maps preserve precise ground-truth labels, enabling the network to learn morphological features under conditions that closely mimic MIRI observations.

To model the spatial structures of cones, I implemented a modified U-Net convolutional neural network. This architecture is widely used in biomedical image segmentation for its ability to integrate global context while preserving fine-scale features. The encoder compresses spatial information, while the decoder reconstructs high-resolution segmentation masks. I adjusted kernel sizes, normalization layers, and activation functions to optimize performance on astrophysical data, and training employed a combined Dice-BCE loss function with weighting

chosen to balance detection of faint cone pixels against the background. The final training set comprised roughly 1,100 samples, including 1,000 synthetic maps, 20 real JWST examples, and 80 “altered real” augmentations. A 90–10 training-validation split prioritized synthetic data for controlled learning while reserving real data to evaluate generalization. Training used the Adam optimizer with a learning rate of 10^{-4} over 30 epochs, with dynamic augmentations including rotation and flipping to encourage invariance to orientation.

Evaluation of the model used standard segmentation metrics. The trained network achieved a Dice coefficient of 0.91, precision of 0.85, recall of 0.99, and Intersection-over-Union (IoU) of 0.84 on the validation set. These metrics indicate that the model correctly identifies nearly all ionization cone pixels while minimizing false positives, which is critical for ensuring accurate morphological statistics. Qualitatively, the model successfully traced asymmetric and partially obscured cones under moderate noise conditions. Remarkably, the network learned to infer the directionality of cones, an emergent property not explicitly encoded during training, suggesting it internalized geometric relationships rather than relying solely on pixel intensities. Occasional failure cases, such as dual-cone structures or extremely diffuse emission, informed refinements through expanded synthetic datasets, demonstrating that performance scales with diversity of training examples. Importantly, these results establish a proof-of-concept for automated, high-fidelity detection of ionization cones across a range of AGN morphologies.

Building on these results, I am currently developing a catalog of emission line ratios for all major MIRI AGN emission lines using Dr. Zhang’s published tables (Zhang et al., 2025, Figure 3). The catalog includes lines such as [Fe II] 5.34 μm, [Mg VII] 5.50 μm, [Ne VI] 7.65 μm, [Ar III] 8.99 μm, [S IV] 10.51 μm, and [O IV] 25.89 μm, among others. Each line is characterized by its wavelength, ionization potential, and critical electron density. This catalog will enable systematic analysis of emission line ratios within the ML-derived segmentation masks, providing a foundation for population-level studies of AGN feedback. While the model and pipeline are fully operational, this catalog represents the immediate next step for translating segmentation outputs into physical measurements suitable for publication. It will support applications to archival GATOS datasets and future JWST MIRI observations, allowing high-fidelity morphological and spectroscopic analysis across large AGN samples.

The computational framework developed in this project represents a methodological advancement for extragalactic astrophysics. By integrating physically motivated synthetic training data with a modified U-Net architecture and a weighted Dice-BCE loss function, the pipeline achieves high-fidelity segmentation of faint and complex ionization structures. Its modularity allows rapid adaptation to multiple emission lines, providing a scalable approach to population-level analyses. When fully applied to archival and upcoming JWST MIRI datasets, this tool will generate comprehensive emission line catalogs and morphological measurements at unprecedented scale. The combination of automation, reproducibility, and sensitivity positions the methodology to produce high-impact scientific publications, including multi-author GATOS studies, and lays the foundation for broader application of deep learning to extragalactic spectroscopy and AGN feedback studies.

Beyond immediate outputs, the long-term vision of this work is twofold. First, the pipeline can be used to extract physically informed predictions from faint AGN where traditional signal-to-noise limitations prevent direct observation of ionization structures. With sufficient computational resources, including potential access to TACC supercomputing clusters through UTSA, the

pipeline could process high-volume datasets in parallel, enabling characterization of extremely faint or distant AGN. Second, as the tool matures, it will be capable of ingesting vast datasets simultaneously, generating statistically robust trends in MIRI AGN morphologies, emission line ratios, and cone characteristics. This scalability promises transformative insights into the population-level behavior of AGN feedback and its role in galaxy evolution.

The skills I have developed through this project encompass both computational and astrophysical domains. I designed and implemented a complete machine learning pipeline, from synthetic data generation to network training and evaluation. I independently wrote preprocessing, training, and visualization scripts, and established logging systems to monitor model behavior. These technical skills are complemented by a rigorous understanding of AGN physics, including radiative transfer, ionization structures, and emission line diagnostics. I have also gained expertise in interpreting model outputs in an astrophysical context, connecting algorithmic predictions with physically meaningful measurements. This combination of computational and scientific proficiency positions me to extend the project into novel research directions, including multi-wavelength analysis, three-dimensional IFU cube segmentation, and population-level studies of AGN feedback.

The project demonstrates independence and initiative, as every scientific contribution—from dataset construction to model implementation—was conducted by myself. I sought guidance on coding techniques from my Ph.D. mentor, John, and consulted Dr. Zhang on specific astrophysical interpretations, but the conceptualization, development, and analysis were executed independently. This level of ownership exemplifies my ability to identify a scientific challenge, apply computational methods, and produce reproducible, physically meaningful results.

In summary, this research combines astrophysics and artificial intelligence to address a longstanding limitation in extragalactic astronomy. By automating the identification of ionization cones, the pipeline enables reproducible, scalable, and high-fidelity morphological analysis of AGN, laying the groundwork for population-level studies and high-impact publications. The immediate outputs include a working model capable of segmenting cones across multiple emission lines, along with a catalog of line ratios derived from Dr. Zhang's data. The longer-term vision encompasses high-volume data processing, physically informed inference on faint galaxies, and integration with large MIRI datasets, positioning this methodology at the forefront of computational extragalactic astrophysics. The combination of rigorous computational development, astrophysical interpretation, and methodological novelty demonstrates my potential as an independent researcher capable of bridging machine learning and observational astronomy.

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Machine Learning Identification of Ionization Cones in Active Galactic Nuclei

William Olson, University of Texas at San Antonio

Introduction

Active Galactic Nuclei (AGN) are among the brightest and most influential environments in the universe, driven by supermassive black holes that can outshine their entire host galaxies. As material falls inward, the released energy shapes the surrounding gas, dust, and stars. This process, called AGN feedback, plays a key role in galaxy evolution and black hole growth.

Ionization cones are one of the clearest markers of AGN feedback. These conical regions trace the radiation escaping from the dusty torus and preserve information about the geometry of the nucleus. Despite their importance, ionization cones are difficult to identify because their [O III] emission is faint, irregular, and often buried in noise. In many cases, manual inspection is still common, which makes the process slow and hard to scale.

After I joined the Mid-Infrared Active Galactic Nuclei Group at UTSA under Dr. Chris Packham and Dr. Lulu Zhang, I was assigned an independent project that aims to automate ionization cone detection using machine learning. My role was to design a complete computational framework that could preprocess JWST/MIRI data, create realistic synthetic training sets, train a segmentation model, and evaluate its performance. I took full responsibility for the project, from preprocessing to training and evaluating the model.

Methods

I first developed a Python-based pipeline to standardize JWST/MIRI Integral Field Unit (IFU) cubes. The pipeline performs normalization, denoising, resizing, and data augmentation, and I added logging features so every step could be reproduced and verified.

Real JWST datasets are limited and inconsistent so I began to implement a synthetic dataset. Each example includes one or more ionization cones represented by two-dimensional Gaussian profiles with randomized opening angles, orientations, intensities, and noise. I added dust attenuation, pixel noise, and partial obscuration to imitate realistic telescope conditions. These synthetic samples provided perfect ground truth labels and helped the model learn the structure of ionization cones rather than memorizing specific patterns.

For the segmentation model, I implemented a modified U-Net. This architecture is effective for images that contain both global structure and fine details. I adjusted kernel sizes, normalization layers, and activation functions to work well with astronomical data. Training used a combined Dice and Binary Cross Entropy loss function to recover faint

pixels without inflating false positives.

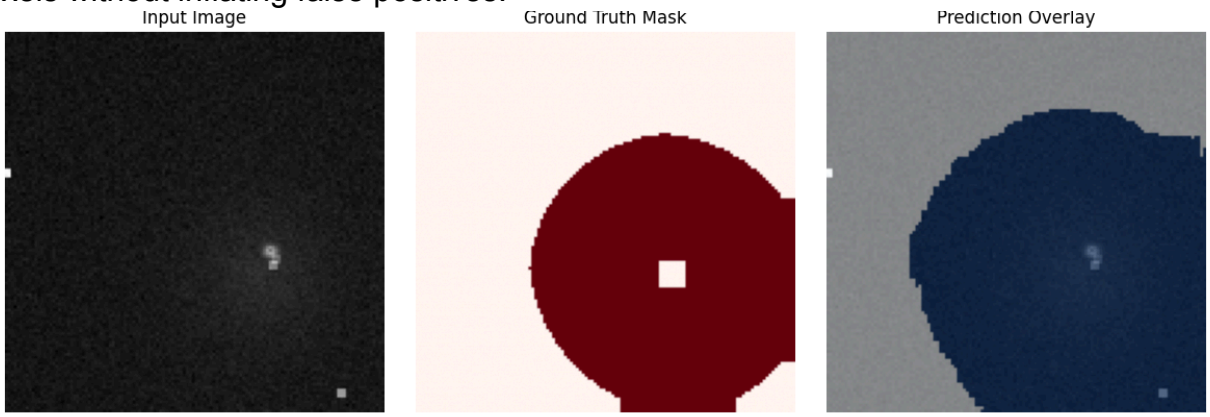


Figure 1. An example of the synthetic narrowband data, the Gaussian-generated mask, and the U-Net prediction.

The full dataset included roughly 1,100 examples: 1,000 synthetic images, 20 real JWST images, and 80 augmented real samples. I used a 90 to 10 training and validation split and trained the model for 30 epochs with the Adam optimizer at a learning rate of 10^{-4}. I also wrote visualization tools that overlay predictions on the true masks and track performance through metrics such as Dice coefficient, precision, recall, and IoU.

In progress

Figure 2. Schematic of the data pipeline and U-Net architecture. Raw FITS cubes are preprocessed, augmented, and passed through the network to produce segmentation masks.

Results

The U-Net showed strong and stable learning behavior. Both training and validation losses decreased quickly within the first ten epochs and leveled off after about fifteen. The final metrics were:

- Dice coefficient: 0.91
- Precision: 0.8
- Recall: 0.99
- IoU: 0.84

These values show that the model successfully detected nearly all ionization cone pixels while avoiding excessive false positives. Visual inspection confirmed that the network handled asymmetric and partially obscured cones even with moderate noise. One interesting outcome is that the model learned to identify the direction of the outflow, which was not explicitly included in the training labels but emerged naturally through the structural patterns.

Not all cases were perfect. Extremely diffuse emission or galaxies with more complex geometries sometimes led to merged or incomplete segmentations. To address this, I expanded the synthetic dataset to include broader morphological variations, which

reduced these errors and improved overall robustness.

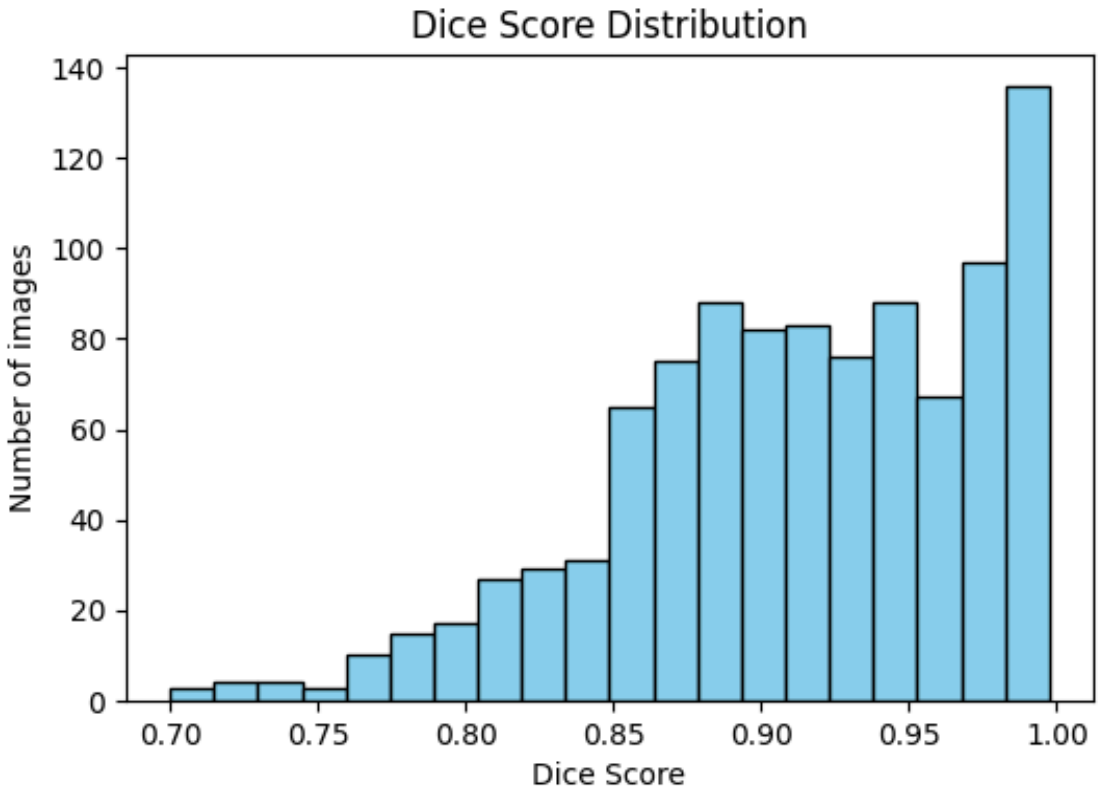


Figure 3. Dice score distribution histogram of the final predictions, showing an average score of 0.91 for the narrowband data.

Future Research

My next step is applying the trained model to full JWST/MIRI datasets, including nearby Seyfert galaxies in collaboration with the GATOS team. A major goal is to determine how well a model trained heavily on synthetic data transfers to real observations.

I am currently developing a second generation of synthetic data that uses more realistic physical assumptions. This includes hydrodynamically inspired outflows, irregular gas structures, variable dust extinction, and emission in multiple spectral lines such as [O III], Hα, and [N II]. Training on multiple lines will allow the model to learn how different ionization states relate to one another. In the longer term, I plan to extend the framework to three dimensional IFU cubes using 3D convolutional networks. This would open the door to analyzing outflow inclinations, velocity fields, and full spatial structure.

The tools I built are already being adopted within the MAE and GATOS collaborations, and I am expected to appear as a co author on an upcoming JWST/MIRI analysis paper. Seeing the framework used by other researchers has shown me how computational methods can scale up scientific discovery.

Figure 4. Conceptual diagram illustrating future plans for multi line and 3D IFU cube analysis.

Conclusion

Before this project, I had never designed an astrophysical data pipeline or trained a deep learning model on real observations. Working through every stage, from FITS preprocessing to model evaluation, helped me learn how to translate astrophysical questions into computational systems that can produce scientific insights.

This project gave me a strong foundation in AGN physics, including radiative ionization, geometry, and the effects of dust. I also strengthened my skills in Python, machine learning, and data visualization. More importantly, I learned how to evaluate models critically, iterate thoughtfully, and connect the model's predictions to physical interpretation.

My background as a track and field athlete influences how I approach research. When I train, I break down each stride to become more efficient, and I use the same mindset when improving a data pipeline or architecture. Machine learning feels like a natural extension of that thought process, because it rewards careful tuning and disciplined improvement.

The Goldwater Scholarship would support my goal of earning a Ph.D. in astrophysics and continuing to work at the intersection of AI and astronomy. I hope to build transparent and reproducible tools that help uncover hidden structures in AGN, detect exoplanets, and analyze the time variable sky. This project has shown me that discovery does not start with the telescope. It starts with how we choose to analyze and interpret the data.

I like this draft

Machine Learning Identification of Ionization Cones in Active Galactic Nuclei

William Olson, University of Texas at San Antonio

Introduction

Active Galactic Nuclei (AGN) are among the universe's most powerful engines, fueled by supermassive black holes that can outshine their host galaxies. As material falls into these black holes, the released energy shapes surrounding gas, dust, and star formation. Understanding how this energy is transferred, known as AGN feedback, is essential for understanding galaxy evolution and black hole growth.

Ionization cones provide one of the clearest signatures of AGN feedback. These conical regions trace radiation escaping from the dusty torus and preserve information about the nucleus's geometry. Their morphology encodes the coupling between the AGN and the interstellar medium. However, identifying ionization cones is challenging. Their [O III] emission is faint, asymmetric, and often buried in dust or instrumental noise, so astronomers frequently rely on manual inspection, which is slow and limits large-scale studies.

After I joined the MAE Group at UTSA under Dr. Chris Packham and Dr. Lulu Zhang, I was tasked with automating ionization cone detection using machine learning. My goal was to create a computational framework to preprocess JWST/MIRI data, generate realistic synthetic training sets, train a segmentation model, and evaluate performance. I took ownership of the project from start to finish.

Methods

I developed a Python-based pipeline to standardize JWST Integral Field Unit (IFU) cubes in FITS format. The pipeline performs normalization, denoising, resizing, and data augmentation, with custom logging for reproducibility.

Because real JWST datasets are limited and inconsistent, I generated synthetic [O III] emission maps for training. Each map contained one or more ionization cones modeled as two-dimensional Gaussian profiles with randomized opening angles, orientations, asymmetries, and noise levels. I added dust attenuation, pixel noise, and partial obscuration to mimic realistic observing conditions. These synthetic datasets provided precise ground-truth labels and encouraged the model to learn the underlying morphological patterns of ionization cones.

For segmentation, I implemented a two-dimensional U-Net convolutional neural network. The U-Net captures both global spatial context and fine-scale details. I fine-tuned kernel sizes, normalization layers, and activation functions for astrophysical images. Training used Dice loss to emphasize accurate segmentation of faint cone pixels.

The final dataset included roughly 1,100 samples: 1,000 synthetic maps, 20 real JWST images, and 80 augmented real samples. I adopted a 90–10 training-validation split and trained the model for 30 epochs using the Adam optimizer (learning rate = 10^{-4}). Dynamic augmentations such as rotation and flipping were applied to improve generalization. I also built visualization tools to overlay predicted masks on ground-truth cones and logged metrics including Dice coefficient, precision, recall, and Intersection-over-Union (IoU).

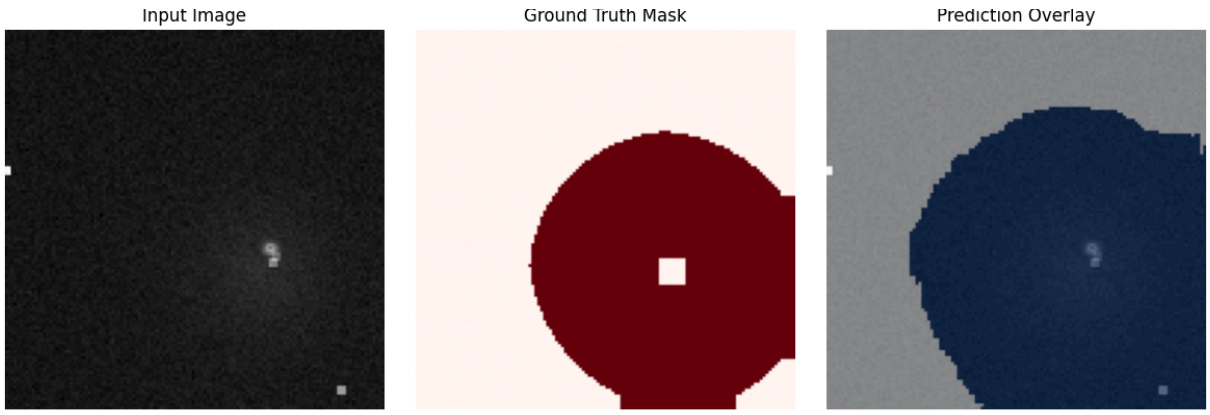


Figure 1. An example of the synthetic narrowband data, the Gaussian-generated mask, and the U-Net prediction.

Figure 2 illustrates the data pipeline and U-Net architecture, showing how raw FITS cubes are preprocessed, augmented, and passed through the network to produce segmentation masks.

Results

The U-Net learned efficiently and stably. Training and validation losses dropped quickly during the first ten epochs and converged by epoch fifteen. The validation Dice coefficient plateaued around 0.91, indicating strong agreement between predicted and true cone regions.

Quantitatively, the final model achieved a Dice coefficient of 0.91, precision of 0.85, recall of 0.99, and IoU of 0.84. This shows the algorithm detected nearly all ionization cone pixels while maintaining low false positives.

Qualitative inspection confirmed that the network accurately traced asymmetric and partially obscured cones, even under moderate noise. Notably, it learned the directionality of cones, an emergent property not explicitly encoded during training.

A few failure cases involved extremely diffuse emission or dual-cone geometries, occasionally merging or truncating structures. Adding synthetic examples with broader morphological variation mitigated these errors, demonstrating that performance scales with dataset diversity.

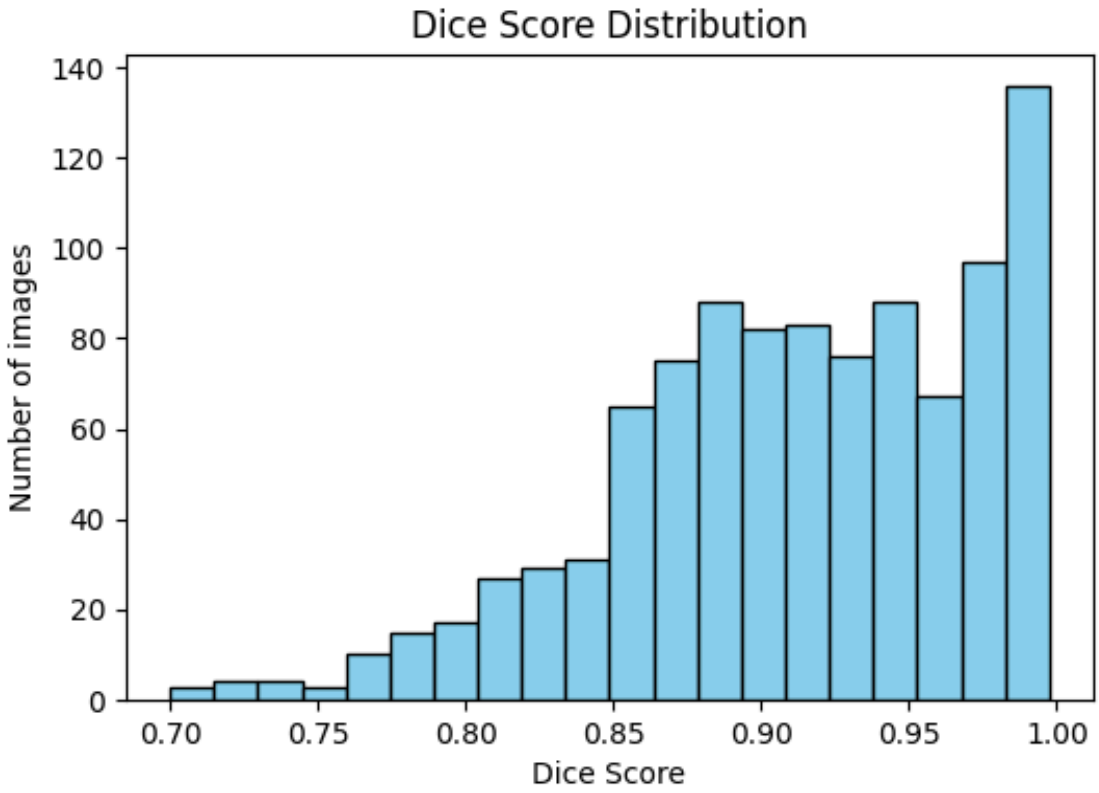


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Future Research

The next step is applying the trained model to full JWST datasets, including nearby Seyfert galaxies in collaboration with the GATOS team, to test how well a model trained on synthetic data transfers to real observations.

I am developing a second generation of synthetic data that incorporates hydrodynamically inspired outflows, irregular gas structures, and variable dust extinction. Multi-line training with [O III], Hα, and [N II] emission will allow the model to capture correlations between ionization states.

Longer-term, I plan to extend the framework to three-dimensional IFU cubes using 3D convolutional networks. This would enable measurement of outflow velocities, inclinations, and full spatial morphology. The tools I built are already being adopted within the MAE and GATOS collaborations, and I am expected to appear as a co-author on a forthcoming JWST/MIRI analysis paper.

Figure 4 illustrates conceptual plans for multi-line and 3D IFU cube analysis.

Conclusion

Before this project, I had never built a complete astrophysical data pipeline or trained a neural network on real observations. The work taught me to translate astrophysical intuition into computational frameworks, enabling a computer to “see” what astronomers see.
I gained a deeper understanding of AGN physics, including radiative transfer, torus geometry, and the effects of dust. I strengthened skills in Python, machine learning, and visualization while learning to critically evaluate model performance and connect predictions to physical interpretation.
My experience as a track and field athlete informs my research approach. Optimizing each stride mirrors how I refine pipelines and architectures. Machine learning extends that philosophy, allowing careful, focused effort to uncover hidden structures.
The Goldwater Scholarship would support my goal of pursuing a Ph.D. in astrophysics and advancing research at the intersection of AI and astronomy. I aim to develop transparent, reproducible computational tools to study AGN, detect exoplanets, and analyze the time-variable sky. This project showed me that discovery does not start with the telescope, but with how we choose to analyze and interpret data.

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Gardner, J. P., et al. The James Webb Space Telescope, Space Sci. Rev., 123, 485, 2006.



Other Essays

Essay 1: Career Goals and Aspirations (Polished)

Opening Hook:
I pursue questions most do not yet know to ask.

Body:
Since childhood, I have been drawn to curiosity and precision which are both dual parts of discovery and efficiency. Physics captivated me early, and computational astrophysics solidified my direction. I envision a career at the intersection of machine learning and astrophysics, developing tools that analyze and interpret complex astronomical phenomena.

My research on Active Galactic Nuclei (AGN) and [O III] emission maps at UTSA demonstrated the power of AI: neural networks detect patterns too subtle or irregular for human eyes, enabling reproducible, large-scale analyses. Algorithms built for astrophysics can also apply to any field requiring pattern recognition, including medicine, climate modeling, and materials science. I aim to conduct research that is translational by design, accelerating discovery across domains.

Long-term, I plan to earn a Ph.D. in astrophysics and lead independent research applying deep learning to exoplanet characterization, AGN morphology, and time-domain astronomy. By creating pipelines capable of automated detection, segmentation, and classification, I can reduce the time between observation and insight, enabling research at larger scales.

Mentorship is integral to my approach. Coaching the Randolph Roadrunners AAU track and mentoring UTSA students allows me to guide others in computational astrophysics, equipping the next generation with skills they can immediately apply in research.

Closing:
By advancing computational astrophysics and mentoring others, I aim to accelerate discovery and create tools that expand research across disciplines.

Essay 2: Defining Research Experience (Polished)

Opening Hook:
I train computers to see what humans cannot.

Body:
My research focuses on deep learning methods to identify ionization cones in AGN using JWST [O III] data. Manual analysis of such data is slow and subjective. I hypothesized that a neural network could detect and segment faint, asymmetric structures automatically, producing reproducible, scalable results.

I implemented a Python-based pipeline to preprocess FITS cubes, including normalization, denoising, resizing, and augmentation. To address the scarcity of labeled observational data, I

generated synthetic [O III] emission maps with physically motivated variations: multiple cones, asymmetries, dust attenuation, and realistic noise. These maps served as ground truth for training a U-Net convolutional neural network adapted for astrophysical imagery. The model learns global structure while preserving fine detail, segmenting cones accurately even under challenging conditions. Training on 1,100 images: 1,000 synthetic, 20 real, 80 augmented, produced a system that detects nearly all true emission features with high precision.

This work required intellectual independence, creativity, and rigor. Observing my father navigate complex research problems without formal training taught me persistence and systematic problem-solving all of which directly inform my approach to computational astrophysics.

Closing:
This project taught me to design systems that make discovery repeatable and scalable, preparing me to extend these methods across astrophysics and related fields.

Essay 3: Challenges and Growth (Polished)

Opening Hook:
Resilience is the engine of my scientific curiosity.

Body:
I was raised in a low-income family, witnessing the consequences of financial mismanagement firsthand. My father pursued independent research in psychology and brain science, demonstrating that rigorous thinking is not bound by formal training. His example shaped my approach: I pursue patterns, test hypotheses, and act decisively on insight.

Leaving the Mormon church presented another formative challenge. Our family faced criticism, skepticism, and strained relationships. Over time, we reconciled, learning to balance integrity with empathy and pursue truth with courage. These skills transfer directly to scientific collaboration, where diverse perspectives and rigorous debate are essential.

Athletics has reinforced discipline and resilience. Daily training in multiple sports from a young age taught me perseverance, self-evaluation, and iterative improvement. Coaching the Randolph Roadrunners and mentoring UTSA students allows me to pass on these lessons while sharpening my own. I approach research as I approach sprinting: analyzing each step, iterating continuously, and striving for measurable improvement.

Closing:
These experiences strengthened my persistence and problem-solving skills, preparing me to pursue independent, computationally driven research in astrophysics.

1. Opening Hooks

- Right now, each essay has a strong hook, but we can make them even more “reviewer-stop-and-listen”.

- For example, Essay 1: “I pursue questions most do not yet know to ask.” is good, but we could tie it directly to astrophysics + AI, e.g., “I teach computers to see patterns the universe hides from human eyes.” Instant connection to your work, curiosity, and computational edge.

- Reviewers remember hooks, so every first sentence should scream uniqueness + capability.

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2. Personal Narrative & Stakes

- You’ve got great life context (Mormon transition, family challenges, low income), but we can tie these experiences directly to scientific traits: curiosity, rigor, independence.

- For instance, your family’s financial ups and downs + dad’s independent research could be phrased as: “I inherited a mindset that combines intellectual audacity with strategic problem-solving—traits that shape every line of code and every astrophysical question I pursue.”

- Right now it’s implied, we can make it explicit, powerfully framed, and “Goldwater-ready”.

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3. Translational Emphasis

- This is your killer differentiator. We say your research is translational, but we can hammer it home more:

    - Show 1–2 concrete examples where your pipeline or approach could benefit another domain.

    - Keep it aspirational but grounded: “The same architecture that identifies ionization cones can segment tumor growth patterns, accelerating discovery in medicine.”

- This frames you as a high-impact thinker, not just another astrophysics student.

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4. Teaching / Outreach

- Right now it's brief, but you already coach AAU and mentor UTSA students.
- We can add one precise, vivid moment, like a student "aha!" moment, to show you already make impact now.
- Goldwater likes "mentorship in action," even if just one line. Could be: "A UTSA student once realized they could automate their own analysis pipeline after our session—watching them do that confirmed for me that teaching multiplies discovery."

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5. Word Choice / Power Sentences

- There are many great sentences, but we can highlight 2–3 per essay that reviewers will underline.
- Example: "Efficiency, rigor, and insight are not incidental—they are my methodology." — that's already strong, but we could make it even punchier: "Efficiency, rigor, and insight are not tools—they are my operating system."
- Goldwater essays often live or die on one or two unforgettable phrases. We can add more of these signature lines.

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6. Conclusion Lines

- Currently strong, but we can make them "Goldwater-bold":
- Push the bigger picture impact of your work + personal traits:
    - Not just "I want a Ph.D."
    - But "I will transform the way humans extract knowledge from data, whether from distant galaxies or the hidden patterns of the world around us."
- The goal: every reviewer leaves thinking, "this kid is going somewhere."

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7. Humor / Personality

- A sprinkle of light, humanizing humor is perfect.

- Right now mostly serious. We could weave in one or two memorable, subtle lines per essay—e.g., a sprinting analogy, or “watching a student light up like the JWST seeing a new galaxy.”

- Humor = memorability, but subtle, controlled, and relevant.

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8. Metrics / Proof

- Your computational research is strong, but we can throw in one brief quantitative highlight in Essays 1 or 2:

    - Example: “My U-Net model achieved a Dice coefficient of 0.91 on synthetic [O III] data, demonstrating that automation can rival human inspection.”

- Goldwater reviewers like proof you can execute at a high level.

TL;DR:

- Sharpen opening hooks to tie immediately to AI + astrophysics.

- Frame personal challenges as traits that drive scientific rigor.

- Emphasize translational impact—what you create will multiply discovery across domains.

- Add one vivid teaching/outreach moment per essay.

- Add 1–2 signature power sentences per essay reviewers will underline.

- Make conclusions bold, aspirational, and “visionary”.

- Sprinkle controlled humor and memorable metaphors.

- Include 1 concrete metric or outcome from research to show execution ability.



SMART DoD

Great — Will, since you're focused on astrophysics + AI, I found **five excellent DoD SMART Scholarship-for-Service Program sponsoring facilities** that would align well with your interests (little red dots, AGN, AI/image-processing). I'll list each facility, what they do, why it suits you (and any caveats). Then you can pick which ones to include in your application ranking.

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1. U.S. Army Combat Capabilities Development Command Army Research Laboratory (DEVCOM ARL)

Why it's a fit:

- DEVCOM ARL is a major Army research hub that explicitly lists involvement in computational sciences, network/cyber/systems, electromagnetics, photonics — all of which tie into your AI + astrophysics/image-processing skill set. (Wikipedia)
- They are a listed SMART sponsoring facility. (DEVCOM Army Research Laboratory)
- From your profile: you study "little red dots, AI in astronomy" and like coding/data pipelines; a facility strong in computational systems and image/data-analysis could allow you to apply those skills in a defense-research context (e.g., signal/image detection, pattern recognition).
  Caveats / questions to ask:
- Their work may skew more toward defense/weapon systems than astrophysics directly, so you'd want to target labs/groups within ARL that focus on image/remote sensing, pattern recognition, or large-data.
- Location: check if relocation aligns with you, and how much of your work would be autonomous research vs being part of a large team.

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2. Air Force Research Laboratory (AFRL)

Why it's a fit:

- The AFRL is regularly cited in SMART-related articles, especially for advanced computing, sensors, space/vehicles, and autonomy. (SmartScholar)

- Given your interest in astrophysics and "little red dots" (could map to space/remote-sensing), AFRL's "Space Vehicles Directorate" or sensor/data analysis groups might be a stellar match.
  Caveats / questions:

- You'll want to identify which directorate (e.g., Space Vehicles, Sensors, Directed Energy) aligns best with your astrophysics+AI background.

- Confirm the intern/research assignment: Will you be analyzing astronomical-style data, or more conventional defense sensor data? If you lean too defense-only, it may feel less aligned with your academic interests.

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3. Naval Air Warfare Center Weapons Division (NAWCWD)

Why it's a fit:

- NAWCWD is actively involved with the SMART program; their facility lead recently earned an award for SMART scholarship efforts. (NAVAIR)

- The work involves sensors, electronics, directed energy — strong overlap with image/data/AI pipelines.

- You could leverage your AI/image-processing interest here and maybe pivot some of your astrophysics/data-analysis methods into a defense context.
  Caveats / questions:

- Location: if at China Lake/Point Mugu, remote location may need adjustment.

- Research may tilt heavily toward weapons/sensors rather than "pure science" (e.g., astrophysics). Make sure to articulate how your skills (image segmentation, AI) apply.

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4. Naval Undersea Warfare Center (NUWC) Division Newport

Why it's a fit:

- NUWC Newport is explicitly reported as hosting SMART scholars and giving them tours and briefings. (NAVSEA)

- They work in sensor systems (undersea, sonar, signal processing) — your AI/image segmentation skills can translate well to signal detection/feature extraction in noisy media (analogous to faint ionization-cone detection!).
  Caveats / questions:

- The "undersea warfare" context is different from astrophysics, but your skillset is transferable if you frame your work in pattern detection, segmentation, faint-signal extraction.

- Ensure that you can negotiate/choose a group that aligns with your interests in computer/AI/data rather than purely mechanical/engineering only.

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5. Naval Facilities Engineering Systems Command (NAVFAC) Southeast

Why it's a fit:

- Though not as glamourous in "space/astrophysics" terms, NAVFAC Southeast deals with engineering systems, infrastructure, modeling — could map well if you're interested in image/data analysis of large-scale systems or remote sensing imagery (e.g., earth observation, maybe analogous to your astrophysics work). (NAVFAC)

- Choosing a slightly different environment might give you a unique niche (you don't *have to* stay strictly astrophysics; your AI/data pipeline skills are valuable broadly).
  Caveats / questions:

- Might feel less directly about "space" or "astronomy"; you'll need to craft your narrative to show how your astrophysics/data skills bring added value.

- Research may lean more civil/engineering than pure science — check for groups doing remote sensing/image analysis, not just structural/engineering.

Bonus thoughts & Strategy

- When you apply, rank **5 facilities** anyway — you might include a mix of “perfect fit” + “stretch” + “backup” options.

- Tailor your personal statement/essay to each facility: highlight how your astrophysics/data pipeline work (ionization-cone detection) maps to their mission (e.g., sensor/remote-image detection, pattern recognition in noisy data).

- During application, **mention your coding/AI experience** and how you can bring that to DoD research (not just astrophysics but data-intensive science).

- Consider geographic/location preferences and lifestyle (relocation, summer internship site) — you’ll sign on for one year of service per year of funding. (DoD Research & Engineering Enterprise)

If you like, I can **pull together a longer list of ~10 DoD SMART facilities** that match “astronomy/data/AI/image processing” specifically — with quick one-line summaries and links — that way you pick the best 5 to put on your application. Want me to dig that?

REU Applications Folder

Cornell

We do not require that you have college research experience, but if you have any, please tell us about it. o Have you done research during the school year, either as part of a class or outside of your coursework? What was the project? What was your responsibility? How many hours a week did you devote to it?

Currently, I shadow the GATOS Collaboration lab at the University of Texas at San Antonio (UTSA), led by Dr. Chris Packham, where we research active galactic nuclei (AGN) and infrared imaging of tori. Through this experience, I've learned techniques such as deconvolution and FortesFit for galaxy imaging, as well as how to map torus structures and refine data to uncover critical phenomena like AGN. Although I only spend one hour a week in the lab, I devote 3-4 hours outside of the lab to coding projects assigned by my mentor. While I have not yet been part of an official project, I am involved in an project in my class taught by Dr. Chris Packham for both scientists and engineers where we are designing a fictional NASA mission—a project that challenges me to apply astrophysical concepts in a creative and practical setting.

Have you participated in a post-high school standalone research program(s), either an REU or something similar lasting 6 or more weeks?

No

Describe your personal and educational background

My friend once said to me, "You can get the boy out of church, but you can't take the church out of the boy." She was joking about my past as a Mormon, but her words got me thinking: maybe she had a point. Growing up Mormon taught me a lot about the importance of community, integrity, and helping others. These weren't just ideas—they were habits. I still admire the way Mormons stand up for what they believe in and support one another so strongly.

When my family left the church, it wasn't because we were angry or rebellious. We just had questions—big ones—that didn't seem to have clear answers. The decision wasn't easy. I said goodbye to friends and mentors who had shaped my early life, and our choice left some family members scratching their heads. But we stayed close, and I came away from that experience with two things: a deep respect for my roots and an insatiable hunger for answers. If I had to pin down what makes me tick, it's that drive to keep asking questions—and not just settling until I get to the truth.

Curiosity drove me up a wall at age nine. I learned that my saliva which is used to dissolve food is made of the same primary atoms as the big, blazing stars that later collapse into frigid, foreboding black holes. I learned this on Cosmos, a scientific television program that explores the nearby sizzling stars, as well as the far corners of the cold universe. Ever since

watching Cosmos, I have been drawn to physics. Now, I love to study and research active galactic nuclei. Studying the giants of reality is one of the best things to happen to me.

That same curiosity found its way into every corner of my life, especially academics. As a kid, I was fiercely competitive. I raced through multiplication tables like they were Olympic events and even put together my first “financial plan” after learning about actuaries from my uncle and grandfather. I thought numbers would always be my path forward, but over time, I realized math felt incomplete to me. It was a great tool, sure, but I wanted something more—something that connected abstract ideas to the mysteries of the real world.

In high school, I finally found physics. At the time, I was competing in both club and school track and field, and physics fit perfectly with my sport. I could explain the torque of a sprinter’s leg, calculate the momentum needed to maximize speed, and analyze where to push hardest in a race to win. Physics gave me a framework to answer my questions about my coach’s training methods while fueling new curiosities, which I eagerly explored with my physics teacher.

That’s when I discovered astrophysics. The universe is a puzzle that blends math, science, and imagination, and every answer we uncover only raises more questions. Now, more than ever I am racing to uncover truths about the world and space we live in.

How does an REU in astrophysics or planetary science align with your educational and/or professional goals?

Certainly! Here’s a **more concise version** that keeps the essence of your goals and how the REU aligns with them:

I am applying to this REU because it aligns directly with my goal of earning a Ph.D. in astrophysics, where I aim to use artificial intelligence (AI) to analyze data from telescopes and detect exoplanets. This program will provide a unique opportunity to enhance my skills in data analysis, programming, and scientific communication, which are essential for my doctoral research.

Through my current experience at the GATOS lab at UTSA, I’ve gained hands-on experience with astrophysical data techniques, and I’m eager to expand these skills into AI-driven instrumentation. Working with leading experts in astrophysics and planetary science at this REU will allow me to build a deeper understanding of high-resolution imaging and apply machine learning to extract insights from astronomical data, advancing my long-term research goals. Ultimately, this REU will equip me with the technical expertise and collaborative experience needed to pursue my future in exoplanetary science and AI-driven research.

We do not require that you know how to code, but if you have any experience, tell us about your computer and coding background. What prompted you to develop those skills and how did you develop them? Have you used them to solve problems, at school or elsewhere?

Currently, my lab mentor is helping me slowly build a background in python by giving me tasks I should complete over time. Right now, I am currently assigned two projects right now. First, I need to and then I'll make many different apps from a source (like PyCharm), and then a browser based app that can discern different planets both through python. Soon, I will begin learning how to use python to analyze data and create graphs so that I can begin genuinely enrolling in my lab at UTSA.

Students who are accepted into this program will work under the supervision of a faculty mentor and their research team. Within a team, you may have a co-mentor or or secondary mentor along with the faculty lead, which can be thought of as a "multi-layered" team. How do you feel about working in a multi-layered team? What do you hope your day to day experience will be like in such an environment?

Personally, I really like "multi-layered" teams. It kind of reminds me of when I worked full time as a custodian. I would learn and report usually back to the assistant manager of the elementary school and he would work with the main manager of the elementary school. That same manager would work with our regional manager, who was in charge of progression and would tell us what we needed to do, needed to get better at, and where we were doing well. I really liked this work environment, because there would always be someone who I could help, or could help me.

Tell us about an instance in your post high school life, academic or not, when you've run into an unexpected roadblock to your goals. How did you proceed, and what happened in the end?

Money. I originally wanted to attend the University of Texas at Austin because I knew the larger student body and increased competition would push me to excel. However, with my parents prioritizing retirement savings, the additional $15,000 in loans made it financially unrealistic. Instead, I chose UTSA, where I could stay close to home, maintain financial stability, and still receive an excellent education.

At first, it felt like a setback, but I quickly realized that success isn't about where you go—it's about what you make of your opportunities. Since enrolling at UTSA, I've been able to dive into astrophysics research, connect with great mentors, and focus on building my future without financial strain.

How do you keep yourself engaged while working on long-term assignments?

Imagine you're part of a team. A less experienced colleague asks you for guidance on how to handle having multiple priorities at the same time. What do you advise them to do? Have you had an experience you'd draw upon for your advice?

It depends on the circumstances. Would it be "same-time" as in handling multiple things in the same time period, or multiple things literally at once? If at once and stressed for time, I would first prioritize what is most important, get that done, and then focus on what comes next. If there are multiple things in the same time period and the peer would be less stressed, I would tell them to focus and block time off to complete everything in a timely manner. Doing everything in a straight sit through would be bad for the mind and morale, so I would include 5-10 minute breaks within the tasks so the peer who asked would be much happier. I would also emphasize how important it is to stick to a schedule, because when a person is stressed for time, procrastinating, and going outside the schedule is a recipe for disaster. As for a specific experience when I had used this advice, I was in calculus two, and I had insisted to myself that it would be better if I took some breaks when doing my homework. It turns out, that these "breaks" just pushed my homework back further to around the days it was due which turned into a scramble to complete the homework assignment. But it turned out, that the homework was super easy! I took breaks to divide up my homework for almost no reason it seemed!

Have you ever been assigned a task about which you had little background or experience? How did you handle it? How did you feel at the beginning and at the end? What has been the biggest change in your life? How did you manage the difficulties that arose from it?

Whenever I'm faced with something completely new, my instinct is to ask questions first and learn as much as I can. Whether from the internet, peers, professors, or books, I seek out knowledge from every available source. But starting from scratch is always a challenge—just like my current experience with learning Python.

Recently, I was thrown into using Debian Linux with almost no prior experience. My only guidance was a few essential commands given to me by a peer, and the rest I had to figure out on my own. At first, it was overwhelming. I didn't understand what half the commands did, and I was frustrated by the lack of clear direction. But instead of letting that discourage me, I adapted. I scoured online forums, experimented with trial and error, and cross-referenced multiple resources until I built a solid understanding.

By the end, I wasn't just comfortable using Debian—I had developed a stronger ability to problem-solve independently. This experience reinforced a lesson I've learned throughout my life: sometimes the best learning happens when you're forced to piece things together from

multiple sources. While a structured guide is ideal, struggling through uncertainty forces you to think critically, adapt quickly, and build resilience.

Describe an instance in the past two years where you failed at accomplishing something that really mattered to you. How did you handle the situation?

The biggest change in my life came when my family transitioned from being active Mormons to non-Mormons. This shift was dramatic, not only because of the change in beliefs but also because of the emotional and social aspects involved. At first, I felt like an outsider in a community I had once been deeply involved in. A specific example that stands out is when I returned to church to be supportive of my extended family, knowing that the Mormon church is generally welcoming to visitors.
While at the service, I was attending "primary" with the younger kids. During a song I didn't know the words to, I tried to be considerate by quietly humming along. My cousin, sitting next to me, asked why I wasn't singing and, when I explained, he unexpectedly told me, "That's why you can't set foot in the temple." That was a painful moment, especially since I had been to the temple before. This incident came at the peak of the turmoil and really highlighted how big the rift was between my new beliefs and those of my old community.
However, over time, the rift healed. There's no bad blood between us now. I've come to appreciate my family, both immediate and extended, in a new way. I learned that the differences in beliefs don't need to define the relationship, and that curiosity and respect can bridge even the most difficult divides.

How do you adapt to new environments and new personalities?

I adapt to new environments and personalities by staying open-minded and curious. When I first encounter someone or a new situation, I try to understand things without judgment. For me, this means being ready to learn and embrace differences, rather than making assumptions. A key part of this approach is making a mental note of important facts about a person or situation. Once I have an understanding, I then ask questions—"Do you like this? Why or why not?"—to dive deeper into their thoughts and feelings. This helps me get a better sense of where they're coming from and builds a stronger connection.
I believe in being adaptable, both in social settings and in challenging environments. Whether it's learning a new skill like Python or adjusting to different group dynamics, I stay focused on how I can contribute and learn from others. My experience of transitioning from a Mormon upbringing to a more diverse perspective in San Antonio taught me how to appreciate different viewpoints and use them to grow. By staying curious and respectful, I can embrace change and

thrive in unfamiliar environments, creating relationships based on a kind and understanding curiosity.

What is the coping skill that is most important to you?

I wouldn't say this coping skill works for everyone, but for me, working hard is the best way to move forward. The more I push myself to think, train, or improve, the easier it is to adapt to new environments, personalities, or even failures. You also have to be mentally tough—like Aaliyah says, "Dust yourself off and try again." With this, I have gotten through some bad times like when my grandpa died a few years ago, when I was depressed and my family was fighting, you name it. Working hard and resilience come together like Pb&J (I'm allergic to peanut butter).

Let me give you an example. I used to be a dedicated student-athlete, balancing morning practices, dual-credit classes, and evening workouts every day. I took great pride in track & field, especially in long jump, where I hit 19 feet my sophomore year. But one moment really shook me: my mom told me, quite firmly, that I would never go pro. I was so frustrated. Sure, going pro wasn't likely, but to hear it said so definitely felt like a crushing blow. Instead of letting it bring me down, I focused on what I could control: training harder, improving my technique, and staying disciplined.

Another example is when I fell short of qualifying for Junior Olympics at South Texas Regionals. I had trained hard and wanted it badly, but I didn't make the cut. It stung, but quitting was never an option. I kept training. I still train. Even now, I work out two hours a day and maintain strict, healthy habits because my goal isn't just to win one race. It's to keep getting faster.

No matter what challenges I face, my response is always the same: work through it and keep going.

If you are selected, you'd be part of a group of 12 REU students. How would contribute to creating an inclusive and supportive community in the past? Can you share some examples drawn from your experience? Personal or professional/educational settings are eligible.

One experience that shaped my perspective on inclusivity was moving to San Antonio after living in Iowa, Idaho, and Utah—places with very different demographics. It was eye-opening to experience new cultures, perspectives, and ways of thinking. Instead of making assumptions, I learned to listen, ask questions, and appreciate people's backgrounds. This mindset has helped me build connections across differences and would allow me to support my REU peers by ensuring everyone feels valued and heard.

Additionally, in track and field, I have taken on a mentorship role, offering advice on technique, training, and mindset to teammates of all backgrounds. I focus on uplifting others and helping them see their own potential, just as I push myself to improve. I would bring this same

encouraging and collaborative approach to the REU—whether that means helping a peer with coding, learning from their expertise, or simply being someone they can talk to.

An inclusive environment thrives when everyone feels supported, respected, and empowered to contribute. I am excited to be part of a team where we challenge each other, grow together, and create an atmosphere of collaboration and curiosity.

I would love to attend this NASA internship because it offers the opportunity to apply my passion for astrophysics and AI-driven research to real-world challenges in space exploration. My experience at the UTSA GATOS (galaxy, activity, torus, and outflow survey) collaboration lab, where I've worked on AGN and infrared imaging using Python and machine learning, has prepared me to contribute to NASA's mission of uncovering the universe's mysteries. I am particularly excited about the chance to work on large datasets and cutting-edge instrumentation, which align with my goal of developing AI tools for exoplanet detection and galaxy evolution.

NASA's interdisciplinary environment and commitment to innovation resonate deeply with my desire to push the boundaries of astrophysics. This internship will allow me to collaborate with leading experts, gain hands-on experience with advanced technologies, and contribute to projects that could shape humanity's understanding of the cosmos. Ultimately, this experience will bring me closer to my goal of earning a Ph.D. and using AI to revolutionize how we explore the universe.

berkely

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February 1, 2026

Breakthrough Listen Internship Selection Committee
Berkeley SETI Research Center
University of California, Berkeley

Dear Selection Committee,

I am excited to apply for the Breakthrough Listen Internship at the Berkeley SETI Research Center. As a Physics major with an Astrophysics concentration at the University of Texas at San Antonio, I am drawn to SETI's mission of combining observational astronomy, data science, and computational methods to search for extraterrestrial signals. The opportunity to contribute to Breakthrough Listen's cutting-edge radio instrumentation and large-scale data analysis aligns perfectly with my academic and research goals.

Through my work with the MAE group at UTSA, I encountered the challenge of extracting meaningful structures from complex astronomical data. While studying infrared observations of active galactic nuclei (AGN), I realized that manual analysis was slow, subjective, and impractical for large datasets. To address this, I developed Python pipelines to preprocess, normalize, and augment observations and generated physically motivated synthetic datasets for machine learning. Training convolutional neural networks to segment features automatically taught me not only technical skills but also the importance of iterating on methods, diagnosing failure modes, and connecting computation directly to astrophysical reasoning. Each step reinforced my ability to navigate open-ended problems and translate curiosity into reproducible results.

Alongside research, I have built strong programming skills in Python, NumPy/SciPy, Astropy, and PyTorch, applying them to data analysis, simulation, and visualization. For example, when my first neural network failed to accurately segment faint AGN features, I had to redesign the synthetic training data, adjust evaluation metrics, and rethink assumptions about emission geometry. That process taught me how to identify bottlenecks, iterate systematically, and connect computational models directly to astrophysical reasoning. Beyond technical skills, mentoring and collaboration have been central to my development. Working with student athletes and serving on the Honors College advisory board strengthened my ability to communicate complex ideas clearly, balance multiple priorities, and guide others through structured learning.

I am particularly inspired by the prospect of working with Dr. Vishal Gajjar and the Breakthrough Listen team to integrate AI-driven analysis with radio observations. This internship represents an opportunity to apply what I have learned while continuing to grow as a computational astronomer. It aligns with my long-term goal of pursuing a PhD focused on AI-enabled astrophysics, building pipelines that scale across diverse datasets and enable discovery where data alone are not enough.

Thank you for considering my application. I am eager to contribute to Breakthrough Listen's efforts and to advance one of humanity's most profound questions: are we alone in the universe?

Sincerely,
William Eric Olson

MIT

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January 11, 2025

REU Program Coordinator
MIT Haystack Observatory
99 Millstone Road
Westford, MA 01886

Subject: Application for MIT Haystack Observatory REU Program

Dear REU Program Coordinator,

I am excited to apply for the Research Experiences for Undergraduates (REU) program at the MIT Haystack Observatory. As a freshman majoring in Physics with a concentration in Astrophysics at the University of Texas at San Antonio (UTSA), I am eager to contribute to Haystack's groundbreaking work and further develop my skills in radio astronomy and instrumentation.

Currently, I am shadowing the G.A.T.O.S. Collaboration lab at UTSA, where I've gained hands-on experience with deconvolution and data refinement techniques to study active galactic nuclei (AGN) and infrared imaging. These practices have strengthened my astrophysical techniques, and I am eager to refine them further through the exceptional opportunities offered by MIT Haystack's REU. I am particularly excited to work under Dr. Thushara G.S. Pillai, whose expertise in high-frequency VLBI techniques for studying radio jets near supermassive black holes resonates deeply with my research aspirations, and my Ph.D. later in life.

This REU represents a key step toward my goal of earning a Ph.D. in AI-driven instrumentation for exoplanet research. I aspire to develop adaptive AI systems that respond to astrophysical stimuli, accelerating the search for life beyond Earth. With my foundation in data refinement and my drive to learn, I am confident I can contribute meaningfully to advancing Haystack's cutting-edge research while growing as a scientist.

Thank you for considering my application. I look forward to the opportunity to contribute to Haystack's research while furthering my journey in astrophysics.

Sincerely,
William Eric Olson

Smithsonian/Harvard

My friend once said to me, "You can get the boy out of church, but you can't take the church out of the boy." She was joking about my past as a Mormon, but her words got me thinking: maybe she had a point. Growing up Mormon taught me a lot about community, integrity, and helping others. These weren't just ideas—they were habits. I still admire how Mormons stand up for their beliefs and support one another so strongly.

That same curiosity found its way into every corner of my life, especially academics. As a kid, I was fiercely competitive. I raced through multiplication tables like they were Olympic events. This eventually led me to putting together my first financial plan after learning about actuaries from my uncle and grandfather. I thought numbers would always be my path forward, but over time, I realized math felt incomplete. It was a great tool, sure, but I wanted something more—something that connected abstract ideas to real-world mysteries.

In high school, I finally found physics. At the time, I was competing in both club and school track and field, and physics fit perfectly with my sport. I could explain the torque of a sprinter's leg, calculate the momentum needed to maximize speed, and analyze where to push hardest in a race to win. Physics gave me a framework to answer my questions about my coach's training methods while fueling new curiosities, which I eagerly explored with my physics teacher.

That's when I discovered astrophysics. The universe is a puzzle that blends math, science, and imagination, and every answer we uncover only raises more questions. Through shadowing the GATOS lab at UTSA under Dr. Chris Packham, I've explored torus/AGN (active galactic nuclei) research and discovered how much I love the process of using data to understand the cosmos. Now, I hope to pursue a Ph.D. in AI-driven instrumentation for exoplanet research, developing machine-learning algorithms trained on archival data to accelerate the search for life beyond Earth.

What sets this REU program apart is its focus on uncovering the mysteries of black holes and their role in shaping the early universe. I'm especially drawn to the work of Dr. Chris Willott, whose research on high-redshift AGNs and black hole formation aligns closely with my interests. His recent publication on JWST's discovery of moderate-luminosity AGNs caught my attention, particularly how it reveals black holes growing faster than expected and their potential role in re-ionizing the universe. While analyzing redshift and black holes is not my primary interest, the data analysis techniques will be invaluable in developing the AI models I plan to create for exoplanet detection as well as tie directly into my research with Dr. Chris Packham.

By participating in this REU, I aim to expand my scientific horizons and tackle challenging questions alongside talented peers and mentors. This experience will sharpen my technical and analytical skills while bringing me closer to my goal of understanding the universe and maybe even helping humanity take its next big step toward finding life beyond Earth.

Rochester

Additional Questions or Materials

This Opportunity requires you to provide additional information.

Question #1

LONG ANSWER (5000 Character Limit)

List of courses currently in progress

General Chemistry I (3hr)

General Chemistry I Lab (3hr)

General Chemistry I Lab Lecture (1 hr)

Honors: The Civic Ethos (2hr)

Physics for Scientists&Engrs I (3hr)

Physics-Scientists& Engrs I Lab (3hr)

SS: Leadership & Science Int. (3hr)

Texas Politics and Society
University Peer Mentorship

Question #2

Computer software or programming experience. Check all that apply.

Python

C/C++

Fortran

IDL

Mathematica

MATLAB

Electronics

Linux/Unix

Java

Other (please elaborate in the Skills and Extracurricular Activities question)

Question #3

LONG ANSWER (5000 Character Limit)

Skills and extracurricular activities

Skills:

Technical: Data analysis, Python programming, radio astronomy techniques, deconvolution, and astrophysical instrumentation
Research: Scientific writing (posters, presentations, speeches, NOT grants or scholarly articles yet), data refinement, and working with infrared imaging (AGN research)
Soft Skills: Leader, critical thinking, problem-solving, teamwork, and adaptability

Extracurricular Activities:

Research: Shadowing the G.A.T.O.S. Collaboration lab at UTSA (AGN and astrophysics research)
Athletics: Track and field, 2Hr practices everyday, persistence to simply get faster, I train others who also want to have fun getting faster (about 5)
Hobbies: “Coding” in Minecraft, reading

Question #4

LONG ANSWER (5000 Character Limit)

Statement of interest: In 250 words or less, please describe your interest and previous background in research, if any, and what you hope to learn through the REU program.

I am excited to apply for the Research Experiences for Undergraduates (REU) program at the University of Rochester. As a freshman majoring in Physics with a concentration in Astrophysics at UTSA, I am eager to contribute to Rochester’s research while deepening my expertise in astrophysical instrumentation.

At UTSA, I am actively involved in the G.A.T.O.S. Collaboration lab, where I work with deconvolution and data refinement techniques to study active galactic nuclei (AGN) and infrared imaging. These techniques have strengthened my astrophysical skill set, and I look forward to refining them further through Rochester's REU. I am particularly interested in high-resolution observational methods, especially those involving radio or infrared data, to study black holes and exoplanetary environments.

This REU is a crucial step toward my goal of earning a Ph.D. in AI-driven instrumentation for exoplanet research. I aspire to develop adaptive AI systems that enhance observational techniques, accelerating the search for life beyond Earth. Dr. Alice Quillen's research on AGN and torus models closely aligns with my work in GATOS, and I would be honored (and very thrilled) to contribute to her research. With my foundation in data analysis and passion for discovery, I look forward to applying my skills and learning from Rochester's REU, where I hope to make meaningful contributions to its cutting-edge research.

Question #5

LONG ANSWER (5000 Character Limit)

Career goals: In 100 words or less, please describe your educational and career plans. Although some uncertainty is expected, we would like to know if you are considering graduate school, medical school, law school, business school, industry, etc.

I will complete my goal of earning a Ph.D. in astrophysics, where I aim to use artificial intelligence (AI) to analyze data from telescopes and detect exoplanets. This is at least where I am right now, but maybe with AGN research both from this REU, and my lab at UTSA, I will change my mind and try my hand at black hole/torus research.

Question #6

LONG ANSWER (5000 Character Limit)

Honors and Awards

South Texas AAU Janice Henderson Memorial Scholarship Track and Field	*$1250*	*2024*
UTSA/NSF APPEX (Academic Personal and Professional Excellence) Research Training Program and Scholarship	*$8000/ Year*	*2024-2028*
Distinguished Presidential Scholarship	*$5000/ Year*	*2024-2028*
Honors College Collaboration Scholarship-Housing	*$1500*	*2024*
NEISD Academic All-District Team (James Madison Highschool)	*$0*	*2020-2024*
James Madison High School Hall of Honor Track & Field Inductee	*$0*	*2024*

Wisconsin

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When my family left the church, it wasn't because we were angry or rebellious. We just had questions—big ones—that didn't seem to have clear answers. The decision wasn't easy. I said goodbye to friends and mentors who had shaped my early life, and our choice left some family members scratching their heads. But we stayed close, and I came away from that experience with two things: a deep respect for my roots and an insatiable hunger for answers. If I had to pin down what makes me tick, it's that drive to keep asking questions—and not just settling until I get to the truth.

Curiosity drove me up a wall at age nine. I learned that my saliva, which is used to dissolve food, is made of the same primary atoms as the big, blazing stars that later collapse into frigid, foreboding black holes. I learned this on Cosmos, a scientific television program that explores the nearby sizzling stars, as well as the far corners of the cold universe. Ever since watching Cosmos, I have been drawn to physics.

That same curiosity found its way into every corner of my life, especially academics. As a kid, I was fiercely competitive. I raced through multiplication tables like they were Olympic events and even put together my first "financial plan" after learning about actuaries from my uncle and grandfather. I thought numbers would always be my path forward, but over time, I realized math felt incomplete to me. It was a great tool, sure, but I wanted something more—something that connected abstract ideas to the mysteries of the real world.

In high school, I finally found physics. At the time, I was competing in both club and school track and field, and physics fit perfectly with my sport. I could explain the torque of a sprinter's leg, calculate the momentum needed to maximize speed, and analyze where to push hardest in a race to win. Physics gave me a framework to answer my questions about my coach's training methods while fueling new curiosities, which I eagerly explored with my physics teacher.

That's when I discovered astrophysics. The universe is a puzzle that blends math, science, and imagination, and every answer we uncover only raises more questions. Now, more than ever, I am racing to uncover truths about the world and space we live in.

At the University of Wisconsin-Madison Department of Physics, I look forward to "questioning," "competing," and discovering, continuing this humble pilgrimage of mine.

Future Goals & Career Plans: Please write a statement describing how your participation in a summer research program at UW-Madison would contribute to your future goals and career plans. Please comment on why you are interested in the UW-Madison program specifically. (limit 500 words)

This UW Madison REU represents a key step toward my goal of earning a Ph.D. in AI-driven instrumentation for exoplanet research. I aspire to develop adaptive AI systems that respond to astrophysical stimuli, accelerating the search for life beyond Earth. With my foundation in data

refinement and my drive to learn, I am confident I can contribute meaningfully to advancing UW Madison's cutting-edge research while growing as a scientist.
The data-driven nature of modern astronomy aligns closely with my interests in computational techniques and astrophysics. I am excited by the opportunity to develop Python tools for parallel data processing and apply machine-learning techniques to analyze vast astronomical datasets. The chance to explore low-surface brightness galaxies using archival data from world-class observatories or reconstruct galaxy formation histories through Hubble Space Telescope star cluster analysis directly supports my goal of applying AI to complex astrophysical problems. Additionally, participating in quantitative galaxy analysis projects through observations from telescopes like the WIYN and William Herschel would expand my understanding of galaxy dynamics and internal structures. Measuring star formation rates and analyzing gas kinematics while exploring phenomena like polar ring galaxies and super-thin disk galaxies would deepen my knowledge of galaxy evolution and broaden my research experience.
This REU's hands-on approach to data analysis, from using tools like DS9 and TOPCAT to participating in observing runs (if available), will help me develop practical research skills that are essential for my future career. Collaborating with UW Madison's faculty, such as Prof. Kotulla, and engaging with projects that blend observation and data science will strengthen my technical proficiency and prepare me for doctoral-level research.
Participating in quantitative galaxy analysis projects through observations from telescopes like the WIYN and William Herschel would significantly expand my understanding of galaxy dynamics and internal structures. I am particularly excited about measuring star formation rates and analyzing gas kinematics, as these processes are fundamental to understanding galaxy evolution. Exploring unique phenomena like polar ring galaxies and super-thin disk galaxies would not only deepen my knowledge but also challenge me to think critically about the diverse mechanisms driving galaxy formation and evolution.
This REU's hands-on approach to data analysis, from using tools like DS9 and TOPCAT to potentially participating in observing runs, aligns perfectly with my goal of developing practical research skills essential for my future career. Collaborating with UW Madison's faculty, such as Prof. Kotulla, and engaging in projects that blend observation and data science will strengthen my technical proficiency and prepare me for doctoral-level research. I am particularly interested in how these projects can push my understanding of astrophysics and how I can apply AI techniques to analyze complex datasets in the future.
While neither Dr. Kotulla's nor Dr. Gallagher's projects focus directly on AI, I see immense potential in integrating machine learning to enhance data analysis and modeling in astrophysics. This REU would provide me with the foundational knowledge and hands-on experience necessary to explore these possibilities further. Ultimately, I believe this experience will not only broaden my research capabilities but also prepare me to tackle the 'bigger idea' questions in galaxy evolution and beyond.

Currently, I am shadowing the GATOS Collaboration lab at the University of Texas at San Antonio (UTSA), led by Dr. Chris Packham, where we study active galactic nuclei (AGN) and infrared imaging of tori. This experience has introduced me to advanced techniques such as deconvolution and FortesFit for galaxy imaging, as well as methods for mapping torus structures and refining data to uncover critical phenomena like AGN. Although I spend only one hour per

week in the lab, I dedicate an additional 3-4 hours outside the lab to coding projects assigned by my mentor, which has allowed me to develop a strong foundation in data analysis and computational techniques.

One of the key skills I've gained is the ability to work with complex datasets and apply computational tools to extract meaningful insights. For example, I've learned how to process and analyze infrared imaging data to study the structure and behavior of AGN tori. This has deepened my understanding of how observational data can reveal the physical processes occurring in distant galaxies. Additionally, my mentor has been guiding me in building a strong background in Python, which is essential for modern astrophysics research. Currently, I am working on two projects: the first involves creating applications using tools like PyCharm, and the second focuses on developing a browser-based app capable of discerning different planets through Python. These projects are helping me develop practical programming skills and a deeper understanding of how to apply computational tools to astrophysical problems.

Looking ahead, I am particularly excited about my upcoming project, which involves training an AI to discern emission lines of active galactic nuclei. This project will allow me to explore the intersection of astrophysics and machine learning, a field I am deeply interested in. I am eager to learn how AI can be used to analyze complex datasets and uncover patterns that might otherwise go unnoticed. This experience will not only enhance my technical skills but also provide me with valuable insights into how AI can be applied to solve real-world problems in astrophysics.

Through this REU, I hope to further develop my research skills and gain hands-on experience in quantitative galaxy analysis. Specifically, I am seeking to strengthen my ability to analyze observational data, measure star formation rates, and study gas kinematics in galaxies. I am particularly interested in exploring phenomena like polar ring galaxies and super-thin disk galaxies, as these systems offer unique insights into galaxy formation and evolution. By working with telescopes like the WIYN and William Herschel, I hope to gain practical experience in observational astronomy and learn how to interpret data from these instruments.

Additionally, I am eager to enhance my proficiency with data analysis tools such as DS9 and TOPCAT, which are widely used in astrophysics research. I also hope to participate in observing runs, if available, to gain firsthand experience in data collection and processing. Collaborating with faculty like Prof. Kotulla and Prof. Gallagher will provide me with valuable mentorship and help me refine my research approach. I am particularly excited about the opportunity to work on projects that blend observation and data science, as this will prepare me for the challenges of doctoral-level research.

Finally, I am seeking to broaden my understanding of how AI and machine learning can be applied to astrophysics. While my current projects are introducing me to these concepts, I hope to explore how AI can be used to analyze large datasets, model complex systems, and uncover new insights into galaxy evolution. This REU will provide me with the opportunity to work with rich datasets and tackle abstract, "bigger idea" questions, which will be invaluable as I prepare for a career in astrophysics research.

TCU

Currently, I am shadowing the GATOS Collaboration lab at the University of Texas at San Antonio (UTSA), led by Dr. Chris Packham, where we study active galactic nuclei (AGN) and infrared imaging of tori. This experience has introduced me to advanced techniques such as deconvolution and FortesFit for galaxy imaging, as well as methods for mapping torus structures and refining data to uncover critical phenomena like AGN.
One of the key skills I've gained is the ability to work with complex datasets and extract meaningful insights. For example, I've begun to learn how to process and analyze infrared imaging data to study the structure and behavior of AGN tori. This has deepened my understanding of how observational data can reveal the physical processes occurring in distant galaxies. Additionally, my mentor has been guiding me in building a strong background in Python, which is essential for modern astrophysics research. Currently, I am working on two projects: the first involves creating my own applications using tools like PyCharm, and the second focuses on developing a browser-based app capable of discerning planet properties by data through Python. These projects are helping me develop practical programming skills and a deeper understanding of how to apply computational tools to astrophysical problems.
Looking ahead, I am particularly excited about my upcoming project, which involves training an AI to discern emission lines of active galactic nuclei. This project will allow me to explore the intersection of astrophysics and machine learning, a field I am deeply interested in. I am eager to learn how AI can be used to analyze complex datasets and uncover patterns that might otherwise go unnoticed. This experience will not only enhance my technical skills but also provide me with valuable insights into how AI can be applied to solve real-world problems in astrophysics.
Finally, I am seeking to broaden my understanding of how AI and machine learning can be applied to astrophysics. While my current projects are introducing me to these concepts, I hope to explore how AI can be used to analyze large datasets, model complex systems, and uncover new insights into galaxy evolution. This REU will provide me with the opportunity to work with rich datasets and tackle abstract, "bigger idea" questions, which will be invaluable as I prepare for a career in astrophysics research.

I am currently shadowing the GATOS Collaboration lab at the University of Texas at San Antonio (UTSA), led by Dr. Chris Packham, where we study active galactic nuclei (AGN) and infrared imaging of tori. I started early my first semester as a Freshman, and have no plan to stop. This experience has introduced me to advanced techniques like deconvolution and FortesFit for galaxy imaging, along with methods for mapping torus structures and processing complex datasets to uncover critical phenomena.
A key skill I've developed is using Python for astrophysical research. I am working on two projects: creating applications with PyCharm and developing a browser-based app to discern planet properties using data analysis. Looking ahead, I am especially excited about an upcoming project where I will train AI to identify emission lines of AGN, exploring how machine learning can reveal hidden patterns in large datasets.
Finally, I am seeking to broaden my understanding of how AI and machine learning can be applied to astrophysics. While my current projects are introducing me to these concepts, I hope to explore how AI can be used to analyze large datasets, model complex systems, and uncover new insights into galaxy evolution.

Undergraduate Researcher, GATOS Collaboration Lab, University of Texas at San Antonio (UTSA)
Advisor: Dr. Chris Packham
Dates: 10/24 – Present
As a member of the GATOS Collaboration Lab at UTSA, I contribute to cutting-edge research on active galactic nuclei (AGN) and infrared imaging of tori. Starting in my first semester as a Freshman, I have been actively involved in analyzing complex astrophysical phenomena, utilizing advanced techniques such as deconvolution and FortesFit for galaxy imaging. My work focuses on mapping torus structures and processing large, intricate datasets to uncover critical insights into AGN behavior.
A key aspect of my research involves leveraging Python for astrophysical data analysis and software development. I am currently leading two projects:

1. Developing applications using PyCharm to streamline data processing and analysis workflows.
2. Eventually creating a browser-based application designed to discern planetary properties through advanced data analysis techniques.

Looking ahead, I am particularly excited about an upcoming project where I will train machine learning models to identify AGN emission lines. This project will explore how AI can reveal hidden patterns in large datasets, furthering our understanding of galaxy evolution.
Through this research, I have developed strong technical skills in Python programming, data analysis, and software development, as well as a deep appreciation for collaborative scientific inquiry. I am eager to expand my expertise in applying AI and machine learning to astrophysics, particularly in analyzing large datasets, modeling complex systems, and uncovering new insights into the universe's most enigmatic structures.

As of right now, I have a long term goal of earning a Ph.D. in AI-driven instrumentation for exoplanet research. I aspire to develop an adaptive/active AI systems that respond to astrophysical stimuli, accelerating the search for life beyond Earth.

My career objective is to earn a Ph.D. in astrophysics and develop an active AI system capable of discerning exoplanets and identifying potential habitable worlds. I aim to combine machine learning and data analysis techniques to actively process and discern large datasets from telescopes, enabling faster and more accurate detection of exoplanets and their properties.
Through this REU, I hope to enhance my skills in data analysis, programming, and scientific communication while working alongside leading experts at TCU. Although my main focus is on exoplanet detection, and I currently research AI to discern AGN emission lines, I am excited to explore how chemical evolution and gaseous astrophysics will help me immensely, in not only learning about other areas, but how they could fit into this grand goal. This interdisciplinary approach will strengthen my ability to create robust AI tools that can analyze complex astronomical data.
In the long term, I aspire to lead innovative research projects that integrate AI and astrophysics, contributing to missions like the Habitable Worlds Observatory and advancing our understanding of the universe's most intriguing questions.

My friend once said to me, "You can get the boy out of church, but you can't take the church out of the boy." She was joking about my past as a Mormon, but her words got me thinking: maybe she had a point. Growing up Mormon taught me a lot about the importance of community, integrity, and helping others. These weren't just ideas, they were habits. I still admire the way Mormons stand up for what they believe in and support one another so strongly.

When my family left the church, it wasn't because we were angry or rebellious. We just had questions (big ones) that didn't seem to have clear answers. The decision wasn't easy. I said goodbye to friends and mentors who had shaped my early life, and our choice left some family members scratching their heads. But we stayed close, and I came away from that experience with two things: a deep respect for my roots and an insatiable hunger for answers. If I had to pin down what makes me tick, it's the drive to keep asking questions, and not settling until I get to the truth.

That same curiosity found its way into every corner of my life, especially academics. As a kid, I was fiercely competitive. I raced through multiplication tables like they were Olympic events and even put together my first "financial plan" after learning about actuaries from my uncle and grandfather. I thought numbers would always be my path forward, but over time, I realized math felt incomplete to me. It was a great tool, sure, but I wanted something more—something that connected abstract ideas to the mysteries of the real world.

That's when I discovered astrophysics. The universe is a puzzle that combines math, science, and imagination, and every answer we uncover just raises more exciting questions. Through shadowing the GATOS lab directed by Dr. Chris Packham at UTSA, I've explored topics like torus/AGN research and discovered how much I love the process of using data to understand the cosmos. Now, I hope to pursue a Ph.D. in astrophysics, focusing on using advanced tools and techniques to uncover the universe's deepest secrets.

By participating in the SURF program, I aim to deepen my understanding of experimental techniques and contribute to one of the most pressing questions in modern astrophysics: What is dark matter? The LZ detector's search for Weakly Interacting Massive Particles (WIMPs) is not just about uncovering the nature of dark matter, it's about piecing together the fundamental structure of the universe. This work resonates deeply with my fascination for astrophysics, where every discovery opens new doors to understanding the cosmos.

The hands-on experience of working in underground clean rooms, mastering clean room protocols, and assisting with the assembly and characterization of the detector will provide me with invaluable skills in experimental physics. These skills will complement my current research in AGN and infrared imaging, bridging the gap between theoretical questions and practical experimentation.

My career objective is to earn a Ph.D. in astrophysics and develop an active AI system capable of discerning exoplanets and identifying potential habitable worlds. I aim to combine machine learning and data analysis techniques to actively process and discern large datasets from telescopes, enabling faster and more accurate detection of exoplanets and their properties. Though the study of dark matter may not perfectly align with what I study or what I want to accomplish, it will be instrumentational in my understanding of gravity, physics, and cosmic formation.

Ultimately, this program will bring me closer to my goal of pursuing a Ph.D. in astrophysics, where I hope to use advanced tools and techniques to explore the universe's deepest mysteries. Whether it's dark matter, or galaxy evolution, I am driven by the desire to ask big questions and uncover the truths that lie hidden in the cosmos.

I have used AI only to help edit my grammar.

Black HIll

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The hands-on experience of working in underground clean rooms, mastering clean room protocols, and assisting with the assembly and characterization of the detector will provide me with invaluable skills in experimental physics. These skills will complement my current research in AGN and infrared imaging, bridging the gap between theoretical questions and practical experimentation.

Ultimately, this program will bring me closer to my goal of pursuing a Ph.D. in astrophysics, where I hope to use advanced tools and techniques to explore the universe's deepest mysteries. Whether it's dark matter, or galaxy evolution, I am driven by the desire to ask big questions and uncover the truths that lie hidden in the cosmos.

Taylored Statements

- **HAWAII - 500w limit**

My dad loves asking questions. He taught me to reason, question, and seek answers. He says, like a construction worker, I need to have a solid foundation of learning and reasoning before I can build ideas. My dad also emphasizes hard work. I am not always a top student: I "failed" kindergarten and recently, received 75 on a college chemistry test. But, I always keep working: I breezed by 1st grade and got a perfect score on the college chemistry final. Combining my work ethic with my curious nature, I continue to establish my foundation of learning and reasoning. Over time, I found that science offers the best framework for seeking answers. Among its mysteries, space captures my imagination the most, with its vastness and complexity.

As a young kid, I was fascinated with Neil deGrasse Tyson's version of *Cosmos*, which deepened my thoughts on space and its properties. Later, in my high school introductory physics courses, I began exploring the fundamental principles of the physical sciences. This sparked my desire to contribute to the astrophysical field. Currently, I shadow the GATOS Collaboration lab at University of Texas at San Antonio (UTSA), where we research active galactic nuclei (AGN) and infrared imaging of tori. By shadowing this lab, I have been introduced to techniques such as deconvolution and FortesFit for galaxy imaging. Also I have learned about mapping torus structures in galaxies and refining data to reveal critical phenomena like AGN. This experience has taught me the importance of precision, collaboration, and curiosity in scientific research.

The reason I am applying to this REU is my ultimate goal of earning a PhD in astrophysics with a focus in using artificial intelligence (AI) to analyze data from telescopic instruments to determine exoplanets. This REU will provide the perfect opportunity to broaden my research experience, and continue to develop the skills needed for my doctoral degree like data analysis, programming, and effective communication.

What sets this REU apart is Dr. Michael Bottom's work on exoplanet instrumentation. His recent collaborative publication, *Stellar Cornography,* explains how stars often "overwhelm" the planets around it with light, making the planets–potential exoplanets–quite hard to see. He goes on to say why it is important to have the proper instrumentation to make these cornographs and how it bolsters exoplanet research. Dr. Bottom's findings not only tie into the infrared imaging in the lab I currently shadow, but also directly into the PhD I currently want to pursue because how a telescope is built affects its data, making AI much more valuable for inferences. I am eager to contribute to this research or similar research at the UHawaii IfA REU.

Participating in this REU will allow me to broaden my scientific expertise, refine my technical skills, and prepare me for my graduate study and my career in astrophysics. This program represents a crucial step in my journey to explore the universe's mysteries while contributing meaningfully to the field and the University of Hawaii.

- **NAU**

My dad loves asking questions. He taught me to reason, question, and seek answers. He says that, like a construction worker, I need a solid foundation of learning and reasoning

before I can build ideas. He also instilled in me the value of hard work. I haven't always been a top student—I failed kindergarten and recently scored a 75 on a college chemistry test—but I always keep pushing forward. I breezed through first grade and earned a perfect score on my chemistry final. Combining taught persistence with my curious nature, I continue to establish my foundation of learning and reasoning. Over time, I found that science offers the best framework for seeking answers. Among its mysteries, space captivates me the most with its vastness and complexity.

As a kid, I was fascinated by Neil deGrasse Tyson's *Cosmos*, which deepened my thoughts on space and its properties. Later, in high school physics, I became intrigued by the fundamental principles of the physical sciences, sparking my desire to contribute to astrophysics. Now, I shadow the GATOS Collaboration lab at the University of Texas at San Antonio (UTSA), where I research active galactic nuclei (AGN) and infrared imaging of tori. This experience has introduced me to techniques such as deconvolution and FortesFit for galaxy imaging, along with mapping torus structures to reveal critical AGN phenomena. Through this work, I have developed a deeper appreciation for precision, collaboration, and curiosity in scientific research.

Also through UTSA, I have begun to be tutored privately by a peer doctorate student in python. He has gone as far as to lend me one of his personal laptops, walk me through how to configure Debian (a Linux distribution), and help me set up my own little python projects. As of my writing of this, I am still new to coding and data analysis through python, but by the end of the semester, I will have a firm grasp on the basics and probably more. Coding is essential to my doctorate I plan to get, so I must master this skill. I cannot stress enough that developing my coding skills is a crucial step toward my future research in astrophysics. As I continue to refine my programming abilities, I am excited to learn and grow with the NAU REU as my foundation.

I am applying to this REU because I hope to earn a Ph.D. in astrophysics, focusing on artificial intelligence (AI) applications for analyzing data from telescopic instruments to detect and characterize exoplanets. This REU offers an ideal opportunity to strengthen my research skills in data analysis, programming, and scientific communication—skills essential for my future graduate studies.

What sets this REU apart is its focus on exoplanetary systems at Lowell Observatory. Specifically, Joe Llama's work (perhaps not specifically in the REU, but his work at Lowell is wonderful), including his recent collaboration in *Planet Hunters TESS. V. A Planetary System Around a Binary Star, Including a Mini-Neptune in the Habitable Zone*, showcases the excitement of discovering exoplanets. His team's research on habitable zones, where water may exist (as well as many other significant findings), aligns with my fascination with the search for planetary conditions that could sustain life. Their methodology of data noise reduction to refine observational data ties directly into my research goals: using AI to accelerate and enhance exoplanet detection methods. I would be eager to contribute to this work or similar projects at the NAU REU.

Participating in this REU will allow me to broaden my scientific expertise, refine my technical skills, and prepare for graduate study and a career in astrophysics. This program represents a crucial step in my journey to push the boundaries of astrophysics, and I am hopeful to contribute to the University of Arizona's research while advancing my skills for a future in exoplanetary science.

- **MIT**

My friend once said to me, "You can get the boy out of church, but you can't take the church out of the boy." She was joking about my past as a Mormon, but her words got me thinking: maybe she had a point. Growing up Mormon taught me a lot about the importance of community, integrity, and helping others. These weren't just ideas—they were habits. I still admire the way Mormons stand up for what they believe in and support one another so strongly.

When my family left the church, it wasn't because we were angry or rebellious. We just had questions—big ones—that didn't seem to have clear answers. The decision wasn't easy. I said goodbye to friends and mentors who had shaped my early life, and our choice left some family members scratching their heads. But we stayed close, and I came away from that experience with two things: a deep respect for my roots and an insatiable hunger for answers. If I had to pin down what makes me tick, it's that drive to keep asking questions—and not just settling until I get to the truth.

That same curiosity found its way into every corner of my life, especially academics. As a kid, I was fiercely competitive. I raced through multiplication tables like they were Olympic events and even put together my first "financial plan" after learning about actuaries from my uncle and grandfather. I thought numbers would always be my path forward, but over time, I realized math felt incomplete to me. It was a great tool, sure, but I wanted something more—something that connected abstract ideas to the mysteries of the real world.

That's when I discovered astrophysics. The universe is a puzzle that combines math, science, and imagination, and every answer we uncover just raises more exciting questions. Through shadowing the GATOS lab directed by Dr. Chris Packham at UTSA, I've explored topics like torus/AGN research and discovered how much I love the process of using data to understand the cosmos. Now, I hope to pursue a Ph.D. in AI-driven instrumentation for exoplanet research, using machine learning to accelerate the search for life beyond Earth.

What sets this REU apart is [specific feature of the REU program, e.g., the emphasis on interdisciplinary collaboration or unique research facilities]. I'm especially drawn to the work of [Professor X], whose research on [specific topic] aligns closely with my interests. Their recent publication on [specific paper/project] caught my attention, particularly [something specific you liked about it]. I would love to contribute to similar projects during this program and gain valuable experience in [specific research skill or topic].

By participating in this REU, I aim to expand my scientific horizons and tackle challenging questions alongside talented peers and mentors. This experience will not only sharpen my technical and analytical skills but also bring me closer to my goal of understanding the universe—and maybe even helping humanity take its next big step toward finding life beyond Earth.

- **UTAH**

My dad loves asking questions. He taught me to reason, question, and seek answers. He says that, like a construction worker, I need a solid foundation of learning and reasoning before I can build ideas. He also instilled in me the value of hard work. I haven't always been a top student—I failed kindergarten and recently scored a 75 on a college chemistry test—but I

always keep pushing forward. I breezed through first grade and earned a perfect score on my chemistry final. Combining taught persistence with my curious nature, I continue to establish my foundation of learning and reasoning. Over time, I found that science offers the best framework for seeking answers. Among its mysteries, space captivates me the most with its vastness and complexity.

As a kid, I was fascinated by Neil deGrasse Tyson's Cosmos, which deepened my thoughts on space and its properties. Later, in high school physics, I became intrigued by the fundamental principles of the physical sciences, sparking my desire to contribute to astrophysics. Now, I shadow the GATOS Collaboration lab at the University of Texas at San Antonio (UTSA), where I research active galactic nuclei (AGN) and infrared imaging of tori. This experience has introduced me to techniques such as deconvolution and FortesFit for galaxy imaging, along with mapping torus structures to reveal critical AGN phenomena. Through this work, I have developed a deeper appreciation for precision, collaboration, and curiosity in scientific research.

Also through UTSA, I have begun to be tutored privately by a peer doctorate student in Python. He has gone as far as to lend me one of his personal laptops, walk me through how to configure Debian (a Linux distribution), and help me set up my own Python projects. As of my writing of this, I am still new to coding and data analysis through Python, but by the end of the semester, I will have a firm grasp on the basics and likely more. Coding is essential to the Ph.D. I plan to earn, so I must master this skill. Developing my coding abilities is a crucial step toward my future research in astrophysics. As I continue to refine my programming skills, I am excited to learn and grow with the University of Utah REU as my foundation.

I am applying to this REU because I hope to earn a Ph.D. in astrophysics, focusing on artificial intelligence (AI) applications for analyzing data from telescopic instruments to detect and characterize large-scale structures in the universe. This REU offers an ideal opportunity to strengthen my research skills in data analysis, programming, and scientific communication—skills essential for my future graduate studies.

What sets the University of Utah REU apart is its focus on observational cosmology and large-scale structure. Specifically, Dr. Zheng Zheng's research on galaxy clustering and cosmological surveys is deeply intriguing. His work on the Dark Energy Spectroscopic Instrument (DESI) Early Data Release represents an exciting frontier in mapping the universe with unprecedented precision. The methodology of using large-scale galaxy surveys to constrain cosmological parameters aligns directly with my research interests in using AI to refine data analysis techniques for astrophysics. I would be eager to contribute to this work or similar projects at the University of Utah REU.

Participating in this REU will allow me to broaden my scientific expertise, refine my technical skills, and prepare for graduate study and a career in astrophysics. This program represents a crucial step in my journey to push the boundaries of astrophysics, and I am eager to contribute to the University of Utah's research while advancing my skills for a future in cosmology and data-driven astrophysics.

- **New Mexico**

I am excited to apply for the Research Experiences for Undergraduates (REU) program at the University of New Mexico (UNM). As a freshman majoring in Physics with a concentration in Astrophysics at the University of Texas at San Antonio (UTSA), I am eager to contribute to UNM's research in long-wavelength radio astronomy and deepen my expertise in astrophysical instrumentation.

Currently, I am shadowing the GATOS Collaboration lab at UTSA, where I research active galactic nuclei (AGN) and infrared imaging of tori. Even more specifically, I have been tasked with helping to design/train an AI that will correctly sift through JWST data on AGN. This experience has introduced me to techniques such as deconvolution and FortesFit for galaxy imaging, along with mapping torus structures to reveal critical AGN phenomena. Through this work, I have developed a deeper appreciation for precision, collaboration, and curiosity in scientific research.

Additionally, I am enhancing my programming skills through private tutoring in Python, focusing on data analysis and coding—skills essential for my future research in astrophysics. As I continue to refine my programming abilities, I am excited to learn and grow with the UNM REU as my foundation.

The main reason I am applying to this REU is my aspiration to earn a Ph.D. in astrophysics, focusing on artificial intelligence (AI) applications for analyzing data from telescopic instruments to detect and characterize large-scale structures in the universe. Through further AGN research and hands-on experience with advanced data analysis techniques, I aim to refine my ability to extract meaningful astrophysical insights from complex datasets. This REU provides the ideal environment to strengthen my expertise in observational techniques and computational methods critical for my future studies.

I am particularly interested in working with the Long Wavelength Array (LWA) project, pioneered by Professor Greg Taylor at UNM. The LWA, covering frequencies from 5 to 88 MHz, offers unique opportunities to study pulsars, transients, and AGN. Professor Taylor's research, including the detection of Fe K lines from the binary AGN in 4C+37.11, aligns with my interest in AGN phenomena and their observational signatures. I am eager to contribute to projects involving the LWA, applying my developing expertise in data analysis and radio astronomy with the goal of gaining intense insight on how to make an AI for data reduction/refinement on both exoplanets, and potential uses for AGN research at both UTSA and UNM.

Participating in this REU will allow me to broaden my scientific expertise, refine my technical skills, and prepare for graduate study and a career in astrophysics. This program represents a crucial step in my journey to push the boundaries of astrophysics, and I am hopeful to contribute to the University of New Mexico's research while advancing my skills for a future in exoplanetary science.

Do not let my lab shadowing experience suggest a lack of teamwork skills. I have worked in both formal and informal team settings. Last summer, I balanced full-time custodial work at an elementary school with club track and scholarship applications. In my custodial role, I reported to the assistant manager, who coordinated with the main manager to evaluate my performance. This experience required communication, adaptability, and responsibility within a structured team environment. Simply put, I can handle the heat.

Outside of the areas mentioned above, I have driven through New Mexico twice during long-distance trips, and I was struck by its beauty. UNM stands out as a university that is not only geographically stunning but also rich in cultural diversity and technological innovation, making it an especially appealing place to study and grow.

- **University of Kentuky**

I am drawn to these projects because they align perfectly with my research interests and academic journey. My experience analyzing AGN through the GATOS Collaboration at UTSA, where I explored infrared imaging and deconvolution techniques, has fueled my curiosity about large-scale structures and their connection to galactic evolution. Additionally, my growing proficiency in Python, data analysis, and statistical tools prepares me to handle the computational demands of projects like IllustrisTNG simulations and AGN variability analysis. I am particularly excited by the opportunity to study galaxy clusters and AGN properties, as these investigations tie directly into my passion for using simulations and data analysis to understand the universe's most complex structures. By participating in these projects, I aim to deepen my astrophysics expertise and contribute meaningfully to cutting-edge research in cosmology and galaxy evolution.

At the University of Kentucky Department of Physics and Astronomy, I look forward to "questioning," "competing," and discovering, continuing this humble pilgrimage of mine.

Cover Letter

William Olson
14243 Swallow Dr.
San Antonio TX, 78217
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+1 (210) 274-3958
1/11/2025

REU Program Coordinator
MIT Haystack Observatory
99 Millstone Road
Westford, MA 01886

Subject: Application for the MIT Haystack Observatory REU Program

Dear REU Program Coordinator,

I am excited to apply for the Research Experiences for Undergraduates program at the MIT Haystack Observatory. As a freshman studying Physics with a concentration in Astrophysics at the University of Texas at San Antonio (UTSA), I am eager to learn from and contribute to [Haystack's] [cutting-edge research in radio astronomy, atmospheric science, and related fields].

My academic and research experiences have fueled my passion for understanding the universe through innovative methods. Currently, I am shadowing the G.A.T.O.S. Collaboration lab at UTSA, where I am learning techniques like deconvolution and data refinement to study active galactic nuclei and infrared imaging. This experience has honed my skills in data analysis, programming, and teamwork. These skills, combined with my curiosity and determination, make me eager to contribute to Haystack's research.

I am particularly drawn to Haystack's advancements in Very Long Baseline Interferometry (VLBI) and radar techniques used to observe black holes revealing new phenomena (event horizon attributes and more). These areas align with my goal of using advanced instrumentation and data analysis to explore active galactic nuclei (AGN) as well as deconvolution to isolate the nuclei from the noise data. I am confident in my foundation with deconvolution and my enthusiasm for learning will help me thrive in this dynamic research environment.

Beyond my technical abilities, I bring a strong work ethic and collaborative spirit. While shadowing the GATOS lab, I followed closely with team members to refine imaging processes and interpret data, which taught me the importance of precision and teamwork in achieving meaningful results. I am excited to bring this adaptability and focus to the [Haystack] REU program.

Thank you for considering my application. I look forward to the opportunity to discuss how my background and goals align with the program's objectives. I am eager to contribute my skills and enthusiasm to [Haystack's] research while continuing to grow as a scientist.

Sincerely,
William Eric Olson

Taylored Letters

- **MIT**

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- **Berkely**

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Sincerely,
William Eric Olson

JWST Summer School

From a young age, I've been driven by an insatiable curiosity to ask big questions and uncover the universe's hidden truths. This passion led me to astrophysics, where I now pursue a Physics degree with an Astrophysics concentration at UTSA. Through the G.A.T.O.S. Collaboration lab, I've gained hands-on experience with deconvolution and data refinement techniques, studying active galactic nuclei (AGN) and infrared imaging. While I don't yet have direct JWST experience, I am deeply inspired by its potential to revolutionize our understanding of the cosmos.

The JWST Summer School is a transformative opportunity to bridge my current research with my long-term goals. I am eager to learn how JWST's advanced instrumentation can enhance our understanding of AGN, torus models, and exoplanetary systems. This program will equip me with the technical expertise and collaborative experience needed to advance my research and prepare for a Ph.D. in astrophysics.

As a driven and curious freshman, I am excited to contribute my unique perspective, absorb new knowledge, and take the next step in my journey to uncover the universe's deepest mysteries. Thank you for considering my application.

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As a driven and curious freshman, I am excited to contribute my unique perspective, absorb new knowledge, and take the next step in my journey to uncover the universe's deepest mysteries. Specifically, I hope to explore how JWST's infrared capabilities can shed light on the structure and evolution of AGN tori, a topic I am actively studying in the GATOS lab at UTSA. By participating in this program, I aim to deepen my understanding of observational techniques and contribute to the groundbreaking discoveries made possible by JWST.
Thank you for considering my application. I look forward to the opportunity to learn, grow, and contribute to the JWST Summer School.

From a young age, I've been driven by an incurable curiosity to ask big questions and uncover the universe's hidden truths. This passion led me to astrophysics, where I now pursue a Physics degree with an Astrophysics concentration at UTSA. Through the G.A.T.O.S. (galaxy, activity, torus, and outflow survey) collaboration lab, I've gained hands-on experience with deconvolution and data refinement techniques, studying active galactic nuclei (AGN) and infrared imaging. While I don't yet have direct JWST experience, I am deeply inspired by its potential to revolutionize our understanding of the cosmos.

The JWST Summer School is a transformative opportunity to bridge my current research with my long-term goals. I am eager to learn how the telescope's advanced instrumentation can enhance our understanding of AGN, torus models, and exoplanetary systems. This program will equip me with the technical expertise and collaborative experience needed to progress my own research and prepare for a Ph.D. in astrophysics.

As a driven and curious freshman, I am excited to contribute my unique perspective, absorb new knowledge, and take the next step in my journey to uncover the universe's deepest mysteries. At the G.A.T.O.S. collaboration lab, I've worked alongside a postdoc who secured JWST archive time to advance our research on AGN tori, dust structures, and modeling techniques. This experience has fueled my desire to leverage the JWST's infrared capabilities to explore these phenomena further. By participating in this program, I aim to deepen my understanding of observational techniques and contribute to the groundbreaking discoveries made possible by the telescope.

Thank you for considering my application. I look forward to the opportunity to learn, grow, and contribute at the JWST Summer School.

BEST VERSION YET??
My friend once said to me, "You can get the boy out of church, but you can't take the church out of the boy." She was joking about my past as a Mormon, but maybe she had a point. Growing up in Mormonism taught me the importance of community, integrity, and helping others—values that forged habits. When my family left the church, it wasn't out of anger or rebellion; we just had big questions without clear answers. That experience left me with two things: a respect for my roots and an insatiable hunger for understanding, a drive that now fuels my pursuit of astrophysics.

This drive has shaped my academic path and fueled my passion for astrophysics. As a Physics major concentrating in Astrophysics at UTSA, I began shadowing the GATOS Collaboration lab under Dr. Chris Packham in my first semester. We study active galactic nuclei (AGN) and infrared imaging of tori, and I've mastered techniques like deconvolution, FortesFit, and Python for data analysis. I'm currently training AI models to identify AGN emission lines further exploring how machine learning can uncover hidden patterns in large datasets. This will hopefully illuminate hidden "tendencies" of cosmological evolution, and help realize the cosmos's explosive past.
Ultimately, my goal is to earn a Ph.D. in astrophysics and develop AI-driven instrumentation for exoplanet research, creating adaptive systems that respond to astrophysical stimuli and accelerate the search for life beyond Earth. The JWST Summer School will allow me to apply my skills in AGN and infrared imaging to its groundbreaking data, advancing my understanding of tori and detection systems. Participating in this program will sharpen my technical skills, bridge my research with cutting-edge observational techniques, and bring me closer to answering the universe's most profound questions: a journey I have committed to.

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Ultimately, my goal is to earn a Ph.D. in astrophysics and develop AI-driven instrumentation for exoplanet research, creating adaptive systems that respond to astro-stimuli and accelerate the search for life beyond Earth. The JWST Summer School will allow me to apply my skills in AGN and infrared imaging to its groundbreaking data, advancing my understanding of tori and detection systems. Participating in this program will sharpen my technical skills, bridge my research with detection software, and bring me closer to answering the universe's most profound questions: a journey I have committed to.

European Astrochem

The dust chemistry of active galactic nuclei (AGN)—especially in high-redshift ‘little red dots’—holds critical clues about black hole accretion and feedback, but spectral obstacles and observational limits hinder progress. The dust torus, likely rich in silicates and carbonaceous grains, exhibits complex radiative transfer behavior that traditional models struggle to decode. This is where AI-driven astrochemistry can revolutionize our approach. By collaborating with this COST Action, I aim to develop machine learning tools (e.g., neural networks for spectral deblending or Bayesian inference for dust properties) to disentangle AGN dust signatures from host galaxies. My goal is to leverage multi-wavelength datasets and the Action’s interdisciplinary network to bridge theoretical predictions with observable chemistry.
Joining this effort would not only advance AGN science but also contribute open-source AI solutions to the broader astrochemistry community—aligning perfectly with COST’s mission of innovation through collaboration.

AGN (active galactic nuclei) like the ‘little red dots’ have been found to be surrounded by dust that absorbs and scatters light. That “dust” limits Current models can’t fully explain what this dust is made of or how it behaves.
I believe AI tools could crack this problem. For example, machine learning might help separate the dust’s signal from other light in AGN spectra or predict its composition faster than old methods. Through this COST Action, I want to work with others to build these tools, share ideas, and use AI to push our understanding of black holes and their dust.
This isn’t just about AGN—it’s about creating open, reusable methods that help everyone studying cosmic chemistry.

4yr plan scholarship

Efficiency is the name of the game for the next thirty years. Artificial Intelligence (AI) is increasingly taking over tedious aspects of research, and this scholarship will help me keep building tools that make science faster, more accurate, and more accessible. In astrophysics, there are calculations like spectral line identification, galaxy morphology classification, and source extraction that can take hours to do manually. By creating AI models to handle these tasks, scientists can spend more time asking deeper questions instead of getting stuck in repetitive work.

In the field of active galactic nuclei (AGN) research, I developed my own AI tool this summer to predict ionization outputs from AGN. I planned the data processing pipeline carefully, trained the model, and made sure it could scale to other areas of AGN science. This experience is giving me the foundation I need for my long-term goal: earning a Ph.D. and becoming a professor who uses AI to speed up astrophysical discovery. My dream is to expand my work toward exoplanet research and possibly using machine learning to flag worlds most likely to harbor life. The big win I see is when AI can make reliable predictions or even conclusions without the need for as many massive, billion-dollar instruments.

My fascination with efficiency began my freshman year when I joined my first research group. I imagined working directly with a telescope, recording observations by hand. My mentor explained that this was the “old, old” way, long replaced by computers. It hit me how far efficiency had already come. Combined with lessons from track and field, I realized that my academic and athletic life both revolved around refining processes to get the best possible result.

Track often feels like swinging a sword in some wuxia novel— endless practice, always whetting the blade and sharpening my stride. When I train correctly, my efficiency improves and my speed increases. It’s like a farmer watching crops grow, knowing every small adjustment today pays off later. I take the same approach in my hobbies, like optimizing competitive Yu-Gi-Oh! decks to maximize consistency. Whether it’s running, gaming, or research, I’m drawn to breaking complex systems into their most effective forms.

I also give my time to help others grow. In the summer, I coach with a club track and field team, focusing especially on kids ages 7–10. I work to build their foundation early so they don’t have to rebuild it later, giving them the option to be fast for life. I also volunteer at the San Antonio Food Bank, where I’ve seen firsthand how teamwork and planning turn limited resources into maximum impact.

If awarded, this scholarship will directly support my research and outreach. I would use it to invest in computing tools and resources to advance AI applications in astrophysics and share them with others. I’ve already attended the CosmicAI conference to exchange ideas with other researchers, and with this scholarship, I can continue spreading the efficiency mindset that drives both my science and my service.

I am very honored to receive the Spring 2025 Dr. Craig Jordan Excellence in Student Success Endowed Award. Thank you for investing in my academic and professional journey.

This scholarship is far more than just $500 to me. It represents momentum akin to a push forward in my work to make astrophysics research more efficient through the use of machine learning. My goal is to create AI tools that can take on time-consuming calculations in astronomy, saving scientists hours of work and potentially reducing the need for expensive,

resource-heavy instruments. In the long run, this approach could save millions of dollars while opening the door to faster, more frequent scientific discoveries.

Your support is not a gamble. I intend to carry the principle of efficiency through every aspect of my career and share it with others. From my undergraduate research on active galactic nuclei to my aspirations of one day applying AI to exoplanet studies, this award fuels my ability to keep building, learning, and contributing.

Thank you for believing in my potential and for supporting the next generation of astrophysicists.

With sincere gratitude,

William Olson

California

LONG ANSWER (5000 Character Limit)

What most interests you about being a part of the LIGO SURF program? What are you hoping to learn about? (200 words or fewer)

Which of these research areas are you most interested in working on? You can read about each topic at:
https://labcit.ligo.caltech.edu/LIGO_web/students/SURF/projects.shtml

Rank in the order of your preference (1=most preferred)

Frequency Stabilization of 2 Micron Lasers Using Optical Delay Self-Heterodyne Interferometry - 2
Sensor Fusion for Improved Length Sensing and Control of the LIGO 40m prototype - 1
Quality Testing Optically Contacted Bonds - 8
Measuring Birefringence and its Fluctuations - 7
Theoretical astrophysics, general relativity, black hole physics - 3
GWs from compact binary coalescences - 4
Understanding LIGO detector behavior - 5
Multi-messenger astronomy - 6

Question #7

LONG ANSWER (5000 Character Limit)

What steps have you taken to pursue research opportunities at your home institution or how did you first become involved in research? Have you had difficulties finding research opportunities? Note that there will be space to describe your research experience in a later question. (150 words or fewer)
0 of 5000 characters

Through financial considerations, I was introduced to the Academic Personal and Professional Excellence (APPEX) scholarship program at UTSA, which encouraged me to actively seek out research opportunities. Through APPEX, I joined Dr. Chris Packham's MAE research group, which collaborates with the international GATOS survey. I was initially tasked with identifying ionization cone structures in simplified 2D images. To approach this, I developed a U-Net–based model trained on a combination of synthetic data and JWST NIRCam observations. Through this work, I recognized that morphology alone was insufficient for reliably identifying ionization cones. This realization motivated a transition to working with spectral data cubes, which contain richer physical information. I then implemented slice-based methods and line-ratio maps to generate improved masks, with the goal of extending this approach to JWST MIRI data.

Question #8

LONG ANSWER (5000 Character Limit)

Please describe a research project in which you used lab skills, analyzed data, or something related. If you haven't worked in a formal lab before, please describe a class project. (200 words or fewer)
0 of 5000 characters

One challenge I encountered during my research was working with spectral data cubes, which required handling higher-dimensional data than I had previously used. To analyze these cubes, I built a pipeline to extract spatial slices at specific wavelengths and normalize them consistently across the dataset. I then generated line-ratio maps to emphasize ionized gas features while suppressing unrelated structure.
This process required careful debugging, as small inconsistencies in slice alignment or normalization produced misleading features in the resulting masks. I learned to systematically isolate sources of error by testing intermediate outputs and comparing synthetic and real observations. Managing noise and resolution differences across wavelengths was particularly challenging and forced me to think carefully about how physical assumptions translated into data processing choices.
Through this work, I gained experience working independently with complex observational data, designing reproducible analysis pipelines, and iteratively refining methods when initial assumptions proved insufficient.

Question #9

LONG ANSWER (5000 Character Limit)

What (if any) past experience do you have doing research in astrophysics, gravitational-wave science, or a related topic? (100 words or fewer)
0 of 5000 characters

I have research experience in observational astrophysics and data-driven analysis through work on identifying ionization cone structures in active galactic nuclei using JWST observations. This work involved analyzing noisy, high-dimensional data, developing reproducible pipelines, and evaluating model performance under imperfect observational conditions. While I have not yet conducted research directly in gravitational-wave science, my experience working with complex datasets, signal extraction, and instrumentation-aware analysis provides a strong foundation for contributing to gravitational-wave detector and data analysis projects.

Question #10

LONG ANSWER (5000 Character Limit)

Tell us about what you do other than schoolwork or research that might be relevant for this opportunity. For example: What jobs do you have or have had in the past? What extracurricular or volunteer activities do you participate in? Do you have any hobbies or interests? (100 words or fewer)

0 of 5000 characters

Outside of coursework and research, I am drawn to activities that reward discipline and optimization. I am a competitive track and field athlete, where progress depends on careful training, iteration, and long-term consistency. I also serve on the Honors College Student Advisory Board, where I am developing a machine-learning–based project to identify factors influencing student engagement at Honors events. In my free time, I enjoy strategic games that emphasize pattern recognition and optimization. Across these activities, I am motivated by improving systems, whether physical, organizational, or computational, through careful analysis and iteration.

Question #11

LONG ANSWER (5000 Character Limit)

How will this program benefit your physics education and career goals? The LIGO SURF program aims to support students on the path to pursuing a higher degree in physics or a related field. We are particularly interested if this program can benefit you in ways that are not available at your home institution. (200 words or fewer)
0 of 5000 characters

My current research experience has focused on developing data-driven methods for extracting physical structure from complex astrophysical datasets. While this has given me a strong foundation in analysis and modeling, my home institution does not offer direct access to large-scale gravitational-wave instrumentation or detector development.
The LIGO SURF program would allow me to gain hands-on experience with interferometric detectors, sensing and control systems, and the process by which precision measurements are translated into astrophysical data. Working within the LIGO environment at Caltech would expose me to experimental challenges that cannot be replicated through coursework or small-scale projects, including noise characterization, calibration, and real-time problem solving on operational systems.
This experience would be a critical step in my preparation for graduate study in physics by complementing my background in data analysis with a deeper understanding of how gravitational-wave data are generated and validated. By learning how detector design and control influence scientific outcomes, I would be better equipped to pursue research at the intersection of astrophysics, instrumentation, and data analysis.

Floridia GRAV

My motivation for applying to the Gravitational Physics International REU grew out of a gradual shift in how I think about data, measurement, and what it means to “see” a physical signal. In my current research, I work with JWST observations of active galactic nuclei, initially approaching the problem from an image-based and machine-learning perspective. The goal was simple on paper: identify physical structures in complex data. In practice, it quickly became clear that the difficulty was not just computational.

As I began working with noisier and higher-dimensional datasets, I noticed how strongly the results depended on choices made long before any model was trained. Calibration methods, resolution limits, and noise treatment could significantly change which features appeared real and which disappeared entirely. That was a turning point for me. Rather than viewing data analysis as the final step in extracting knowledge, I became increasingly interested in the upstream physics that determines what information is recoverable in the first place.

This way of thinking aligns naturally with gravitational physics. Gravitational-wave science sits at the intersection of theory, instrumentation, and data analysis, where understanding noise sources and measurement limits is not optional but central. Signals are subtle, detectors are exquisitely sensitive, and meaningful discoveries depend on knowing how the instrument shapes the data as much as how the data are analyzed afterward. That challenge is exactly what draws me to the field.

What particularly excites me about this REU is its international structure. Gravitational physics is fundamentally collaborative, and learning within multiple research environments would expose me to different approaches to precision measurement, detector characterization, and data interpretation. Experiencing these perspectives firsthand would deepen my understanding of how large-scale scientific collaborations function and how experimental decisions propagate into scientific results.

At this stage in my education, I am preparing for graduate study and am actively seeking opportunities that expand my foundation beyond my current research environment. This REU offers a chance to engage directly with gravitational physics while strengthening my understanding of measurement, noise, and instrumentation. For me, it represents a natural next step: building a stronger foundation before continuing upward, and learning how signals are shaped before asking what stories they tell.

Floridia GRAV List

1. Probing neutron star interiors using gravitational waves
   This project excites me because it connects the signals from mergers directly to the physics inside neutron stars. The post-merger waveform contains information about the equation of state, which is still uncertain. The idea of analyzing simulated and real data to extract meaningful astrophysical insight is fascinating, and I am drawn to refining models that push both detectors and theory.

2. Ensemble gravitational-wave detections: more than the sum of the parts
   I am interested in what can be learned from a population of events rather than individual detections. Studying patterns across multiple mergers to reveal deviations from general relativity or to understand how stellar-mass black holes formed is compelling. It combines computational work with astrophysics, which fits my interest in using data to answer fundamental questions.

3. Gravitational waves from neutron star magnetic explosions
   Exploring gravitational waves from magnetar flares is a new frontier. Investigating how magnetic field topology and neutron star equations of state affect the emission is intriguing. I enjoy projects where I can model complex astrophysical systems and link them to observables, and this project gives me that opportunity.

4. Squeezed light in a cryogenic opto-mechanical system
   This project tackles a fundamental sensitivity limit in gravitational-wave detectors: quantum noise. Using frequency-dependent squeezed light to push beyond that limit combines optical simulation, FPGA and Python programming, and data analysis. I am drawn to applying computational and experimental skills to a hands-on challenge at the forefront of detector technology.

5. Full calibration of LISA Pathfinder dynamics
   I am interested in working with real space-based data and understanding how a detector's performance affects the signals we see. Redoing the system calibration and exploring cross-talk between degrees of freedom offers a mix of modeling, data analysis, and problem-solving. This project gives a deep dive into how precision instrumentation and astrophysical measurement interact.

STScI

Absolutely, we can remove all em dashes and replace them with commas, parentheses, or separate sentences to keep it polished and professional. Here's the same rewritten set of answers with clean punctuation and no em dashes:

Q5. Academic or non-academic work that prepared you for SASP

My preparation for SASP reflects a gradual shift from learning how research works to learning how scientific judgment is formed. Early exposure to research culture through the Academic Personal and Professional Excellence (APPEX) program at UTSA emphasized long-term planning, mentorship, and the expectation that progress emerges through iteration rather than linear success. Framing research as a process of refining questions rather than completing tasks shaped how I engage with open-ended problems.

Initial research experiences focused on developing reliable scientific habits. A collaborative environmental science project assessing water quality at Lake Boerne highlighted the importance of experimental design, consistent measurement, and uncertainty awareness. Interpreting environmental data required careful attention to methodological choices and their consequences, reinforcing that conclusions are only as strong as the assumptions beneath them. Communicating results through a poster presentation further emphasized clarity and accountability in scientific work.

These habits carried forward into astrophysics research within Dr. Chris Packham's group, where working with real astronomical data revealed new layers of complexity. Noise, resolution limits, and modeling assumptions became central constraints rather than technical details. Early exposure to image deconvolution demonstrated how analytical choices propagate into scientific interpretation, motivating deeper questions about scalability, consistency, and physical meaning.

As datasets grew larger and more complex, the limitations of manual and morphology-driven approaches became increasingly apparent. This prompted a shift toward developing automated, physically motivated methods for identifying AGN ionization cones, first in imaging data and now extending into spectroscopic cube analysis. This progression emphasized the importance of embedding physical structure directly into computational workflows rather than relying on pattern recognition alone.

Together, these experiences have prepared me for SASP by strengthening my ability to reason carefully with complex datasets, adapt methods as questions evolve, and connect computation directly to physical interpretation within astronomy.

Q6. Previous research experience (most recent first, narrative bullets)

Ionization Cone Identification and Characterization in AGN – UTSA (May 2025 to Present)
As AGN samples expanded, it became clear that identifying ionization cones by eye was neither scalable nor reproducible, particularly when subjectivity influenced classification. Addressing this limitation motivated the development of an automated, physically constrained approach to cone identification. Early image-based methods evolved toward spectroscopic cube analysis, where spatial and spectral information jointly inform ionization diagnostics relevant to infrared data, with future application to JWST MIRI observations. Iterative refinement, Bayesian evaluation, and failure analysis reinforced the importance of diagnosing why models succeed or fail rather than optimizing performance alone.

AGN Image Deconvolution with FortesFit and CAT3D-WIND – UTSA (Oct 2024 to Mar 2025)
Interpreting deconvolved astronomical images highlighted how sensitive results can be to modeling assumptions and parameter choices. Exploring these dependencies strengthened an appreciation for uncertainty, model dependence, and the iterative nature of astrophysical data analysis. Experience with FITS files, DS9 visualization, and preprocessing established technical foundations later applied to higher-dimensional datasets.

LIRA: Lunar Infrared Array Mission Concept – UTSA (Jan 2025 to May 2025)
Designing a lunar infrared observatory concept emphasized feasibility as a scientific constraint. Integrating science goals with operational and technical limitations required systems-level reasoning and coordination across disciplines, reinforcing how large-scale astronomy depends on coherent planning as much as innovative ideas.

Lake Boerne Water Quality Assessment – UTSA (Sep 2024 to Oct 2024)
This early research experience emphasized data integrity, measurement consistency, and scientific communication. Presenting results through a poster established habits of clarity and accountability that informed later work in astrophysics.

Q7. Coding experience (Thinking-centered version)

My coding experience developed as a response to scientific limitations rather than as a standalone technical pursuit. Programming became a way to test physical ideas, explore assumptions, and identify failure modes within increasingly complex datasets. Python is my primary language, used for data analysis, visualization, and model development, particularly with astronomical FITS files.

As manual approaches to identifying ionization cones proved insufficient, computational methods became necessary to enforce consistency and physical grounding. Developing workflows for image and spectral cube analysis required generating synthetic datasets informed by astrophysical assumptions, constructing evaluation tools that expose weaknesses, and iterating designs based on diagnostic insight rather than performance metrics alone.

Across projects, coding has functioned as an experimental space where assumptions are made explicit and revised when they fail. This approach emphasizes interpretability, reproducibility, and adaptability as core principles rather than technical optimization.

Q8. Immediate and long-term academic/career plans (Judgment-forward)

I plan to pursue a PhD in astrophysics focused on developing computational tools that extract physically meaningful structure from complex observational data. I am drawn to problems where the primary challenge is not data scarcity, but interpretation, particularly in high-dimensional datasets such as infrared spectral cubes.

Training at UTSA emphasized iterative problem-solving and evidence-based refinement, particularly through APPEX and research within Dr. Chris Packham's group. Working at the intersection of astrophysics and computation reinforced the importance of designing models around physical constraints so that results reflect underlying processes rather than statistical coincidence.

Looking ahead, I aim to extend my current work on ionization cones into infrared spectral diagnostics using JWST MIRI data. SASP aligns directly with this trajectory by offering immersion in archival datasets, observatory-scale data products, and the research culture that shapes modern space-based astronomy.

Q9. Extracurricular activities and interests (Tone-aligned)

Outside of formal research, I engage in activities that reinforce habits essential to scientific work, including mentorship, communication, and sustained discipline. Programs such as APPEX emphasized professional development and resilience, while research presentations strengthened my ability to communicate technical ideas clearly and responsibly.

Beyond academics, regular physical training and athletics provide structure and consistency that support focus, time management, and persistence across long-term projects. These activities complement my academic work by reinforcing the discipline and mindset required for sustained research.

If you want, I can **also do the SASP benefit essay next**, making it forward-looking and tied to **the specific MIRI and cube work**, completely avoiding repetition of your Goldwater-style statements. That one benefits a lot from this "thread-of-judgment" style too.

Do you want me to do that now?

My interest in astronomy has gradually shifted from asking what data can show to asking how meaning is reliably extracted from increasingly complex observations. As astronomical datasets have grown in size and dimensionality, especially with facilities like the James Webb Space Telescope, the challenge is no longer access to data but developing analysis methods that are scalable, reproducible, and physically interpretable. Participating in the Space Astronomy Summer Program would place me in an environment where these challenges are addressed directly through hands-on research, mentorship, and exposure to the inner workings of a space-based observatory.

My current research focuses on identifying ionization cones in active galactic nuclei using machine learning techniques informed by astrophysical structure. This work has recently transitioned from two dimensional imaging to working with spectral cubes, where spatial and spectral information must be treated together rather than independently. Moving into cube based analysis has highlighted how closely scientific interpretation depends on calibration choices, data reduction pipelines, and assumptions embedded in software tools. SASP offers direct exposure to these processes through projects involving data reduction, instrument calibration, and archival workflows, making it an ideal setting to deepen both my technical and scientific understanding.

Working at STScI would allow me to learn how raw data from Hubble and JWST are transformed into science-ready products, and how calibration teams characterize instruments to enable precise measurements. This experience is particularly valuable for my long term goal of extending ionization cone detection methods to mid infrared observations using MIRI data. Understanding how spectral cubes are constructed, validated, and archived would strengthen my ability to design machine learning models that respect the physical and instrumental structure of the data rather than treating observations as abstract inputs.

Equally important is the opportunity to work closely with STScI researchers and staff who operate at the intersection of astronomy, software development, and instrumentation. SASP's mentorship structure would provide insight into how professional astronomers formulate research questions, balance scientific ambition with feasibility, and iterate on analysis strategies when results do not behave as expected. Observing how large projects are initiated and carried through at an institute level would help me transition from independent undergraduate research to the scale and rigor expected in graduate study.

The program's emphasis on lectures, cohort interaction, and scientific communication also aligns strongly with my development goals. Astronomy is a collaborative field, and learning how to communicate results clearly across disciplinary boundaries is essential for producing impactful science. Presenting work within a cohort and engaging with peers pursuing diverse

projects would broaden my perspective and strengthen my ability to place my own research within a larger scientific context.

Ultimately, SASP represents a critical step between undergraduate research and doctoral training. It offers an immersive introduction to space based astronomy as it is practiced at a leading research institute, not just in theory but in daily workflow. By combining hands-on project work, mentorship, and exposure to archival and calibration infrastructure, the program would equip me with the technical skills, scientific judgment, and professional perspective needed to pursue research that extracts physical insight from complex astronomical data.

Short Response #2: Personal strengths that have contributed most to my success

My success in academic and research settings has been shaped less by any single technical skill and more by a set of habits developed through working with open ended problems. Curiosity, adaptability, and persistence have been essential as my responsibilities have increased and as problems have shifted from well defined coursework to research questions without predetermined solutions.

Curiosity has consistently driven how I approach scientific work. Early research experiences made it clear that many interesting astrophysical problems are limited not by the availability of data, but by how effectively structure and meaning can be extracted from it. Encountering complex observational datasets raised questions about scalability, reproducibility, and interpretation that traditional methods struggled to address. Rather than treating these limitations as obstacles, I learned to treat them as entry points for investigation. This mindset led naturally toward computational approaches that emphasize structure, assumptions, and physical reasoning rather than surface level pattern recognition.

Adaptability became increasingly important as projects grew more complex and less scripted. Research rarely followed a linear path, and early attempts often failed to produce meaningful results. Instead of abandoning these efforts, progress depended on diagnosing why an approach failed and deciding which assumptions required revision. This was particularly evident in my work developing machine learning methods for astrophysical analysis, where model performance depended strongly on how physical constraints were encoded into data and training workflows. Learning to revise strategies based on evidence rather than expectation became a central strength, allowing me to remain productive even when initial plans proved ineffective.

Persistence has been equally important in navigating ambiguity. Unlike coursework, where success is measured by completion, research demanded sustained engagement with uncertainty. Progress often required extended periods of iteration, testing, and reevaluation without immediate validation. Developing comfort with this process allowed me to continue refining ideas even when outcomes were unclear. Over time, persistence transformed uncertainty from a source of frustration into a signal that a problem was worth pursuing more carefully.

These strengths were reinforced outside traditional research settings through my work with the UTSA Honors College Advisory Board. I was recruited to serve on the board, which collaborates with faculty and administrators to support academic programming and institutional initiatives. In this role, I applied machine learning methods to help analyze participation and engagement across honors events. Unlike research datasets designed for analysis, the available information was incomplete, noisy, and shaped by evolving program goals. Effective progress required identifying which questions could be answered responsibly, adapting methods to limited data, and communicating results clearly to nontechnical stakeholders. This experience strengthened my ability to apply technical tools flexibly and reinforced the importance of aligning analysis with purpose rather than complexity.

Across both academic and nonacademic settings, a common theme has been learning to let problems guide methods rather than the reverse. Whether working with astronomical observations or institutional data, success depended on careful problem formulation, willingness to revise assumptions, and patience with iterative progress. These strengths have allowed me to contribute meaningfully to collaborative environments while continuing to grow toward greater independence.

Together, curiosity drives the questions I pursue, adaptability shapes how I respond when approaches fail, and persistence sustains progress through uncertainty. These qualities have supported my development as a researcher and prepared me to thrive in environments where learning, collaboration, and discovery are inseparable.

NRAO

NRAO/GBO Summer Student On-line Application Form

Contact Information
First Middle Last
Name
Email
Telephone
Web Page URL
US Citizen/Perm Resident? Yes No
Contact Information on March 1, 2026
Email
Telephone
Mailing address
Address (cont.)
City
State/Province
Zip/Postal code
Country

Educational Information (Please list your current college/university first)
College/University Major / Minor Dates Attended Degree and date received/expected

Fall 2026 Status: Sophomore Junior Senior Senior+
1st yr Grad School 2nd yr+ Grad School
Cumulative GPA: on a scale of

Research Area Preferences (enter 1,2,3 next to your first, second, and third preference)

Solar System 1Galaxies and AGN
Exoplanets and Stellar Systems Cosmology and high redshift
3 Star Formation / ISM Instrumentation/Engineering
Pulsars, FRBs, Radio Transients 2Scientific Computing / Software Engineering
Any Other

1. Galaxies and AGN
This is non-negotiable. It aligns perfectly with your ionization cone work, cubes, MIRI, and JWST experience.

2. Scientific Computing / Software Engineering
This signals that you are not just “an astronomy student who codes,” but someone who can contribute to pipelines, archives, and analysis tools. STScI, NRAO, and GBO love this combo.

3. Star Formation / ISM or Cosmology and high redshift
Pick Star Formation / ISM if you want closer physical overlap with emission lines and cubes.
Pick Cosmology and high redshift if you want breadth and archive-driven science.

If forced, I'd lean Star Formation / ISM because it keeps you close to spectroscopy and physical diagnostics.

If you have a preference to work with a specific advisor or advisors, please enter their names here:

Work Site Preference (check all that apply)

Socorro, NM Green Bank, WV Charlottesville, VA Other

3. Work Site Preference

Check all that honestly work for you.

Given your background:

✅ Charlottesville, VA (STScI-adjacent culture, archives, computing)

✅ Green Bank, WV (excellent for data pipelines, instrumentation awareness)

⚠️ Socorro, NM only if you are genuinely open to radio-heavy work

If you are unsure, check Charlottesville + Green Bank. That's a strong, coherent signal.

Programs

Check all that you would like to apply to. Choosing one does not affect your eligibility for others. More information on all student programs can be found here.

REU or undergrad researcher All undergraduates may select this option to apply for either the Research Experience for Undergraduate (REU) program or an undergrad research position.
West Virginia Space Grant Intern For students interested in engineering or education internships in Green Bank. Applications are welcome from students attending WV colleges or universities or WV residents attending out-of-state colleges.
PING The Physicists Inspiring the Next Generation (PING) program involves a research experience in Green Bank plus a one-week commitment to mentor 9th grade students during a summer camp on site.
Grad researcher Graduate students and all others who have completed their undergrad degree as of May 2026 should choose this program.

Work Experience

Computer languages & packages. Please list all that you have a working knowledge of (e.g. Python, CASA, AIPS, IDL, Java, HTML, C, etc.).
Language/Package Months Experience Class/project description

Language / Package	Months	Description
Python	18–24	Data analysis, FITS handling, machine learning, pipeline development for astrophysical research
NumPy / SciPy	12–18	Numerical analysis, synthetic dataset generation
Astropy	12	FITS I/O, coordinate handling, astronomy-specific analysis
PyTorch	8–10	UNet-based image segmentation, model training
DS9	6–12	Astronomical image visualization and inspection

You are allowed to estimate. Everyone does.

Other:

Scientific and Related Employment. List your relevant employment experience since high school, including assistantships.
Dates Employed Employer name and address
Job Title Job Description
WTRSP (Research Work Study) 01/2026 to Present

Full Time NEISD Custodian 05/2024 to 08/2024

to

to

Honors and Awards:
NSF APPEX Scholar, University of Texas at San Antonio, $8,000 per year (2024–2028).
Competitive, merit-based scholarship supporting students engaged in research-focused academic development.

Distinguished Presidential Scholarship, University of Texas at San Antonio, $5,000 per year (2024–2028).
Awarded for academic excellence and leadership potential.

Honors College Collaboration Housing Scholarship, University of Texas at San Antonio, $1,500 (2024).
Recognized for participation in interdisciplinary collaboration and community-based academic initiatives.

South Texas AAU Janice Henderson Scholarship, $1,250 (2024).
Awarded for academic achievement and leadership in athletics.

Honors College Student Advisory Board Stipend, University of Texas at San Antonio, $500 per semester (2025–2026).
Selected to contribute to Honors College programming and student engagement initiatives.

James Madison Hall of Honor, Track and Field (2024).
Inducted in recognition of athletic achievement and leadership.

Other Relevant Experience:
Through the Society of Physics Students at UTSA, I have taken part in peer-led technical discussions, research skill workshops, and informal mentoring that emphasized collaborative problem solving and clear scientific communication. These experiences reinforced the importance of explaining complex concepts at an appropriate level, whether discussing observational astronomy, data analysis techniques, or computational methods. In parallel, I have contributed machine learning support for Honors College events through the Honors College Advisory Board, where technical tools were developed or adapted under real logistical constraints rather than idealized classroom settings. Together, these experiences strengthened my ability to work across disciplines, communicate technical ideas clearly, and translate abstract methods into practical outcomes.

Hobbies, Sports, etc.

Outside of research, I am actively involved in coaching and athlete development. I coach with the Randolph Roadrunners AAU Track and Field team and work individually with student athletes at UTSA. In my coaching, I focus on explaining movement and training concepts through basic physics and biomechanics, while accounting for each athlete's academic and daily workload so that training complements rather than overwhelms the rest of their responsibilities. I also follow a structured, progressive training program modeled after Olympic and Division I approaches, which has reinforced habits of long term planning, consistency, and reflective improvement.

Personal Essay

Please write a short essay (1-2 pages, 500-1000 words) about your background, interest in astronomy, and why you are a good candidate for the program(s) to which you are applying.

My interest in astronomy grew from a broader habit of questioning that developed early in my life. When my family left the Mormon church, it was not driven by rebellion, but by unanswered questions. Navigating that transition required patience, careful listening, and a willingness to sit with uncertainty rather than resolve it prematurely. Observing thoughtful conversations among

family members who held deeply different views taught me to separate emotional reaction from reasoned judgment. That habit of inquiry, evaluating ideas on their merits and remaining comfortable without immediate answers, ultimately became the foundation of how I approach scientific problems.

That mindset found its natural home in astrophysics. I was drawn to the field not only by its scale and ambition, but by the way it combines mathematical structure with incomplete information. In astronomy, data are often indirect, noisy, and interpreted through models that must be questioned as carefully as they are constructed. What captivated me was not simply learning what is known about the universe, but grappling with how knowledge is extracted from imperfect observations.

At the University of Texas at San Antonio, my academic development was shaped early by participation in the Academic Personal and Professional Excellence Research Training Program. Through APPEX, research became something intentional rather than incidental. Goal setting through Individual Development Plans, sustained mentorship, and the expectation that progress is iterative introduced me to research as a long-term process rather than a sequence of assignments. This framework prepared me to manage open-ended questions and to revise strategies when evidence demanded it.

Those principles became central once I began research in Dr. Chris Packham's group, where I encountered the challenges of working with real astronomical data. While studying Active Galactic Nuclei using JWST NIRCam observations, it became clear that many scientifically important structures are still identified by eye. Ionization cones, which trace the geometry and orientation of AGN outflows, are faint, asymmetric, and often obscured. Manual identification is slow and subjective, and it does not scale to the growing volume of space-based data. The problem was not a lack of observations, but a lack of consistent, reproducible methods for extracting physical structure from them.

Addressing this limitation required moving beyond traditional analysis. I began exploring whether machine learning could be used not as a black box, but as a physically informed tool for segmentation and interpretation. The absence of labeled training data forced a deeper engagement with the underlying astrophysics. Physical assumptions about AGN geometry, orientation, asymmetry, and dust attenuation had to be translated into synthetic emission maps with known ground truth. Early failures highlighted how sensitive results were to dataset design, loss functions, and evaluation metrics. Progress depended on diagnosing why models failed and deciding which assumptions to revise, rather than simply tuning hyperparameters.

Through this process, I developed a complete pipeline that trained a convolutional neural network to segment ionization cones automatically. More importantly, the project reframed how I think about computation in astronomy. Coding became a form of experimentation, and models became hypotheses about structure rather than tools for pattern recognition alone. The experience confirmed my interest in a research career focused on building scalable, physically interpretable analysis methods.

As my work progressed, it also became clear that two-dimensional imaging alone does not always provide clean reasoning paths for models or for humans. This realization motivated a shift toward spectroscopic data cubes and emission line diagnostics that encode physical transitions more directly. Developing a bridge from imaging to cube-based analysis using line ratios reinforced my interest in observatory-scale data pipelines and archival science, where careful calibration, reproducibility, and physical context are essential.

Beyond formal research, I have sought opportunities to apply similar reasoning skills in broader contexts. Through service on the Honors College Student Advisory Board, I have contributed to projects that use data-driven tools to improve student engagement and event planning. In athletics, coaching track and field athletes has reinforced the value of clear communication and systems thinking. Training plans must balance stress and recovery, just as research workflows must balance ambition with feasibility. Explaining complex mechanics using simple physical principles has strengthened my ability to teach and mentor, an aspect of academia I value deeply.

I intend to pursue a PhD in astrophysics and a research-focused academic career developing computational tools to study galaxy evolution and AGN feedback. I am particularly interested in problems where scientific progress is limited not by data availability, but by our ability to extract physical meaning from complex observations. My long-term goal is to design transparent, reproducible pipelines that allow large datasets from facilities like NRAO and its archives to be used efficiently and responsibly.

The NRAO/GBO Summer Student Program aligns strongly with these goals. Its emphasis on hands-on research, observatory infrastructure, and large-scale data products offers an environment where computational methods are developed in direct conversation with instrumentation, calibration, and archives. Working within such a setting would deepen my understanding of how scientific tools are built, validated, and sustained, while allowing me to contribute meaningfully to ongoing research efforts.

From early experiences learning to navigate uncertainty to current work at the intersection of astrophysics and machine learning, I have learned that progress depends as much on how questions are framed as on the tools used to answer them. I am eager to continue developing that perspective through the NRAO/GBO Summer Student Program, contributing careful, motivated work while learning how large astronomical facilities turn data into discovery.

How did you first learn about the NRAO/GBO Summer Student program?

Faculty/Advisor NRAO/GBO Website NSF Website
REU Poster Friend AAS Meeting
Facebook AAS Newsletter CSMA Newsletter
SACNAS Newsletter Other (please specify): AAS Website

If you have any special instructions or comments, please enter them here.

Letters of Recommendation/References

Arrange to have up to three faculty members or advisors complete letters of reference by February 1, 2026.

Inform them that letters of reference must be submitted using the online form at http://www.nrao.edu/php/students/references.php.

If you are requesting fewer than three letters, please fill in "n/a" to the text fields below for the second and third references.

NRAO/GBO will not send notifications to your letter of reference writers. You must send them the link above.

Letter Writers:
First Name Last Name Institution Phone Email

Personal Information

Completing the following section is optional, but provides important information that will be used for reporting statistics to the National Science Foundation.

Which one of these categories best describes your ethnic/racial status? (If more than one category applies, use the category that most closely reflects your community.)

American Indian or Alaskan Native Asian
Black or African American Hispanic or Latino
Pacific Islander White, not of Hispanic Origin

Gender (Which pronouns do you use?):

Check here if you do not wish to provide some or all of the above information.

TRANSCRIPTS: Attach your academic transcipt below. Note that:

Only PDF files are accepted.
All of your transcripts must be in one PDF file.
Unofficial transcripts are ok.

yale

My long-term goal is to pursue a PhD in astrophysics focused on developing computational pipelines that extract physically meaningful information from complex astronomical datasets. I am particularly interested in combining machine learning with observational data to study galaxy evolution and active galactic nuclei (AGN) feedback. A summer at Yale offers the opportunity to engage directly with high-quality data, advanced analysis techniques, and collaborative research that aligns perfectly with these goals.

At the University of Texas at San Antonio, I have been working with JWST-style infrared observations of AGN, designing Python pipelines and training convolutional neural networks to segment ionization cones. In the early stages, my models often failed to produce meaningful results because I had not yet learned how to translate physical intuition into computational design. By iterating on synthetic datasets, revising assumptions about AGN geometry and emission, and analyzing why the models failed, I gained a deep understanding of both the astrophysics and the computational methods. This process strengthened my problem-solving skills and taught me how to manage open-ended research challenges with persistence and precision.

I have also learned the value of collaboration and communication in research. Working closely with my mentor and peers, I had to explain complex concepts, justify design choices, and incorporate feedback into my pipelines. Adapting computational tools to observational constraints forced me to think creatively and systematically. A summer at Yale would provide similar opportunities to work closely with experienced researchers and peers, allowing me to apply and expand these skills in a new, highly interactive research environment.

I am particularly drawn to projects that integrate data-intensive analysis with hands-on experience in astronomical instrumentation and archives. Contributing to ongoing research at Yale would give me exposure to the full workflow of modern astrophysics, from observation and calibration to analysis and interpretation. Working with complex datasets under mentorship will challenge me to navigate measurement noise, unexpected results, and practical limitations. These experiences will sharpen my ability to validate models, ensure reproducibility, and think critically about every step of a research project. These are exactly the skills I need to tackle open-ended problems in galaxy evolution and AGN feedback.

A summer at Yale represents a crucial step toward achieving my long-term goals. It would allow me to combine computational methods, physical reasoning, and observational practice in a structured, collaborative setting. The experience will enhance my ability to tackle open-ended astrophysical problems and prepare me for graduate research, where I aim to develop transparent AI-driven tools that advance our understanding of the universe.

NASA internship

I am applying to work on the Habitable Worlds Observatory because my research already sits at the intersection of astrophysics, computation, and observational realism. I build synthetic emission-line images and spectral cube datasets in Python and train machine learning models to trace physical structures in noisy data. This work requires constructing end-to-end pipelines that incorporate modeling, instrument effects, and quantitative validation. Those same skills translate directly to transit spectroscopy and evaluating how changes in mirror size, wavelength coverage, or spectral resolution affect atmospheric retrieval. I am especially motivated by the opportunity to contribute to mission-level trade studies that shape how we detect and characterize Earth-like planets. Working with scientists at GSFC and JPL would allow me to apply my technical background to a problem that directly informs the future of exoplanet discovery.

Honors Essay.

In May 2025, I stood in front of a poster at the NSF-Simons Cosmic AI Institute's Cosmic Horizons conference and explained my research on active galactic nuclei (AGN), the luminous cores of distant galaxies powered by supermassive black holes, to scientists who had spent careers studying the same objects. I was nervous. Not the kind of nervous that makes you want to leave, but the kind that tells you the distance between where you are and where you want to be is real, and that you are standing exactly at its edge. I had spent months building a machine learning pipeline to detect ionization cones, cone-shaped regions of gas energized by AGN outflows, in galaxy images. The work involved failures that forced me to learn not just how to code but how to think like a researcher. Standing at that poster, I was not yet the astrophysicist I intend to become. But I was already closing the distance.

That tension, between where I am and what I am building toward, defines my Honors experience at UTSA. The Honors College did not ask me to arrive ready. It asked me to move with intention, and that invitation changed everything.

Academically, my path has been shaped by a commitment to doing science that holds up. As an undergraduate researcher in the UTSA Mid-Infrared Astronomy and Education (MAE) Group under Dr. Chris Packham, I have worked on problems where no labeled training data existed, where standard tools were insufficient, and where progress required building from first principles. When my early UNet (a convolutional neural network architecture designed for image segmentation) models failed to detect ionization cones meaningfully, I did not treat it as a setback to minimize. I treated it as information. I revised my assumptions about AGN geometry, redesigned my synthetic datasets, and rebuilt the pipeline until the results were physically interpretable and reproducible. Honorable research is not research that succeeds quickly. It is research that does not cut corners when cutting corners would be easy. That standard is one I carry into every project, and the Honors College gave me the environment to develop it seriously.

Resilience, for me, is not a trait I describe. It is a method I practice. The version of me that walked into Dr. Packham's lab in August 2024 did not know Python at a research level, had never worked with FITS (Flexible Image Transport System) data files used in astronomy, and had never written a machine learning model. The version of me that presented at Cosmic Horizons nine months later had built all of it, iteratively, through failure. I did not wait until I felt ready. I diagnosed what was missing, acquired it, and moved forward. That same instinct carried me through the Astro-AI Boot Camp at NSF-Simons and into my current work developing spectroscopic cube analysis methods that extend beyond imaging alone. The distance between ignorance and capability is closed only by moving through it deliberately.

Outside the lab, I have tried to close that same distance for others. As a sprint coach with the UTSA Sprint Group and the Randolph Roadrunners, a community club track team, I work with athletes who have talent but lack the technical foundation to develop it fully. I know that gap personally. I was a sprinter at James Madison High School who had to seek out the right systems on my own. Coaching is not charity. It is the recognition that the infrastructure for development should not be something athletes have to find by accident. I build training environments that are structured, honest, and demanding, because that is what I needed and did not always have. The same principle extends to my work on the Honors College Student Advisory Board, where I have contributed to data-driven approaches to student engagement,

applying the same analytical instincts from my research to the problem of building a stronger Honors community.

My vision for the remainder of my Honors experience is an extension of everything above. I intend to pursue a PhD in astrophysics, focused on developing transparent and physically interpretable computational pipelines for studying galaxy evolution and AGN feedback, the process by which black holes influence the growth of their host galaxies. I want to build tools that make large-scale astronomical datasets more accessible, not just technically but scientifically, so that future researchers spend less time on what the data cannot tell them and more time on what it can. The Honors College has taught me that this kind of contribution requires more than intelligence. It requires integrity in the work, resilience in the process, and a genuine commitment to the people who will come after you.

I stood at that poster nervous but determined. That is still where I stand. The distance is still real. And I am still closing it.

USRA

A completed application form. Only completed application forms will be considered.

A one page (< 600 words) statement stating the applicant's career goals, and relevant past accomplishments and education in the field of space research, aerospace related engineering, or space science education. Please include the applicant's name at the top of the page.

A half-page (< 325 words) description of particular examples of extra-curricular activities that demonstrate leadership or having a broader impact. These may include K-12 activities, community outreach, mentoring, a space science or engineering club/organization, advancing opportunities in STEM, or other ways of giving back to the community. Note that the quality and impact of the activity are more important than the number of activities. Limit is no more than three examples. Please include the applicant's name at the top of the page.

Two letters of recommendation with at least one from a professor, mentor, or school official at the college level. We recommend the letter writers be personally familiar with the applicant's accomplishments, performance, skills, character, and/or experience. We also recommend, where possible, that at least one letter be from a research/project advisor at postdoc level or higher. NOTE: Application forms should be submitted with enough lead time for references to submit letters of recommendation before the August 19 deadline. See instructions below for details.

An official college transcript showing two or more years of college credits including GPA. The transcript should include grades from the Spring 2026 semester. If the two or more years include credits transferred from another university, college, or community college, a transcript from that institution should also be submitted.

A Resume/CV of no more than 2 pages. It should include a list of any college or university you have attended (including your current one), with the dates attended and degree earned (if applicable). It should also include school and community honors and employment history (including dates). Also include a list any leadership positions you may have held in a college/university organization or in a significant project. You may also include research or engineering projects you have been involved with and any publications and/or conference presentations.

William Olson

I want to become an astrophysicist who builds the tools that let scientists actually keep up with what our observatories are collecting. Astronomy is entering an era where the bottleneck isn't observing the universe, it's making sense of what we've already observed. Every year, facilities like JWST produce more data than any team of researchers could inspect by hand. My goal is to develop machine learning methods that close that gap, not by replacing scientific judgment, but by giving researchers a way to find the structure hidden inside datasets too large to search manually.

I am pursuing a Bachelor of Science in Physics, with minors in Mathematics and Astrophysics, at the University of Texas at San Antonio. My coursework has given me the physical foundation this work depends on, but it's my research that has shown me how that foundation actually gets used.

In the Mid-Infrared Astronomy and Exoplanet (MAE) Research Group, I study active galactic nuclei (AGN), the extremely bright centers of galaxies powered by supermassive black holes, using observations from the James Webb Space Telescope. My research focuses on identifying ionization cones, the cone-shaped regions of gas illuminated by radiation escaping the black hole, using machine learning models built around the physics of how those structures actually form. That's the piece I keep coming back to: a model that fits patterns without understanding the physics behind them will eventually fail on the case that matters most. Building it the other way around, physics first, has been the harder path but the one I trust.

This past summer, I extended that approach to a different part of astronomy through an NSF Research Experiences for Undergraduates (REU) appointment with the Breakthrough Listen Initiative at UC Berkeley, evaluating technosignature search algorithms against Green Bank Telescope data. The science questions in radio astronomy are nothing like the ones I ask about AGN, but the underlying problem is identical: how do you build a method reliable enough to trust when you're searching for something rare inside an enormous amount of noise. Working on both problems in the same year taught me that this question, not any specific technique, is what I actually want to spend a career on.

Those two projects have shaped what I want to become: a researcher working at the boundary of astrophysics, machine learning, and instrumentation, building systems that help scientists get more out of the data we already have. Whether the target is a distant AGN, an exoplanet's atmosphere, or a technosignature candidate, the tools should be the same in principle, and I want to be the one building them.

Outside the lab, I've presented this work at university, national, and international conferences, and I coach track and mentor students interested in research and college. I don't see these as separate from my research. Science doesn't move forward on discoveries alone; it needs people willing to explain the work and bring others into it, and I take that responsibility seriously.

The URSA Scholarship would support the next stage of that path as I prepare for graduate study in astrophysics. I want to keep building the kind of tools that let researchers ask bigger questions of the data they already have, and to keep getting better, in the lab and in front of a room, at explaining why that work matters.

William Olson

I want to become an astrophysicist who builds the tools that let scientists actually keep up with what our observatories are collecting. Astronomy is entering an era where the bottleneck isn't observing the universe, it's making sense of what we've already observed. Every year, facilities like JWST produce more data than any team of researchers could inspect by hand. My goal is to develop machine learning methods that close that gap, not by replacing scientific judgment, but by giving researchers a way to find the structure hidden inside datasets too large to search manually.

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These experiences have shaped what I want to become: a researcher working at the boundary of astrophysics, machine learning, and instrumentation, building systems that help scientists get more out of the data we already have. Whether the target is a distant AGN, an exoplanet's atmosphere, or a technosignature candidate, the tools should follow this data-driven principle: build methods that allow scientists to discover structure hidden within complex observations.

I have presented this work at university, national, and international conferences. These experiences have strengthened my ability to communicate scientific ideas and collaborate with researchers across different fields.

The URSA Scholarship would support the next stage of that path as I prepare for graduate study in astrophysics. I want to continue developing tools that allow researchers to extract discoveries from increasingly complicated observations, while becoming the kind of scientist who can bridge theory, computation, and communication. Ultimately, I want to help build the future of astronomy: one where our ability to understand the universe grows alongside our ability to observe it.

William Olson

Last year, I contacted my former physics teacher at James Madison High School and asked if I could speak to students enrolled in a Dual Credit Physics course. I wanted to explain what pursuing a Ph.D. really looks like: coursework, mentorship, research, and years spent on science. When I was in high school, I had never met anyone pursuing a Ph.D., let alone someone only a few years older who was already on that path. I wanted those students to see that a research career was something they could realistically pursue.

I have found a similar opportunity to mentor through volunteer sprint coaching. Coaching has taught me that improvement rarely comes from giving better instructions alone. Different athletes respond to different approaches, and correct and meaningful progress comes from understanding what is holding each person back, be it technique, confidence, or experience. Learning to adapt my communication to the individual has made me a better mentor and has played a big role in how I collaborate with other students and researchers.

Communicating science has become just as important to me as conducting it. One of my advisors at UT San Antonio often reminds our group that "we are not mages in ivory towers." As someone who enjoys fantasy, the idea really resonated with me. Scientific discoveries only matter when other people can understand them, question them, and build upon them. Whether I am speaking to a classroom of high school students or explaining my research to someone outside my field, I want people to understand not only what I do, but why it matters.

As I continue toward a career in astrophysics, I hope to contribute not only through research, but also through mentorship, outreach, and scientific communication. Discovery is a collaborative process, and I will help create an environment where more students can see themselves as future scientists and where new ideas are shared open for others to build upon them.

I want to become an astrophysicist who builds the tools that let scientists keep pace with what our observatories are collecting. Modern astronomy is entering an era where our ability to collect observations is advancing faster than our ability to interpret them. Every year, facilities like JWST produce more data than any team of researchers could inspect by hand. My goal is to develop machine learning methods that close that gap, not by replacing scientific judgment, but by giving researchers a way to find the structure hidden inside datasets too large to search manually.

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These experiences have clarified the kind of researcher I want to become: one working at the boundary of astrophysics, machine learning, and instrumentation, building systems that help scientists get more out of the data we already have. Whether the target is an active galactic nucleus, an exoplanet atmosphere, or a potential technosignature, the objective remains the same: develop methods that allow scientists to recognize meaningful structure hidden within increasingly complex data. I have presented this work at university, national, and international conferences, including the Kyoto Torus Workshop in Japan. Presenting to researchers with different scientific backgrounds has taught me that a method is only useful if others can understand it, question it, and build upon it.

The URSA Scholarship would support the next stage of that path as I prepare for graduate study in astrophysics. I want to continue developing tools that help researchers extract discoveries from increasingly complex observations while becoming the kind of scientist who

can bridge theory, computation, and communication. My goal is to spend my career building methods that allow scientists to keep pace with the universe they are observing.

✅ Step 2 (10 minutes): Replace placeholder research

Instead of

- AI for Astronomy
- Galaxies
- Stellar Physics
- Exoplanets

I'd change them to your *actual* research:

1. AGN
2. Physically based ML
3. Technosignature Searches
4. Exoplanets

Those four basically summarize your research identity.

✅ Step 3 (15 minutes): Finish Projects

Replace the template projects with things like:

- AGN Ionization Cone Pipeline
- Breakthrough Listen MidRes Analysis
- Honors College Event Analyzer

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I am pursuing a Bachelor of Science in Physics, with minors in Mathematics and Astrophysics, at the University of Texas at San Antonio. My coursework has given me the physical foundation this work depends on, but it's my research that has shown me how that foundation is used. After completing my undergraduate degree, I plan to pursue a Ph.D. in astrophysics, developing computational methods and machine learning tools that expand the scientific return of current and future observatories.

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Tab 37